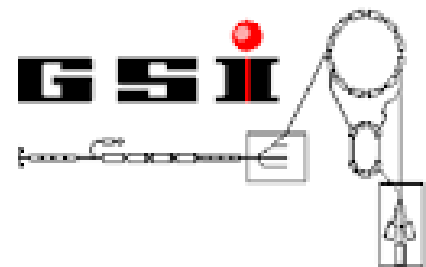




XV International Conference on New Frontiers in Physics

19 - 30 August 2026, OAC, Kolymbari, Crete, Greece

Formation-Time Suppression of Jet Radiation in the sQGP



HFHF

Olga Soloveva (GSI & HFHF &
ITP Goethe University Frankfurt)
I. Grishmanovskii, G. Ingrosso, T. Song,
E. Bratkovskaya, C. Greiner

2609.xxxx

21 August 2026

Properties of QGP: transport coefficients

"Numerical simulations are now essential to make contact between theory and experiment" - J. Kapusta

! One has to specify transport and microscopic properties as well as EoS for theoretical simulations of HICs (hydro / transport approaches)



EoS(ϵ, n)
 $\sigma(\sqrt{s}, m_q, m_q, T, \mu_B)$
 $m(T, \mu_B)$

On practice: effective models
for QGP

Transport simulations with QGP phase:

Catania transport – QuasiParticle Model

F. Scardina, S. K. Das, V. Minissale, S. Plumari, and V. Greco,
PRC 96, 044905 (2017).



– Dynamical **QPM** for partonic phase

W. Cassing, E. Bratkovskaya, PRC 78 (2008) 034919
P. Moreau, O. S, L. Oliva, T. Song, W. Cassing,
E. Bratkovskaya, PRC 100 (2019) , 014911;
O. S, P. Moreau, L. Oliva, V. Voronyuk, V. Kireyeu, T. Song,
E. Bratkovskaya, Particles 3 (2020), 178-192

AMPT – PNJL EoS (Mean field potentials)

K.J. Sun, C. M. Ko, and Z.-W. Lin, PRC 103(2021)

Hybrid simulations with QGP:

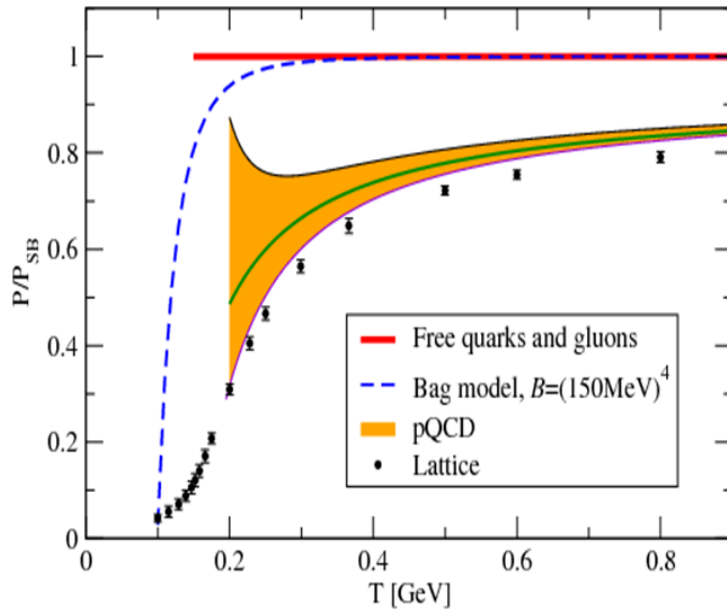
vHLLE/Music+UrQMD/SMASH

Iu.A. Karpenko, P. Huovinen, H. Petersen and M. Bleicher
PRC 91 (2015), 064901.
S. Ryu, J.F.Paquet, C. Shen, G.S. Denicol, B. Schenke
PRL 115 (2015), 132301



Degrees of freedom of QGP

IQCD: QGP EoS at finite μ_B



Non-perturbative QCD ← pQCD

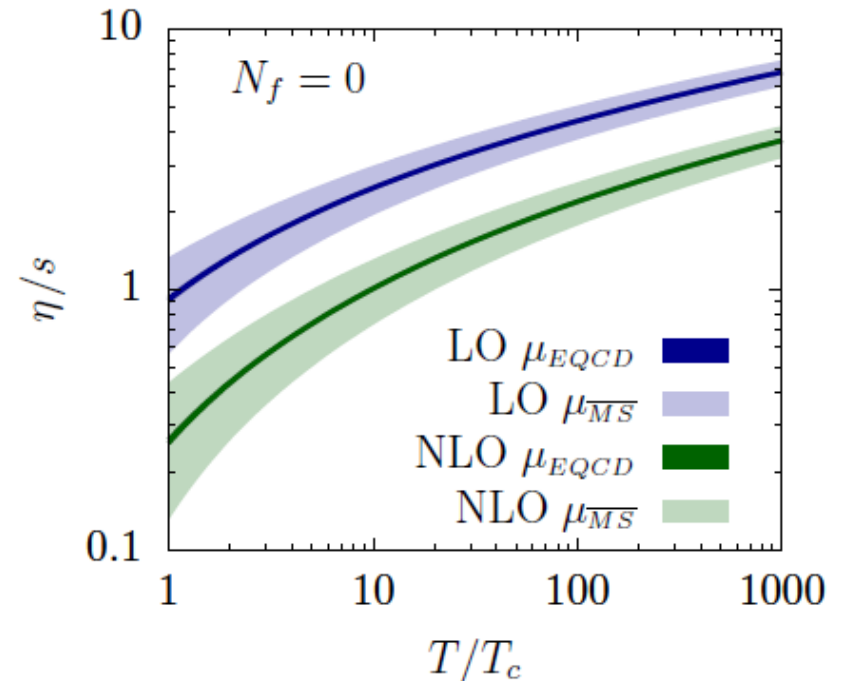
pQCD:

- ☐ weakly interacting system
- ☐ massless quarks and gluons



pQCD: (Yang-Mills) shear viscosity η

J. Ghiglieri, G.D. Moore, D. Teaney, JHEP 1803 (2018) 179



Thermal (non-perturbative) QCD:

- ☐ strongly interacting system
- ☐ massive quarks and gluons

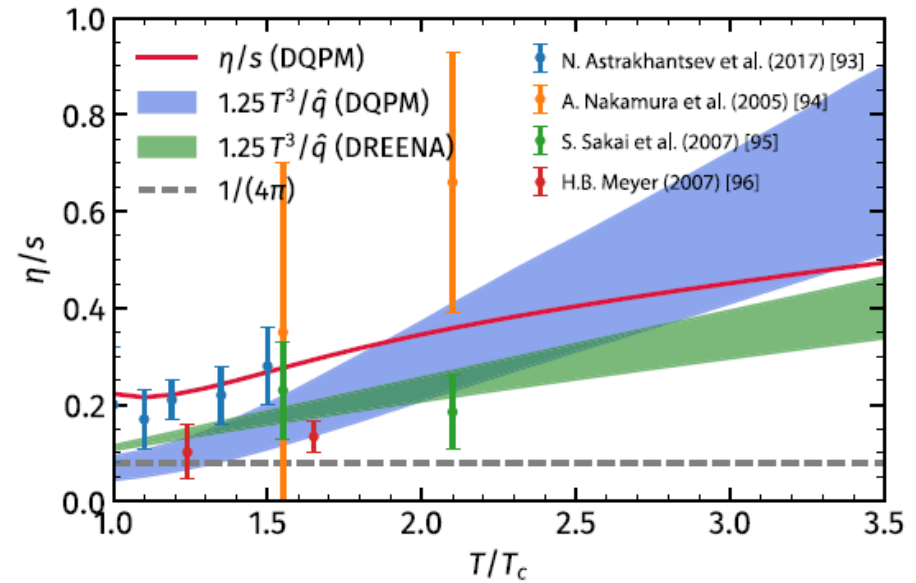
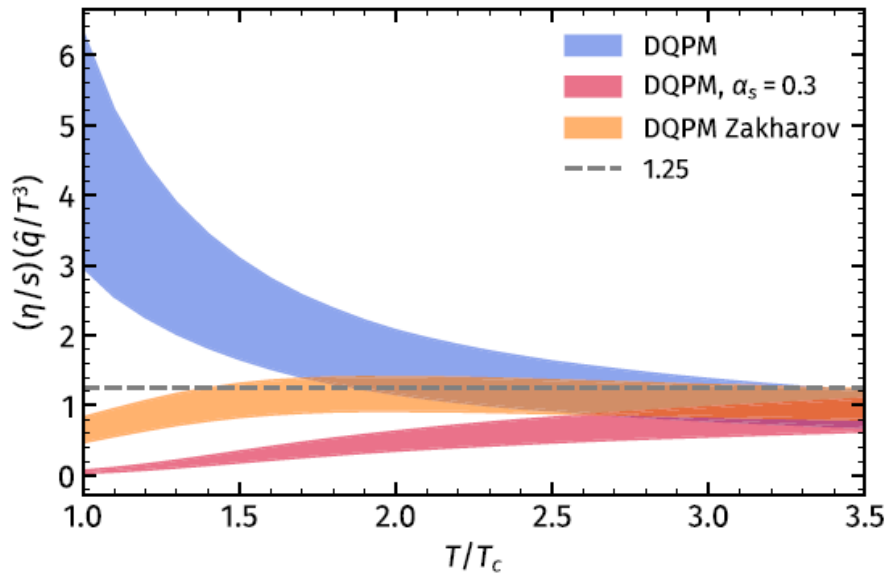
→ Quasiparticles = effective degrees-of-freedom

Relation between η/s and $\hat{q}$

In the **weak coupling limit**:

B. Müller, PRD 104, L071501 (2021)

$$\eta/s \approx 1.25 \frac{T^3}{\hat{q}}$$



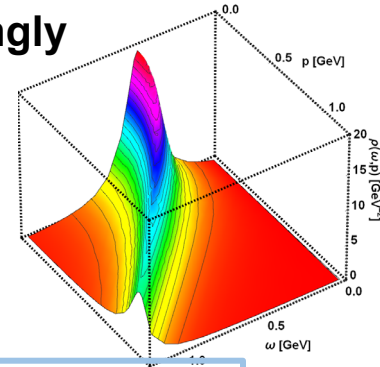
- **Sensitive** to the choice of the **strong coupling**
- **Valid** in the **weak coupling regime** (at high temperatures)
- **Violated** in the **strong coupling regime** (at low temperatures)

Dynamical Quasi-Particle Model (DQPM)

DQPM – effective model for the description of **non-perturbative** (strongly interacting) QCD based on **IQCD EoS**

Degrees-of-freedom: strongly interacting **dynamical quasiparticles** - quarks and gluons

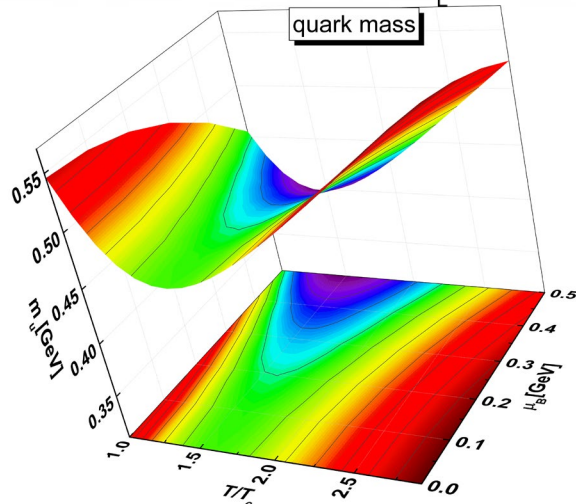
$$\rho_j(\omega, \mathbf{p}) = \frac{\gamma_j}{\tilde{E}_j} \left(\frac{1}{(\omega - \tilde{E}_j)^2 + \gamma_j^2} - \frac{1}{(\omega + \tilde{E}_j)^2 + \gamma_j^2} \right)$$



$$\text{resummed propagators: } \Delta_i(\omega, \mathbf{p}) = \frac{1}{\omega^2 - \mathbf{p}^2 - \Pi_i} \quad \& \quad \text{self-energies: } \Pi_i = m_i^2 - 2i\gamma_i\omega$$

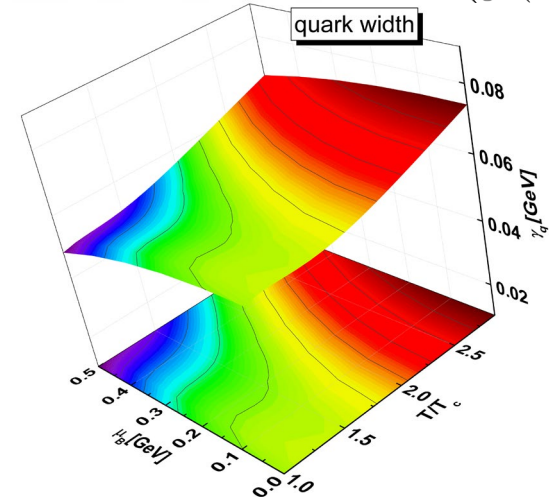
$\text{Re } \Pi_i$: thermal mass (M_g, M_q)

$$m_{q(\bar{q})}^2(T, \mu_B) = C_q \frac{g^2(T, \mu_B)}{4} T^2 \left[1 + \left(\frac{\mu_B}{3\pi T} \right)^2 \right]$$



$\text{Im } \Pi_i$: interaction width (γ_g, γ_q)

$$\gamma_j(T, \mu_B) = \frac{1}{3} C_j \frac{g^2(T, \mu_B) T}{8\pi} \ln \left(\frac{2c_m}{g^2(T, \mu_B)} + 1 \right)$$



DQPM: Parton properties

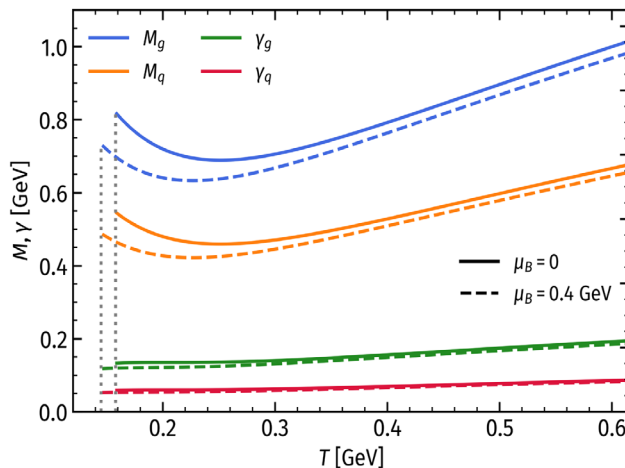
Realization concept:

- introduce an **ansatz** (HTL; with few parameters) for the (T, μ_B) dependence of masses/widths
- evaluate the **QGP thermodynamics** in equilibrium using the Kadanoff-Baym theory
- fix DQPM parameters by comparison of the DQPM entropy density to **IQCD** at $\mu_B=0$
- Masses** and **widths** of quasiparticles depend on T and μ_B

$$m_g^2(T, \mu_B) = C_g \frac{g^2(T, \mu_B)}{6} T^2 \left(1 + \frac{N_f}{2N_c} + \frac{1}{2} \frac{\sum_q \mu_q^2}{T^2 \pi^2} \right)$$

$$m_{q(\bar{q})}^2(T, \mu_B) = C_q \frac{g^2(T, \mu_B)}{4} T^2 \left(1 + \frac{\mu_q^2}{T^2 \pi^2} \right)$$

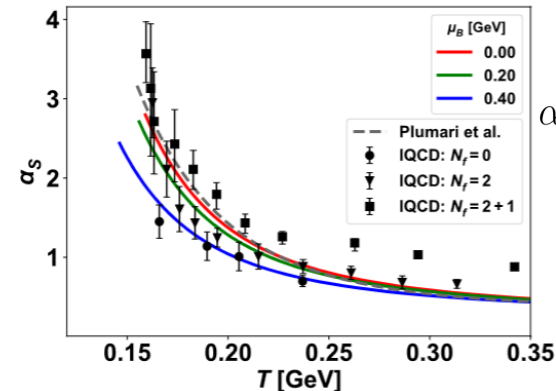
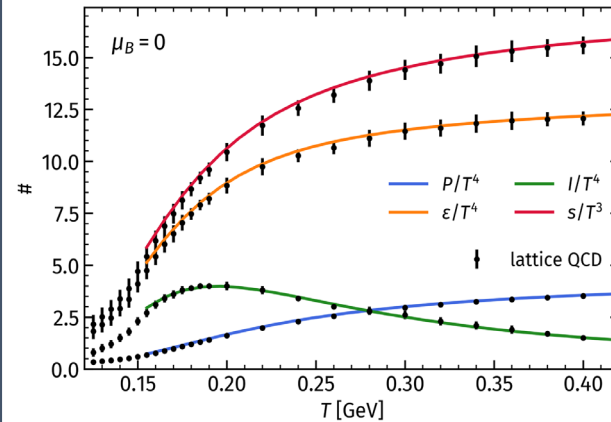
$$\gamma_j(T, \mu_B) = \frac{1}{3} C_j \frac{g^2(T, \mu_B) T}{8\pi} \ln \left(\frac{2c_m}{g^2(T, \mu_B)} + 1 \right)$$



- Strong coupling** (g) is defined from **IQCD entropy density** at $\mu_B=0$, using

$$g^2(s/s_{SB}) = d((s/s_{SB})^e - 1)^f$$

$$s_{SB}^{QCD} = 19/9 \pi^2 T^3$$



$$\alpha_s = g^2(T, \mu_B)/(4\pi)$$

→ DQPM allows to explore QCD in the **non-perturbative regime** of the (T, μ_B) **phase diagram**

Partonic interactions: matrix elements

DQPM partonic cross sections → **leading order diagrams**

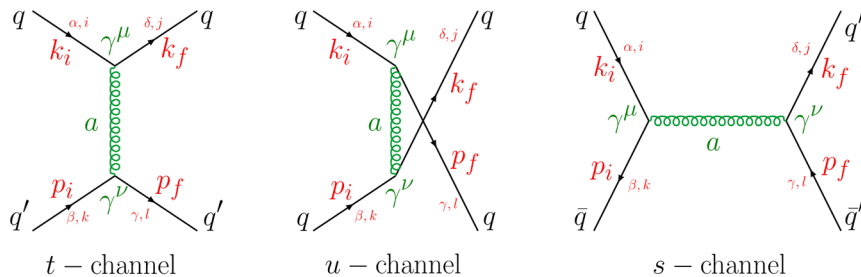
□ **Propagators** for massive bosons and fermions:

$$\text{boson propagator} = -i\delta_{ab} \frac{g^{\mu\nu} - q^\mu q^\nu / M_g^2}{q^2 - M_g^2 + 2i\gamma_g q_0}$$

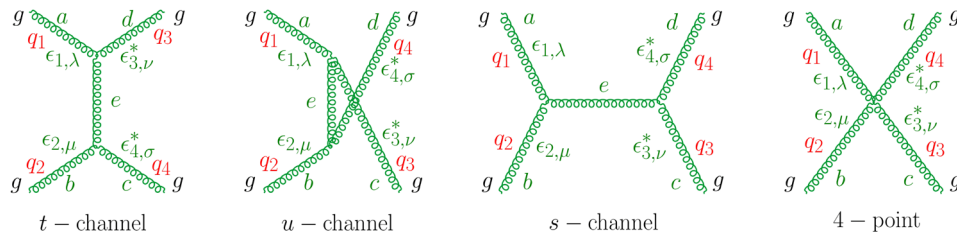
□ **(Quasi-) elastic channels:**

qq' → qq' scattering

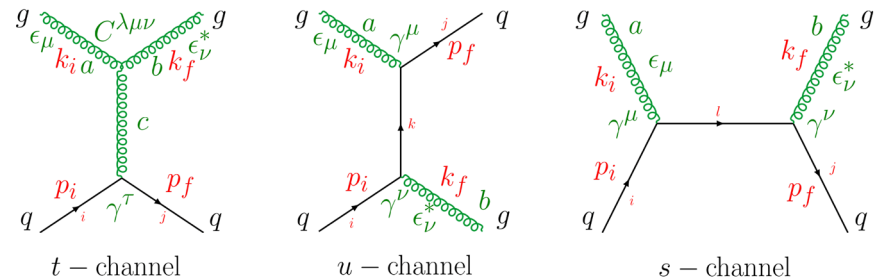
$$\text{fermion propagator} = i\delta_{ij} \frac{\not{q} + M_q}{q^2 - M_q^2 + 2i\gamma_q q_0}$$



gg → gg scattering



gq → gq scattering



□ **Inelastic channels:**

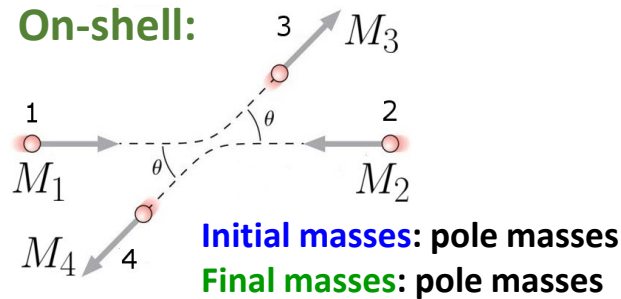
$$q + \bar{q} \rightarrow g$$

$$g \rightarrow q + \bar{q}$$

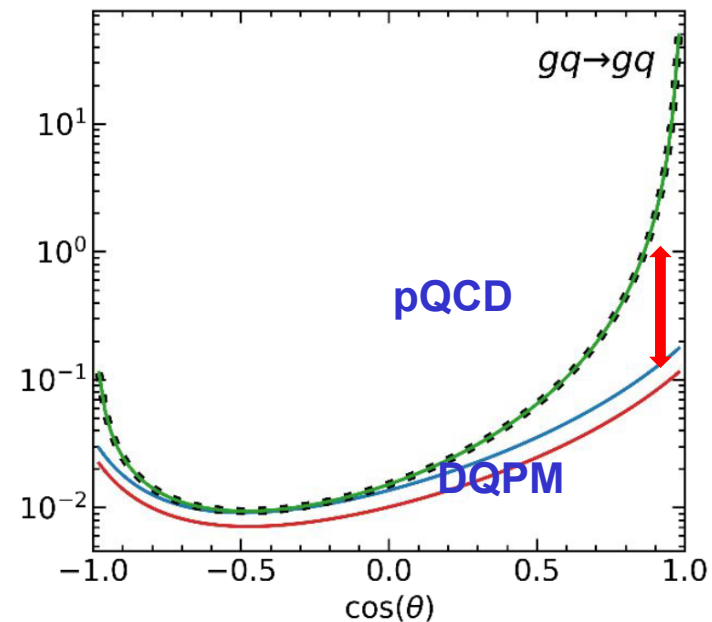
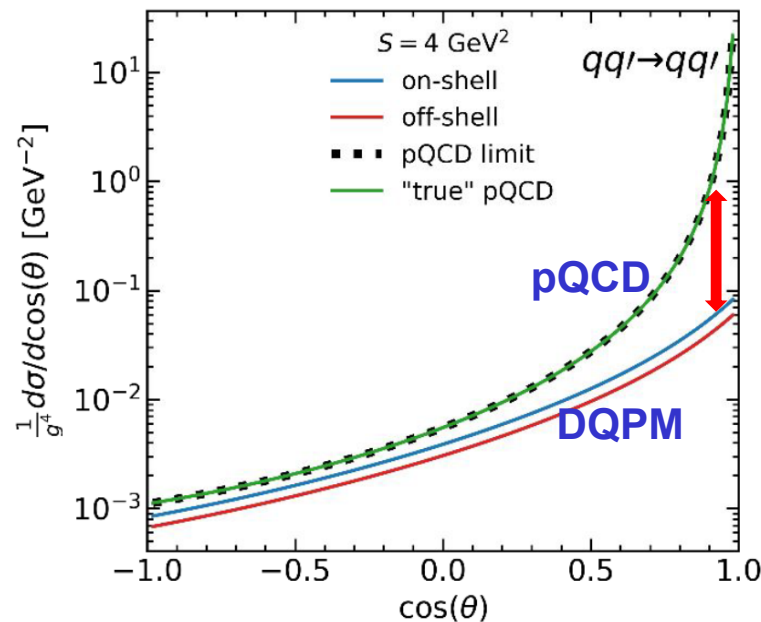
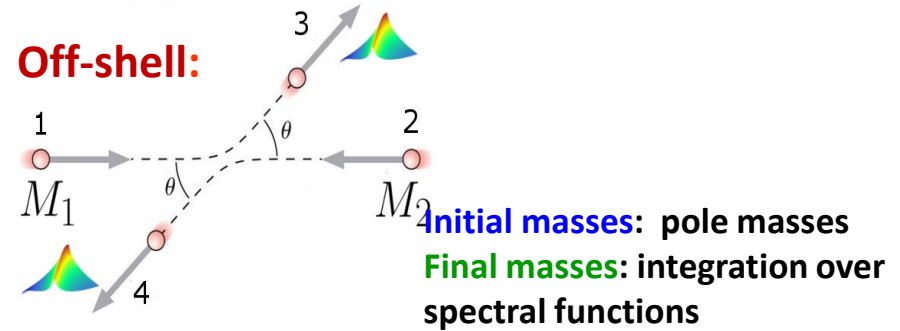


Differential cross section

On-shell:



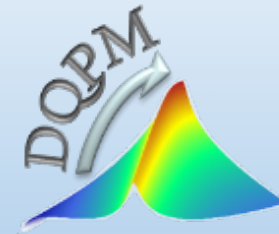
Off-shell:



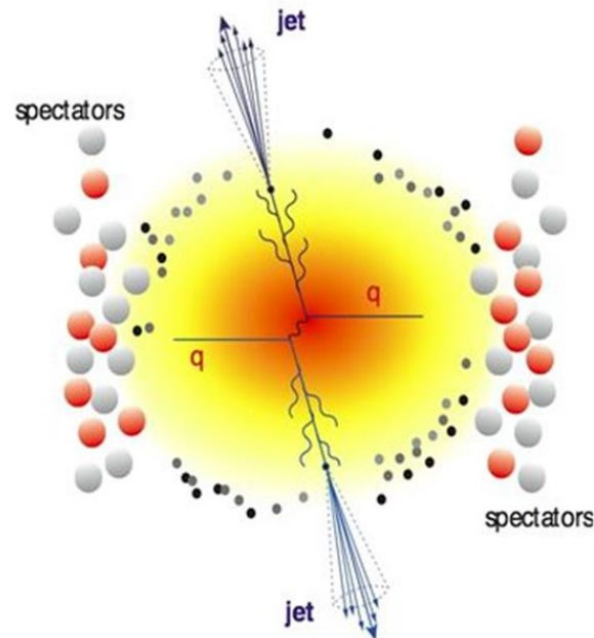
Plot by Ilia Grishmanovskii

- DQPM: $M \rightarrow 0, \gamma \rightarrow 0 \rightarrow$ reproduces pQCD limits
- Differences between DQPM and pQCD : less forward peaked angular distribution leads to more efficient momentum transfer

Probing the properties of the sQGP with jet partons

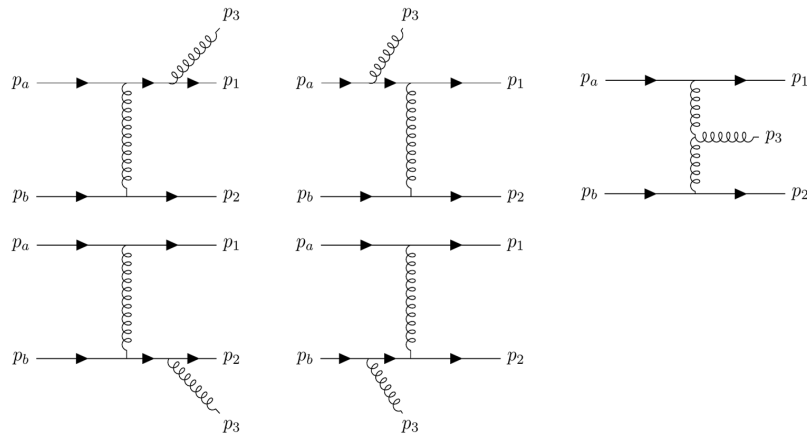


Extension of DQPM (T, μ_q)
elastic ($2 \rightarrow 2$) + inelastic ($2 \rightarrow 3$) scattering

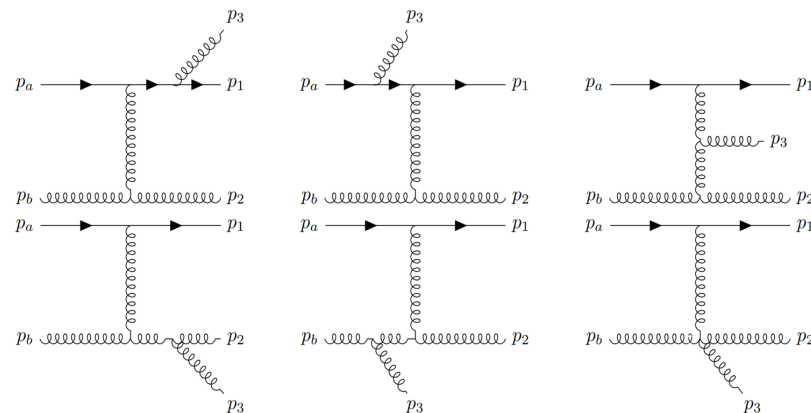


Partonic inelastic 2→3 interactions

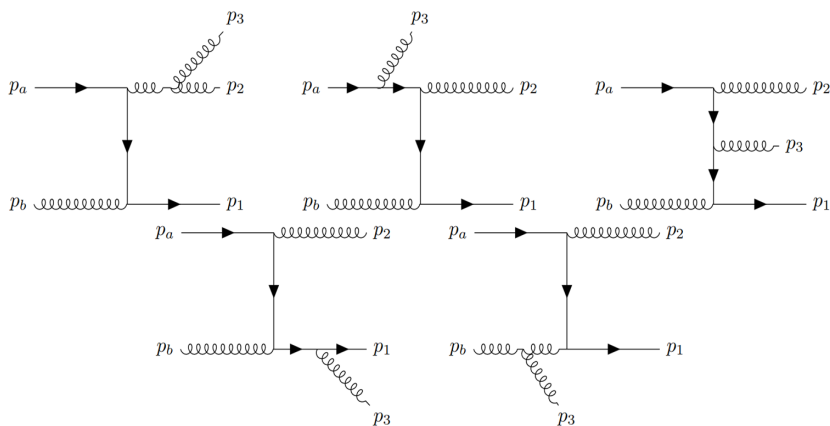
quark + quark



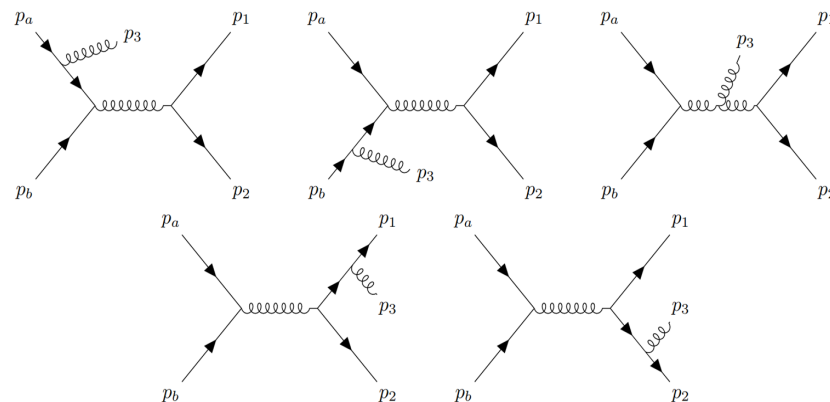
quark + gluon (t-channel)



quark + gluon (u-channel)



quark + gluon (s-channel)



- ❑ No approximations applied
- ❑ All interference terms included
- ❑ Emitted gluon is massive in DQPM

I. Grishmanovskii, O.S. et al., PRC 109, 024911 (2024)

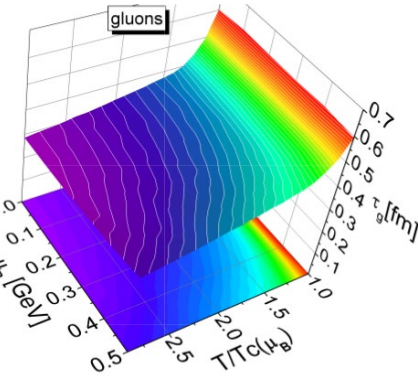
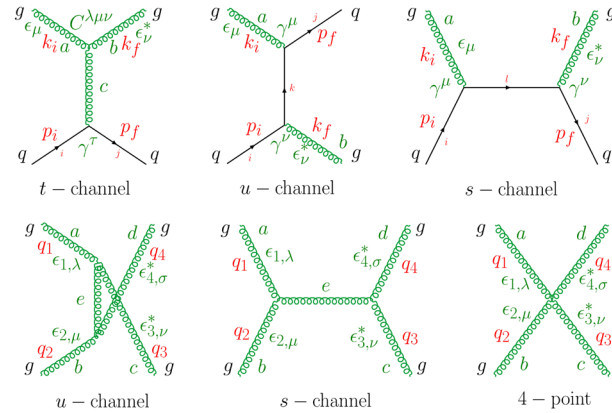
Transport coefficients at finite μ_B

$$\eta^{\text{RTA}}(T, \mu_B) = \frac{1}{15T} \sum_{i=q, \bar{q}, g} \int \frac{d^3p}{(2\pi)^3} \frac{\mathbf{p}^4}{E_i^2} \tau_i(\mathbf{p}, T, \mu_B) d_i(1 \pm f_i) f_i$$

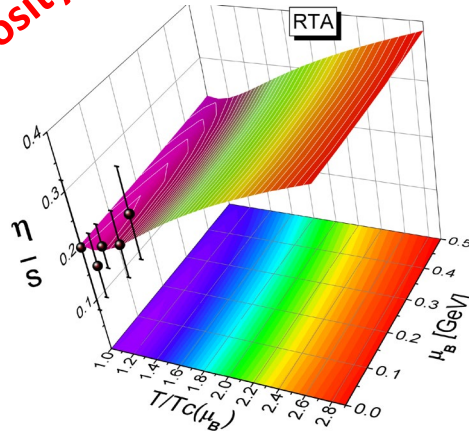
Relaxation times

$$\tau_i(\mathbf{p}, T, \mu_B) = \frac{1}{\Gamma_i(\mathbf{p}, T, \mu_B)}$$

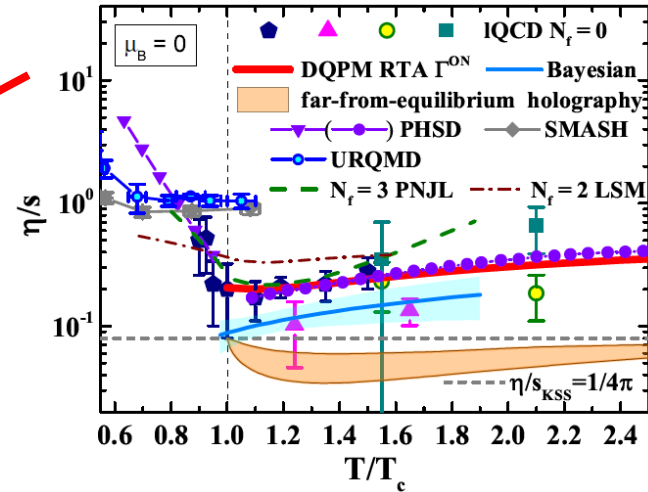
2 $\leftrightarrow$ 2 scatterings



Specific shear viscosity



μ_B

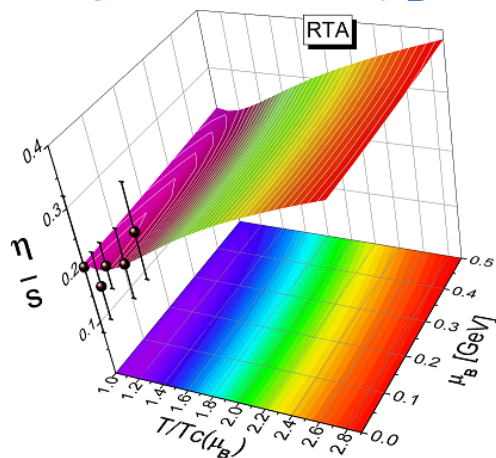


O. S., P. Moreau and E. Bratkovskaya, PRC 101 (2020), 045203

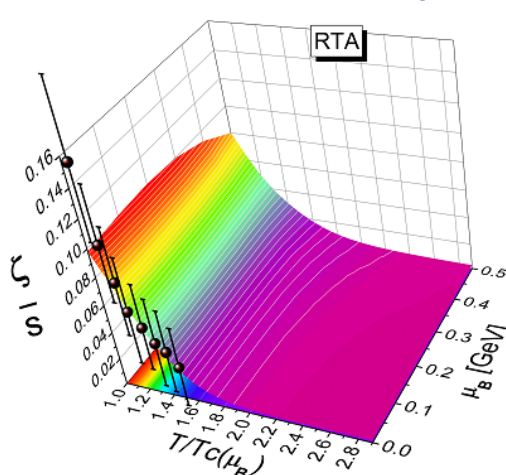
- Good agreement with IQCD predictions and Bayesian estimates
- Light increase with μ_B in the crossover region for viscosities and electric conductivity

Transport coefficients at finite μ_B

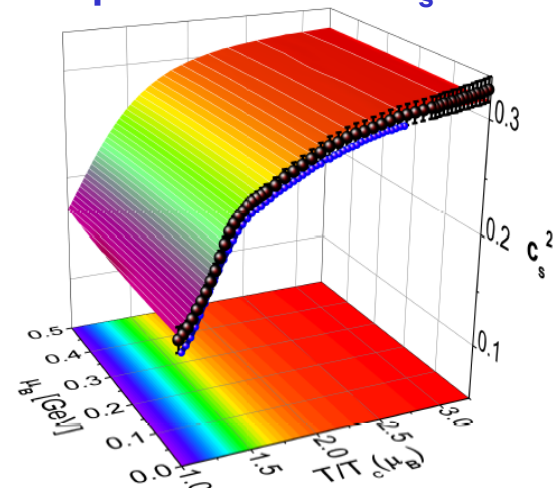
η/s versus (T, μ_B)



Bulk viscosity ζ/s

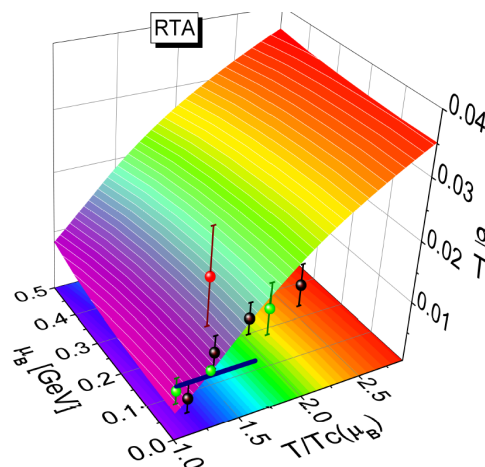


Speed of sound c_s^2

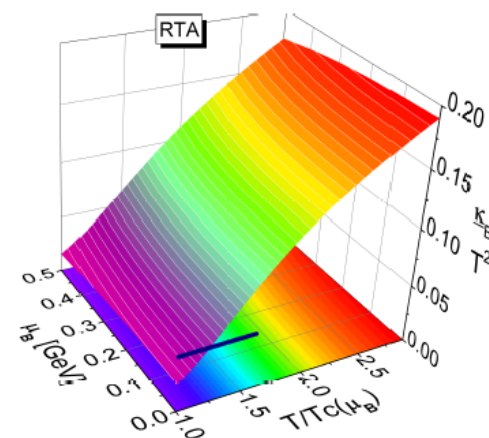


P. Moreau et al., PRC100 (2019) 014911;
O. Soloveva et al., PRC110 (2020) 045203

Electric conductivity σ_e/T



Baryon diffusion coefficient κ_B/T^2



Full diffusion coefficient matrix:

$$\begin{pmatrix} j_B^\mu \\ j_Q^\mu \\ j_S^\mu \end{pmatrix} = \begin{pmatrix} \kappa_{BB} & \kappa_{BQ} & \kappa_{BS} \\ \kappa_{QB} & \kappa_{QQ} & \kappa_{QS} \\ \kappa_{SB} & \kappa_{SQ} & \kappa_{SS} \end{pmatrix} \cdot \begin{pmatrix} \nabla^\mu \alpha_B \\ \nabla^\mu \alpha_Q \\ \nabla^\mu \alpha_S \end{pmatrix}$$

J. A. Fotakis, O.S. et al., PRD 104 (2021),
034014

- Weak dependence on μ_B in the crossover region

Transport coefficients at finite μ_B

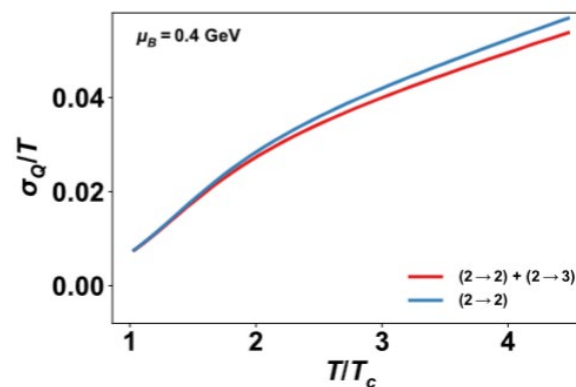
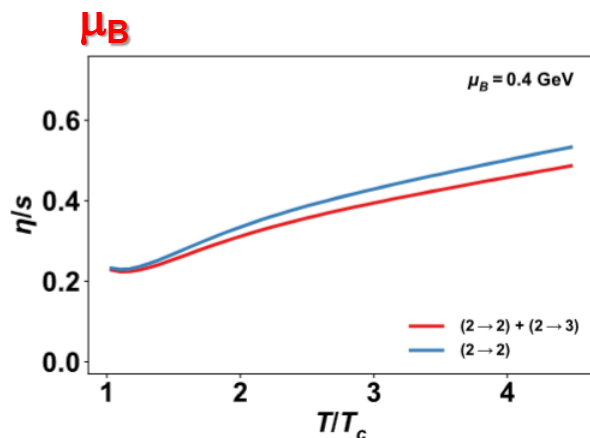
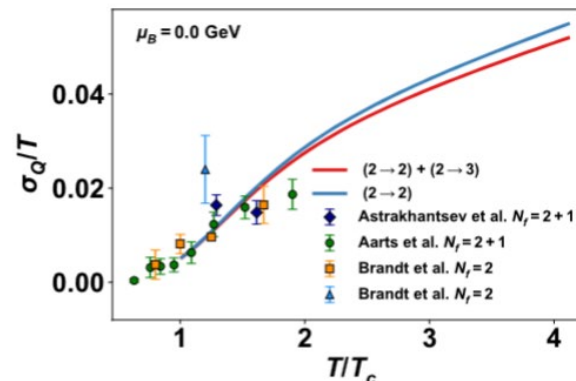
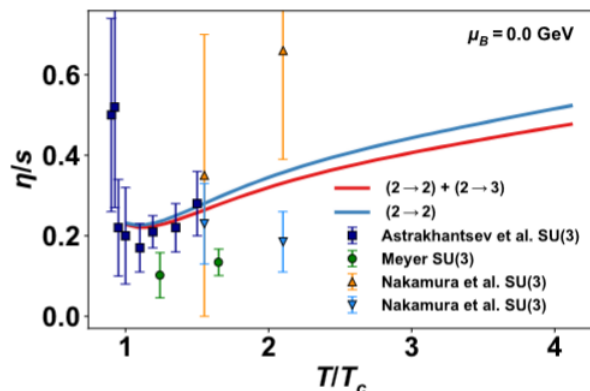
$$\eta^{\text{RTA}}(T, \mu_B) = \frac{1}{15T} \sum_{i=q,\bar{q},g} \int \frac{d^3p}{(2\pi)^3} \frac{\mathbf{p}^4}{E_i^2} \tau_i(\mathbf{p}, T, \mu_B) \quad d_i(1 \pm f_i)f_i$$

Specific shear viscosity

Relaxation times

$$\tau_i(\mathbf{p}, T, \mu_B) = \frac{1}{\Gamma_i(\mathbf{p}, T, \mu_B)}$$

$2 \leftrightarrow 2 + 2 \leftrightarrow 3$
scatterings



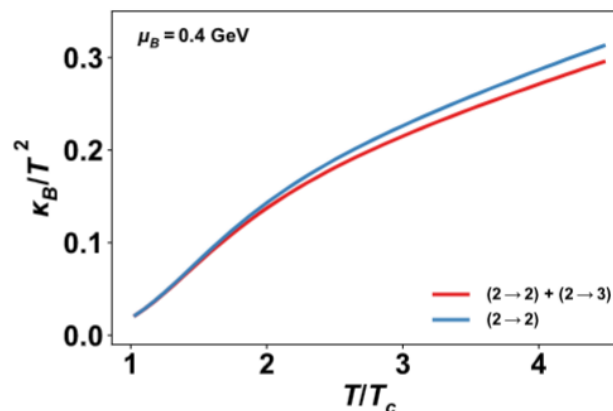
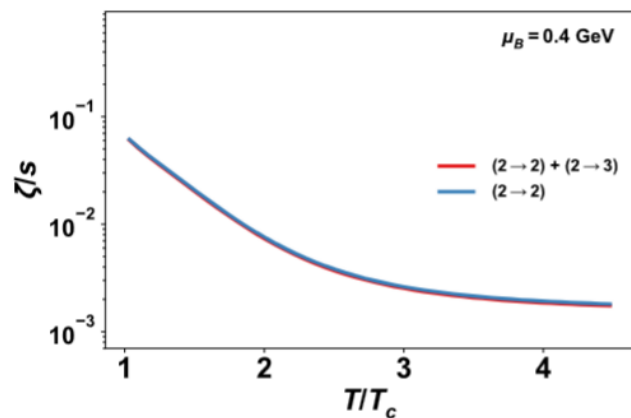
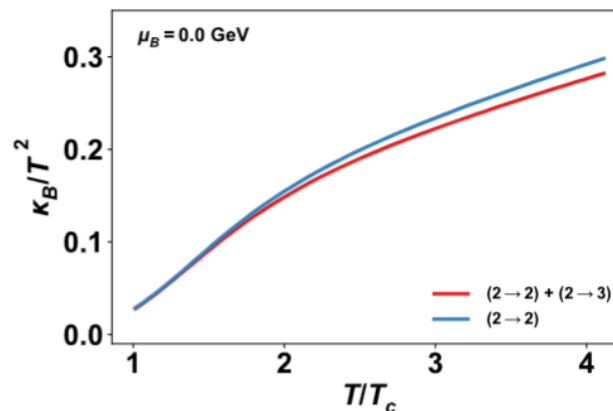
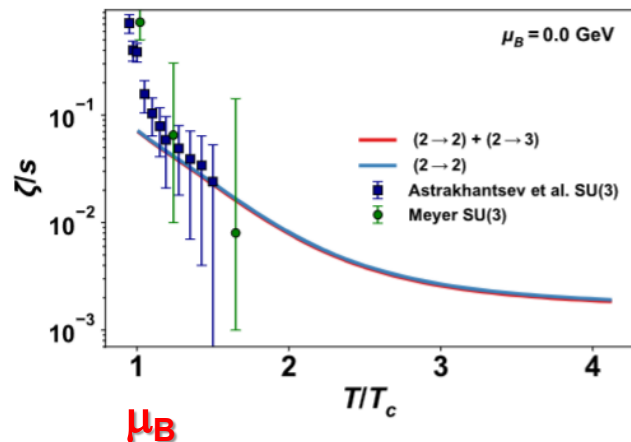
G. Ingrosso, O. S., I. Grishmanovskii and E. Bratkovskaya, arxiv:2606.13363 PRD

- Good agreement with IQCD predictions and Bayesian estimates
- Light increase with μ_B in the crossover region for viscosities and electric conductivity

Transport coefficients at finite μ_B

$$\eta^{\text{RTA}}(T, \mu_B) = \frac{1}{15T} \sum_{i=q,\bar{q},g} \int \frac{d^3p}{(2\pi)^3} \frac{\mathbf{p}^4}{E_i^2} \tau_i(\mathbf{p}, T, \mu_B) d_i (1 \pm f_i) f_i$$

Specific bulk viscosity



Relaxation times

$$\tau_i(\mathbf{p}, T, \mu_B) = \frac{1}{\Gamma_i(\mathbf{p}, T, \mu_B)}$$

$2 \leftrightarrow 2 + 2 \leftrightarrow 3$
scatterings

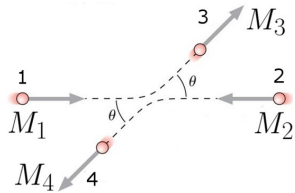
G. Ingrosso, O. S., I. Grishmanovskii and E. Bratkovskaya, arxiv:2606.13363

- Good agreement with IQCD predictions and Bayesian estimates
- Light increase with μ_B in the crossover region

Jet transport coefficients

On-shell:

- integration over momenta
- masses = pole masses

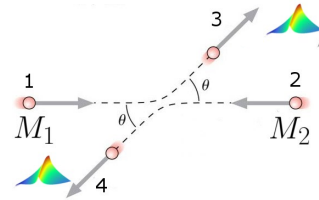


$$E^2 = m^2 + p^2$$

$$\begin{aligned} \langle \mathcal{O} \rangle^{\text{on}} &= \frac{1}{2E_i} \sum_{j=q,\bar{q},g} d_j f_j \int \frac{d^3 p_j}{(2\pi)^3 2E_j} \\ &\times \int \frac{d^3 p_1}{(2\pi)^3 2E_1} \int \frac{d^3 p_2}{(2\pi)^3 2E_2} \\ &\times (1 \pm f_1)(1 \pm f_2) \mathcal{O} |\overline{\mathcal{M}}|^2 (2\pi)^4 \delta^{(4)}(p_i + p_j - p_1 - p_2) \end{aligned}$$

Off-shell:

- integration over momenta
- + two additional integrations over medium parton energies



$$\frac{1}{2E} \rightarrow \int \frac{d\omega}{(2\pi)} \rho(\omega, \mathbf{p}) \theta(\omega)$$

$$\begin{aligned} \langle \mathcal{O} \rangle^{\text{off}} &= \frac{1}{2E_i} \sum_{j=q,\bar{q},g} d_j f_j \int \frac{d^4 p_j}{(2\pi)^4} \rho(\omega_j, \mathbf{p}_j) \theta(\omega_j) \\ &\times \int \frac{d^3 p_1}{(2\pi)^3 2E_1} \int \frac{d^4 p_2}{(2\pi)^4} \rho(\omega_2, \mathbf{p}_2) \theta(\omega_2) \\ &\times (1 \pm f_1)(1 \pm f_2) \mathcal{O} |\overline{\mathcal{M}}|^2 (2\pi)^4 \delta^{(4)}(p_i + p_j - p_1 - p_2) \end{aligned}$$

$$\langle \mathcal{O} \rangle_u = \sum_q \langle \mathcal{O} \rangle_{uq} + \sum_{\bar{q}} \langle \mathcal{O} \rangle_{u\bar{q}} + \langle \mathcal{O} \rangle_{ug} = \langle \mathcal{O} \rangle_{uu \rightarrow uu} + \langle \mathcal{O} \rangle_{ud \rightarrow ud} + \langle \mathcal{O} \rangle_{us \rightarrow us} + \langle \mathcal{O} \rangle_{u\bar{u} \rightarrow u\bar{u}} + \langle \mathcal{O} \rangle_{u\bar{d} \rightarrow u\bar{d}} + \langle \mathcal{O} \rangle_{u\bar{s} \rightarrow u\bar{s}} + \langle \mathcal{O} \rangle_{ug \rightarrow ug}$$

- Scattering rate Γ :
- Transverse momentum transfer squared per unit length $\hat{q}$:
- Energy loss per unit length dE/dx :
- Drag coefficient A :

$$\mathcal{O} = 1$$

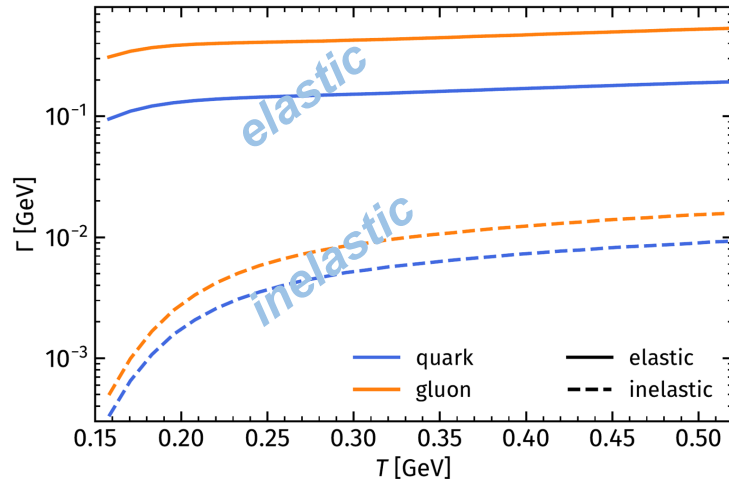
$$\mathcal{O} = |\vec{p}_T - \vec{p}_T'|^2 \rightarrow \langle \mathcal{O} \rangle = \hat{q}$$

$$\mathcal{O} = (E - E') \rightarrow \langle \mathcal{O} \rangle = dE/dx$$

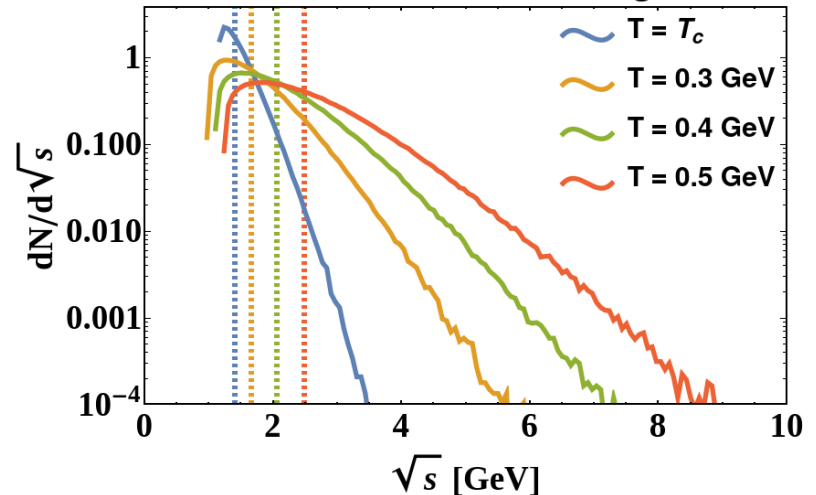
$$\mathcal{O} = p_L - p_L'$$

Partonic interaction rates in equilibrated sQGP

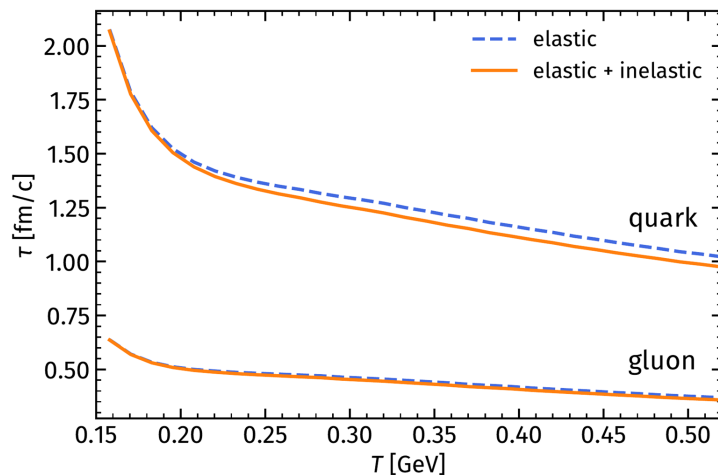
Interaction rate



on-shell u – u scattering



Relaxation time ($\tau \sim 1/\Gamma$)

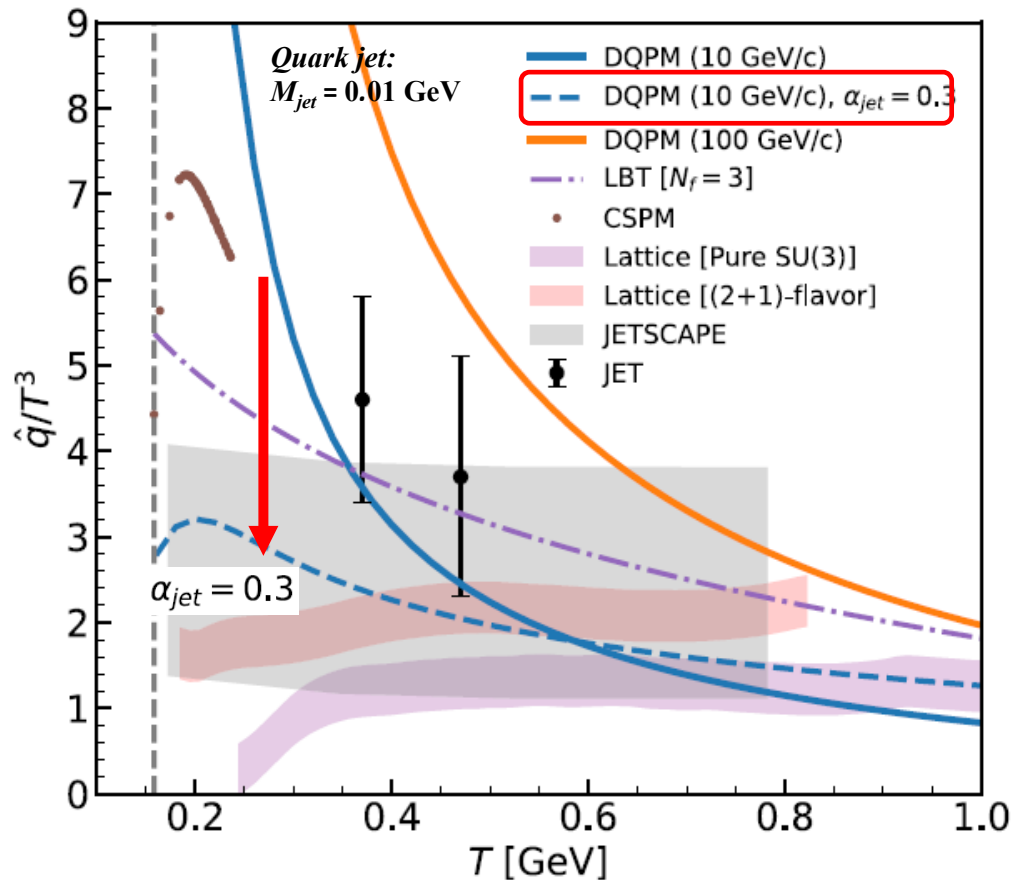


In thermalized QGP **low** energies are **favored** where the elastic scatterings dominate

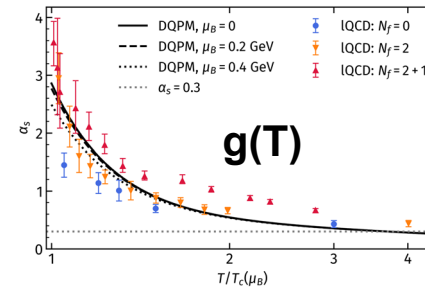
- The partonic **interaction rate** and **relaxation time** are primarily governed by **elastic** scattering
- **Inelastic processes** – with massive gluon emission – are **suppressed** in the **thermalized** QCD medium

I. Grishmanovskii, OS et al., PRC 109, 024911 (2024)

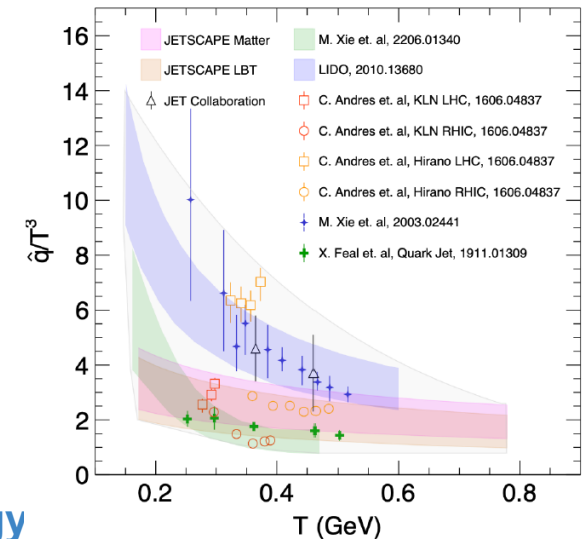
Jet transport coefficients in sQGP



I. Grishmanovskii, OS et al., PRC 109, 024911 (2024)



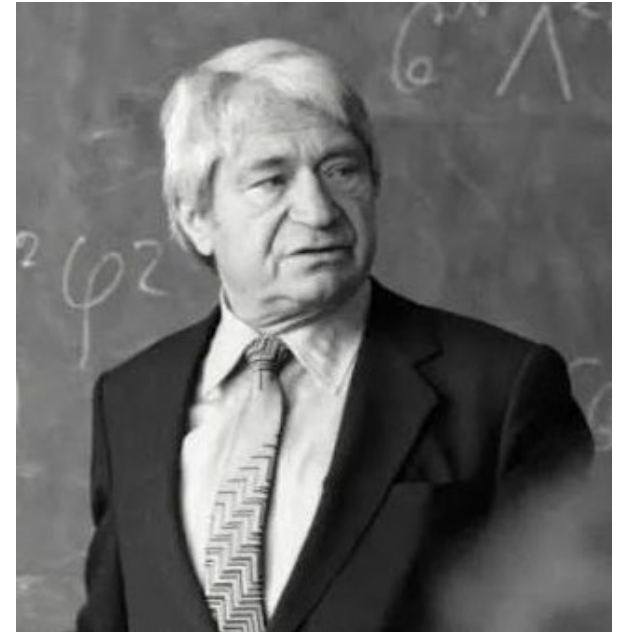
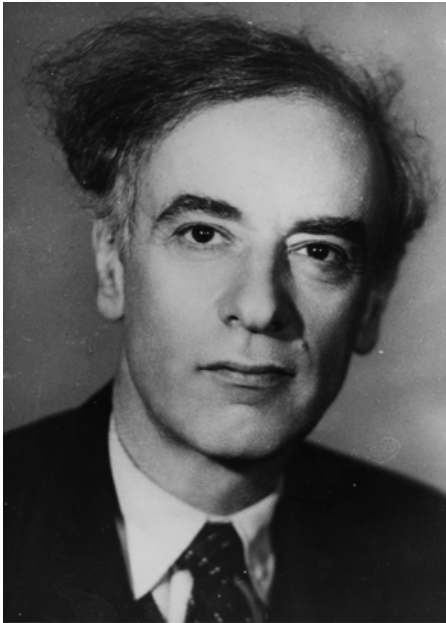
LA, Lee, Winn, Prog.Part.Nucl.Phys. 127 (2022) 103990



→ **Agreement** with the other models at **low jet energy**

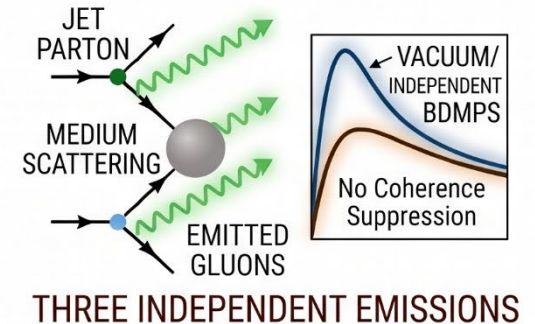
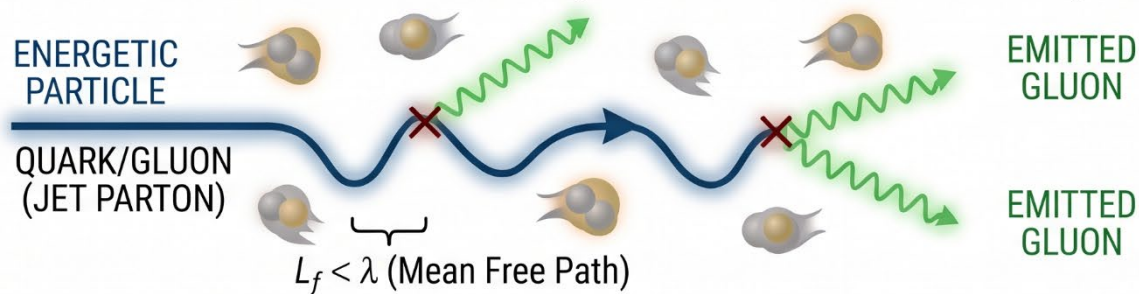
→ Rapid **rise** with **decreasing** medium **temperature** due to strong increase of $g(T)$

Landau-Pomeranchuk-Migdal effect

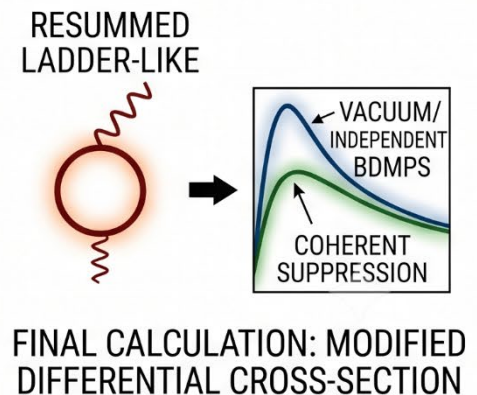
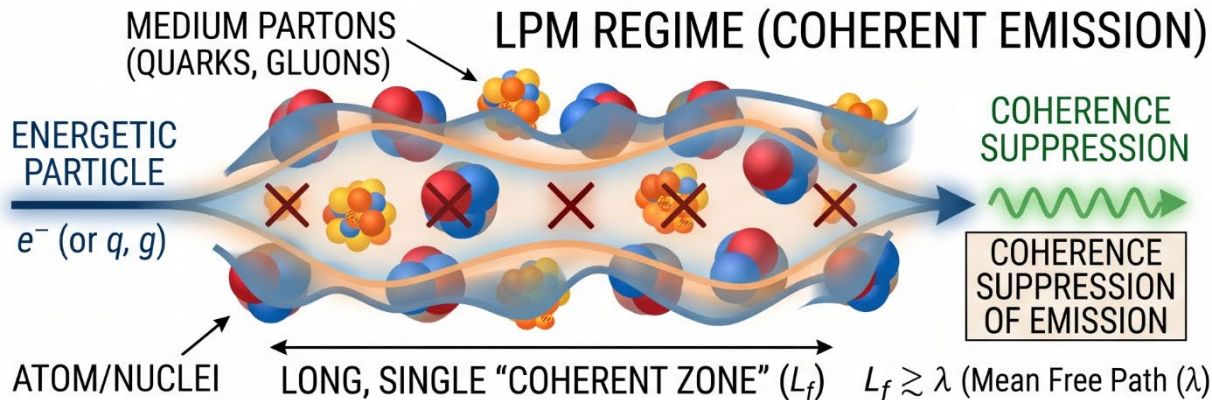


LPM effect: coherent vs independent emission in QGP

BETHE-HEITLER REGIME (INDEPENDENT EMISSION)



LPM REGIME (COHERENT EMISSION)



LPM: Local formation-time

$$\tau_f = \frac{2x(1-x)E_j}{k_{\perp}^2 + x^2 M_j^2 + (1-x)m_3^2},$$

$$\hat{q}_{\text{rad}}^{\text{LPM}} = \sum_i \int d\Gamma_{ji \rightarrow jig}^{\text{DQPM}} S_{\text{LPM}} q_{\perp}^2.$$

mode 0: $S_{\text{LPM}} = 1$,

mode 1 (smooth): $S_{\text{LPM}}^{\text{smooth}} = \min\left(1, \frac{\lambda_{\text{el}}}{\tau_f}\right)$, **Smooth:** long-formation-time configurations remain, but with reduced weight.

mode 2 (sharp): $S_{\text{LPM}}^{\Theta} = \Theta(\lambda_{\text{el}} - \tau_f)$. **Sharp:** all configurations with $\tau_f > \lambda_{\text{el}}$ are rejected.

$\tau_f \ll \lambda_{\text{el}}$: the gluon forms before the next scattering, Bethe–Heitler-like emission is approximately valid.

$\tau_f \gtrsim \lambda_{\text{el}}$: several interactions occur during formation; emission amplitudes interfere. Adding independent $2 \rightarrow 3$ probabilities overestimates long-formation-time radiation.

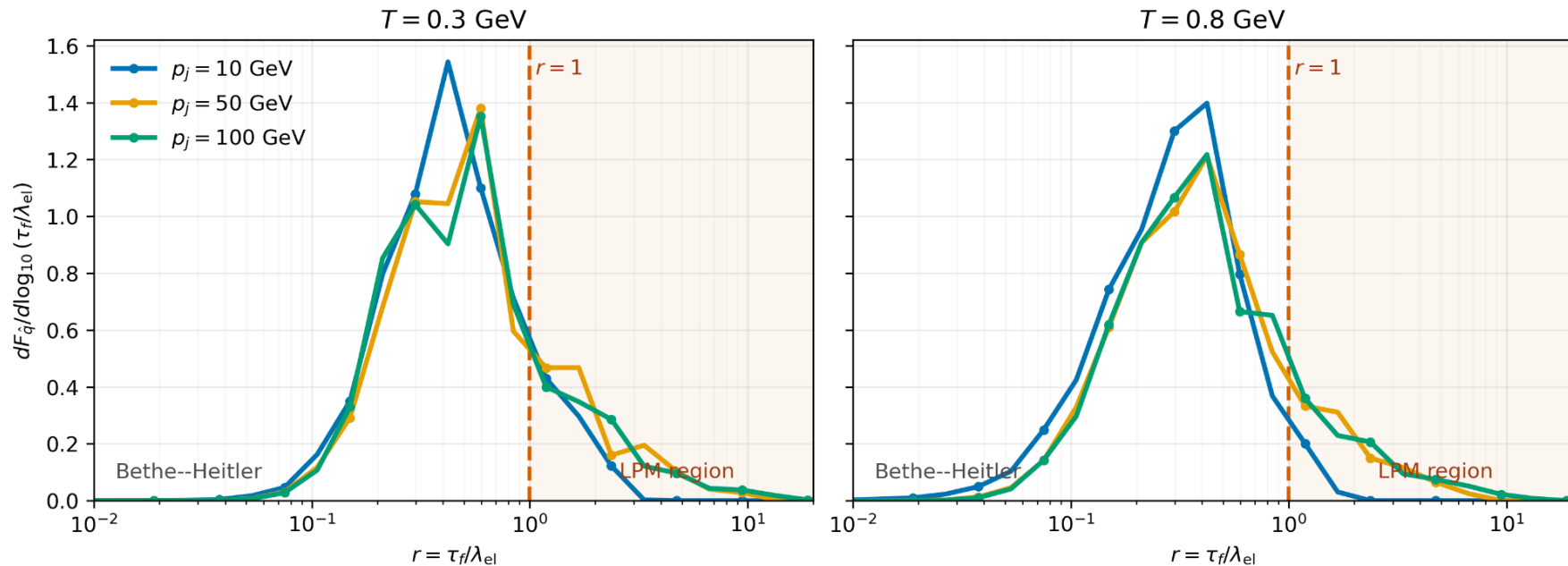
- The LPM effect is not merely a reduction of a cross section
- It reflects the loss of independence between scatterings occurring within one formation time
- We use the ratio $\tau_f/\lambda_{\text{el}}$ as a local diagnostic of whether the sampled emission is likely to overlap with subsequent scatterings

Formation time: Physical meaning

$$N_{\text{coh}} \sim \frac{\tau_f}{\lambda_{\text{el}}} = r.$$

$$S_{\text{LPM}} \sim \frac{1}{N_{\text{coh}}} \sim \frac{\lambda_{\text{el}}}{\tau_f} \quad (r > 1)$$

DQPM formation-time distributions: $\hat{q}$ weighted

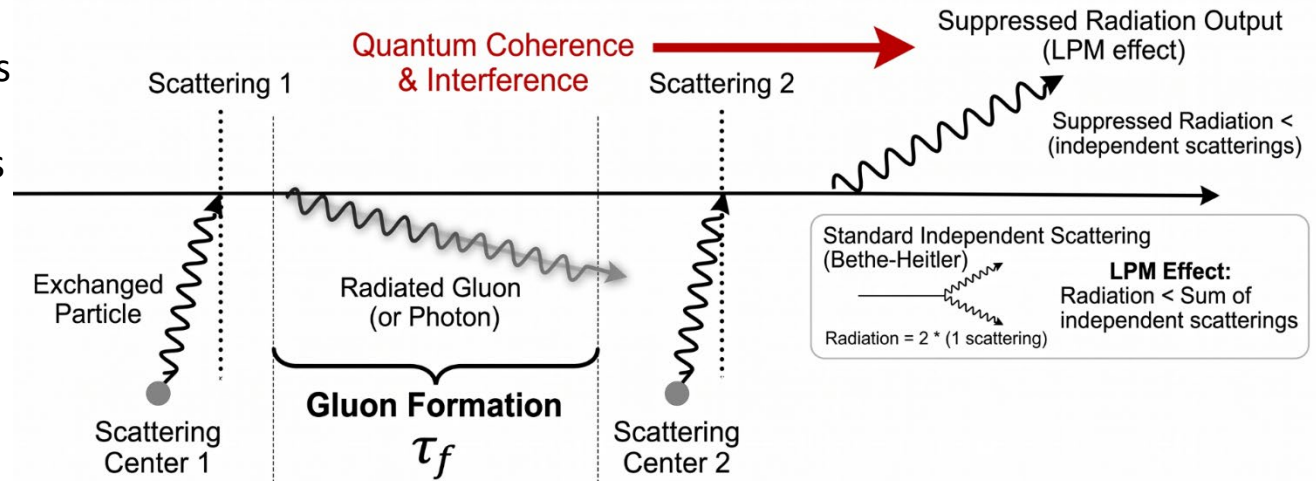


integrated transport coefficients may be only **moderately modified** even when a visible long- τ_f tail exists

Since the dominant observable weight can still originate from $\tau_f < \lambda_{\text{el}}$

Why 2 - 3 reactions are insufficient?

- $\tau_f \ll \lambda_{el}$: the gluon forms before the next scattering; Bethe–Heitler-like emission is approximately valid.
- $\tau_f \gtrsim \lambda_{el}$: several interactions occur during formation; emission amplitudes **interfere**. Therefore, simply adding independent $2 \rightarrow 3$ probabilities **overestimates** long-formation-time radiation.



Connection to BDPMS - Z

$$k_{\perp}^2 \sim \hat{q} \tau_f, \quad \tau_f \sim \frac{2\omega}{k_{\perp}^2},$$

$$\tau_f^{\text{BDMPs}} \sim \sqrt{\frac{2\omega}{\hat{q}}}.$$

emitted gluon accumulates transverse momentum during its formation; many soft interactions are coherently resummed;

τ_f and $k_{\perp}$ are dynamically linked; the result is a medium-induced radiation spectrum rather than a local veto on individual $2 \rightarrow 3$ events.

BDMPs-Z and Zakharov formulate the LPM problem through a full quantum multiple-scattering treatment

DQPM local implementation

Samples one complete $2 \rightarrow 3$ event

Uses sampled $k_{\perp}$ in τ_f

Local $S(\tau_f/\lambda_{\text{el}})$

Produces corrected $\hat{q}_{\text{rad}}, dE/dx$

Massive off-shell quasiparticles

BDMPs-Z

Resums multiple interactions during formation

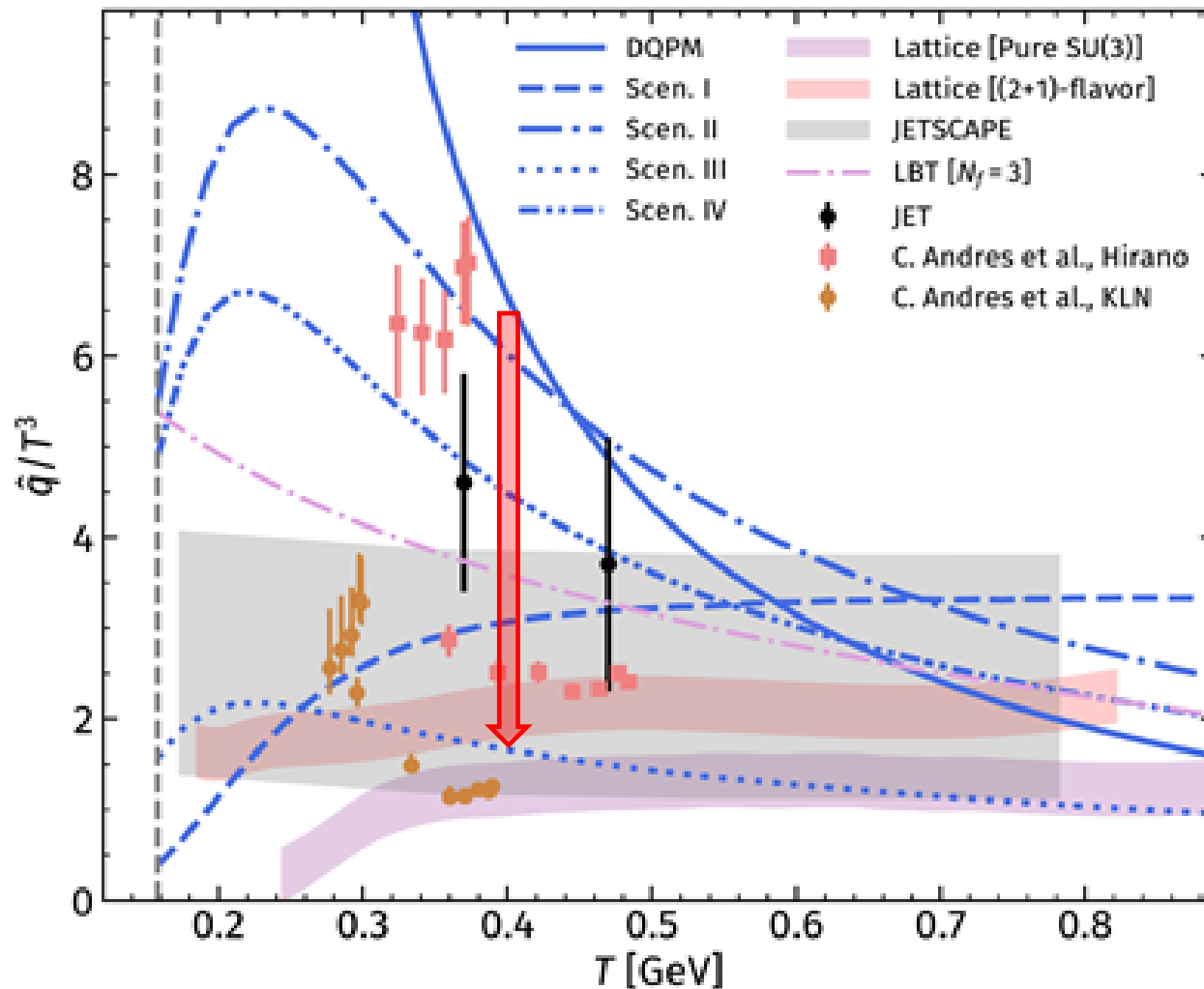
Builds $k_{\perp}$ dynamically through broadening

Quantum interference at amplitude level

Produces medium-induced gluon spectrum

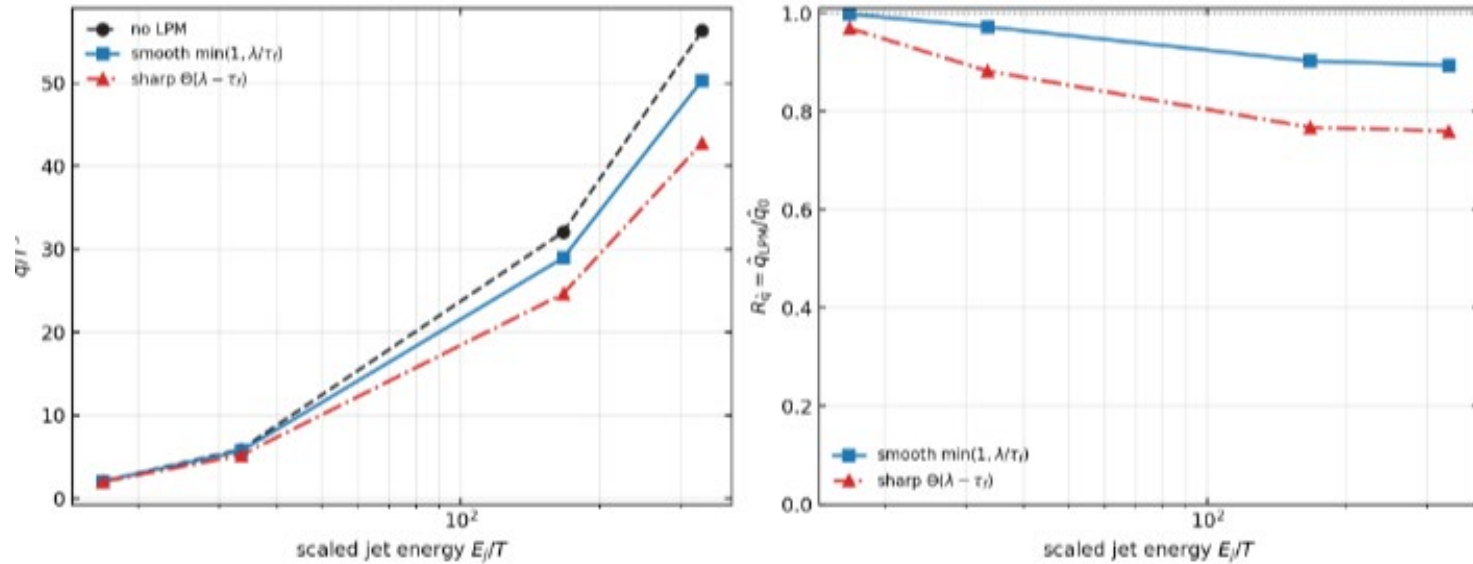
Usually formulated in high-energy/eikonal limits

Standard massive DQPM result

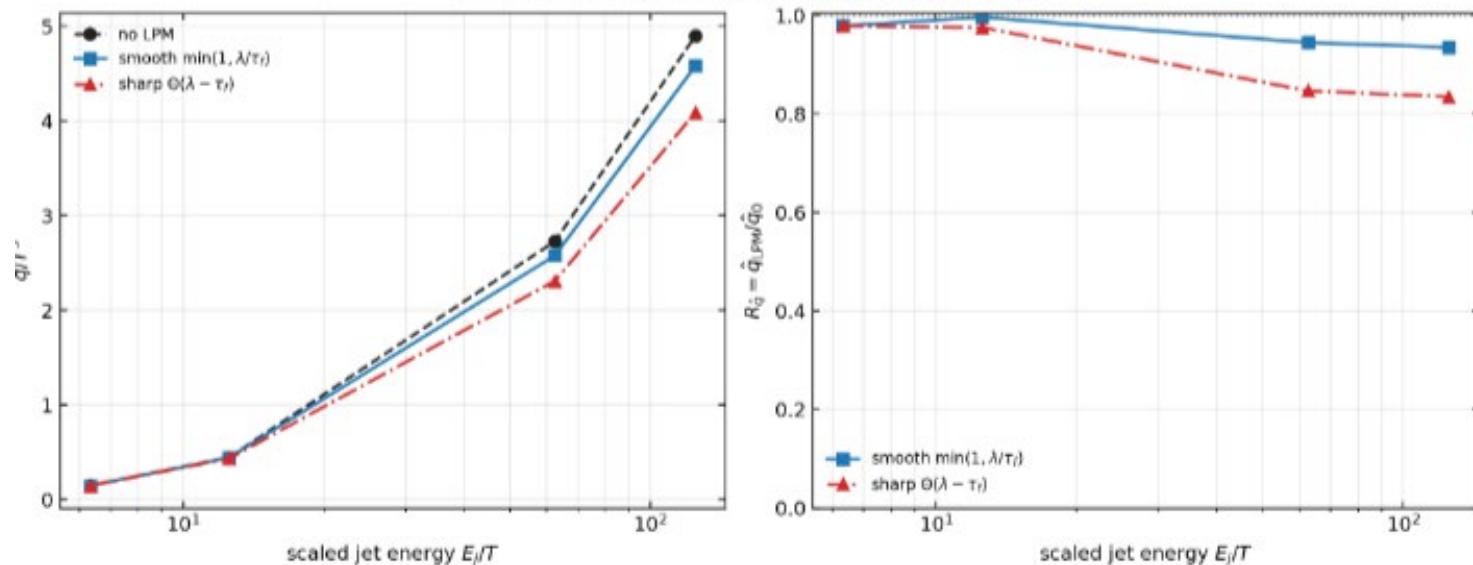


High energy diagnostics

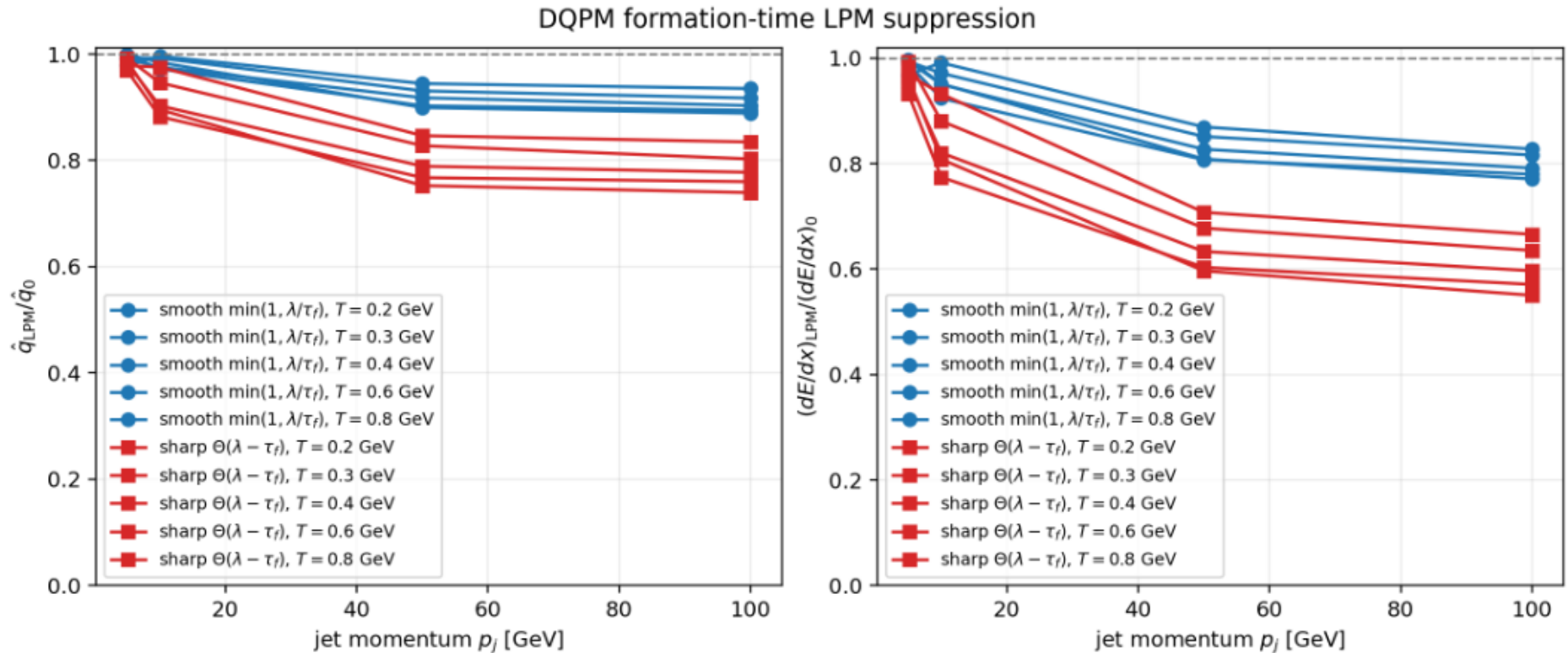
Scaled DQPM broadening, $T = 0.3 \text{ GeV}$, $\mu_B = 0 \text{ GeV}$



Scaled DQPM broadening, $T = 0.8 \text{ GeV}$, $\mu_B = 0 \text{ GeV}$

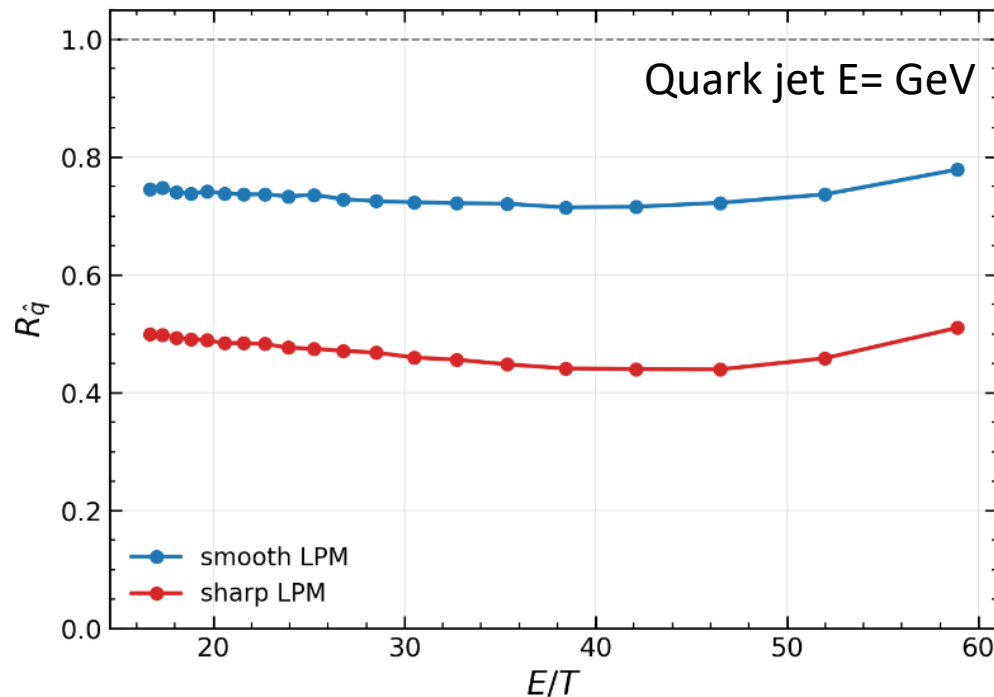


Massless-emission diagnostics



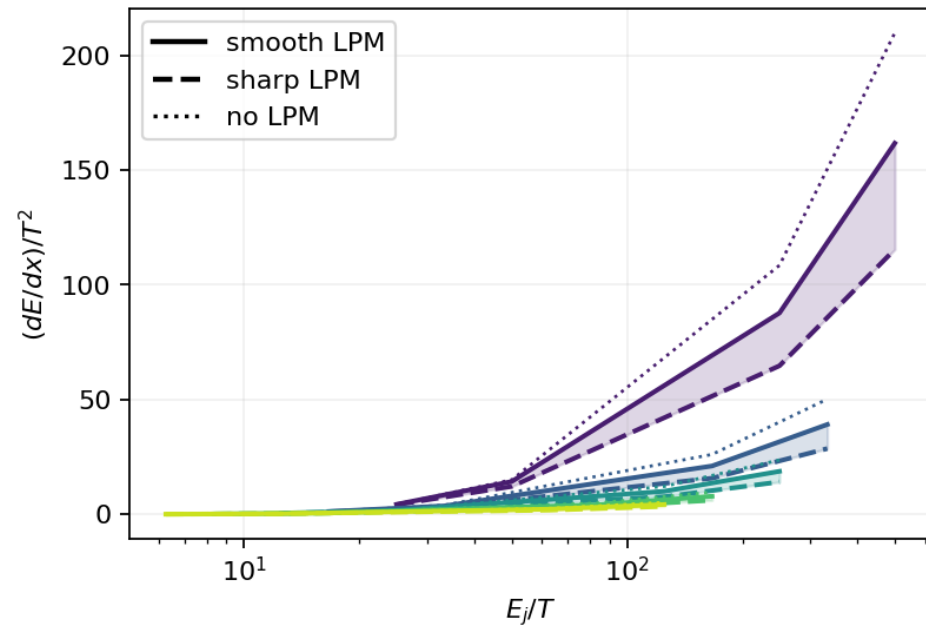
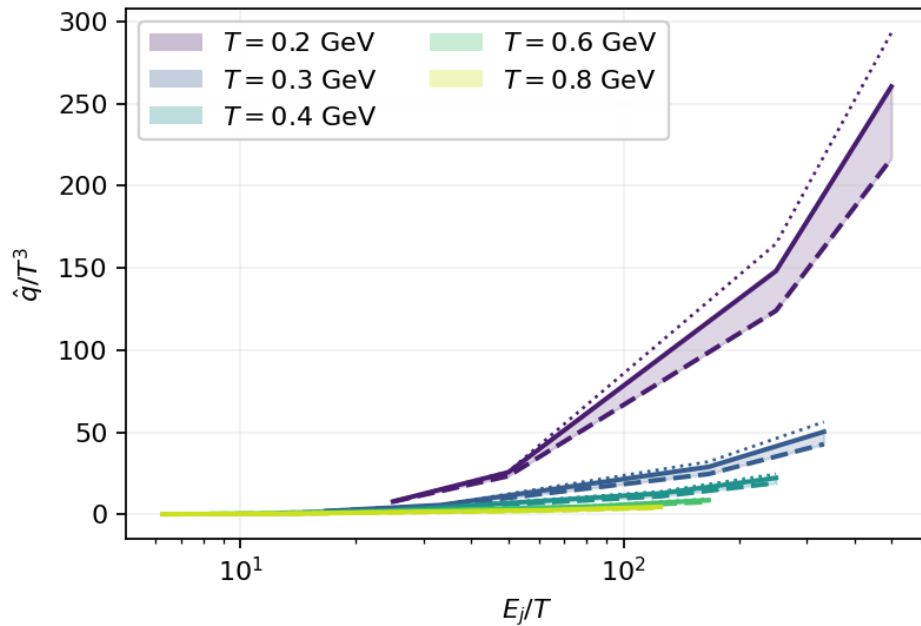
- suppression increases with E/T
- higher-energy emissions can have longer formation times
- more radiative weight enters the region $\tau_f \gtrsim \lambda_{\text{el}}$

High energy and massless-emission diagnostics

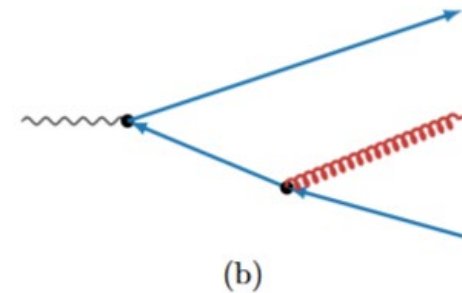
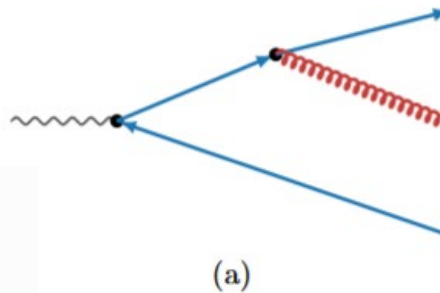


- removing the emitted-gluon mass increases τ_f
- the result becomes more sensitive to the coherence prescription
- gluon jets show a stronger modification than quark jets

Coherence increases with jet energy



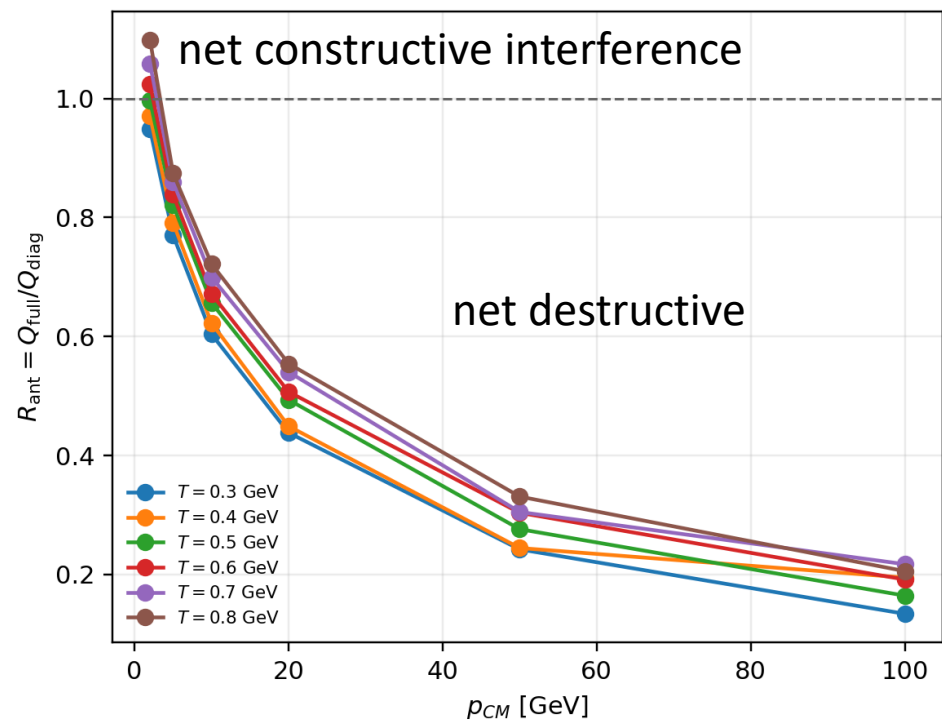
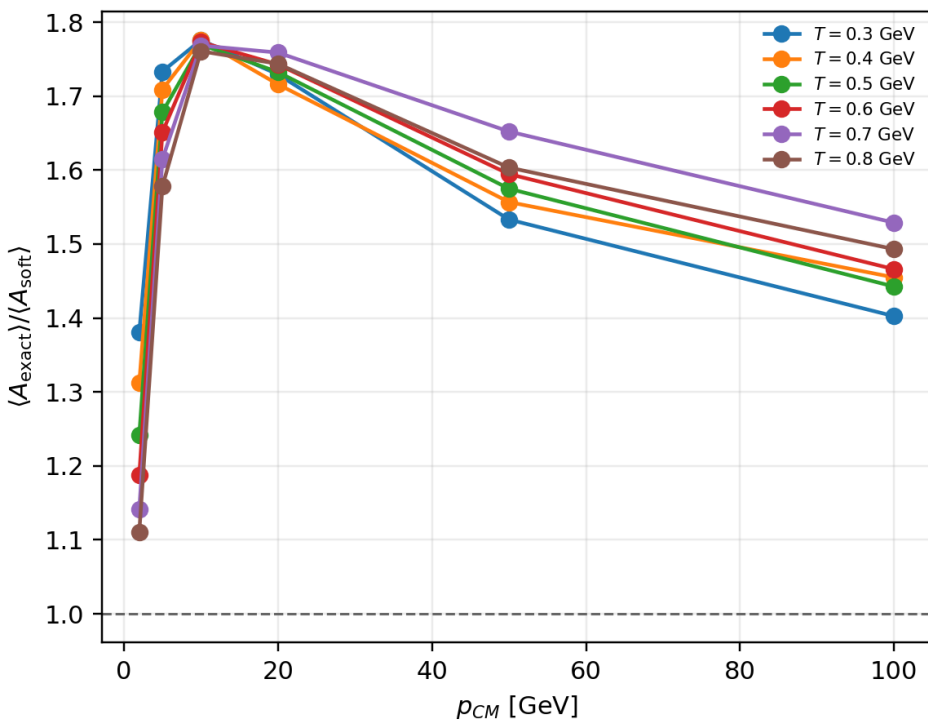
Antenna



$$\mathcal{M} = \sum_i \mathcal{M}_i,$$

$$|\mathcal{M}|^2 = \underbrace{\sum_i |\mathcal{M}_i|^2}_{Q_{\text{diag}}} + \underbrace{\sum_{i < j} 2 \text{Re}(\mathcal{M}_i \mathcal{M}_j^*)}_{Q_{\text{int}}}.$$

Exact DQPM antenna scan, $\mu_B = 0$ GeV



Summary

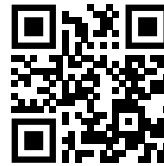
- Local LPM suppression has been embedded consistently into the off-shell DQPM $2 \rightarrow 3$ kernel
 - Coherence effects increase with jet energy and affect dE/dx more strongly than $\hat{q}$
 - The standard massive setup gives moderate suppression; the massless-emission and gluon-jet cases are much more sensitive
 - The method is BDMPS-Z-inspired but remains less dynamical than the SUBA-Jet(SUBATECH) phase-accumulation approach
 - Antenna interference is substantial and motivates a future unified treatment in thermal transport and off-shell transport approach (Parton-Hadron-String-Dynamics) PHSD
-

Outlook:

- Implement the LPM-modified $2 \rightarrow 3$ rates in PHSD
- Benchmark study against BDMPS-Z and SUBA-Jet
- Move beyond the local suppression factor
- Jet transport in an anisotropic and flowing medium

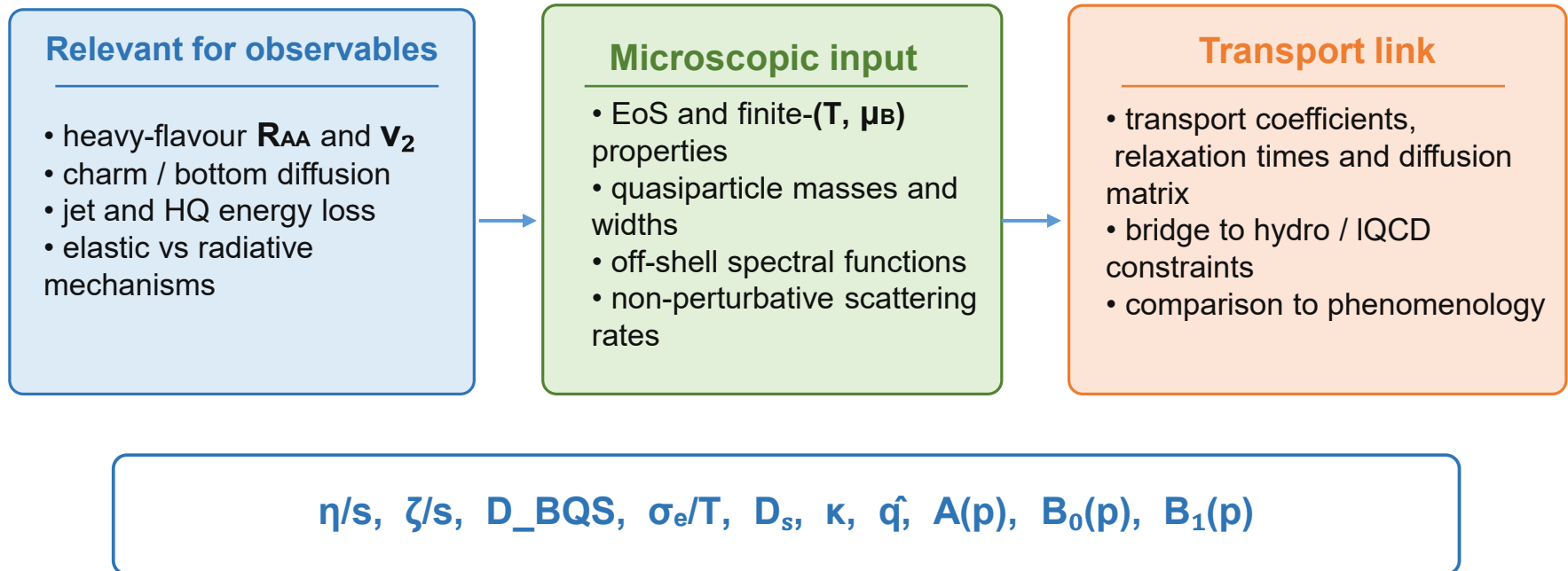
Thank you for your attention!

- olgasolovev.github.io



Back up slides

How do microscopic QGP degrees of freedom control heavy-quark and jet observables?



**PHSD/DQPM
advantage:**

the same microscopic quasiparticle picture can be tested from transport coefficients to final-state heavy-flavour and jet observables.

Heavy quarks and jets are inverse probes of the microscopic structure of the strongly interacting QGP.

Antenna interference within the exact DQPM matrix element

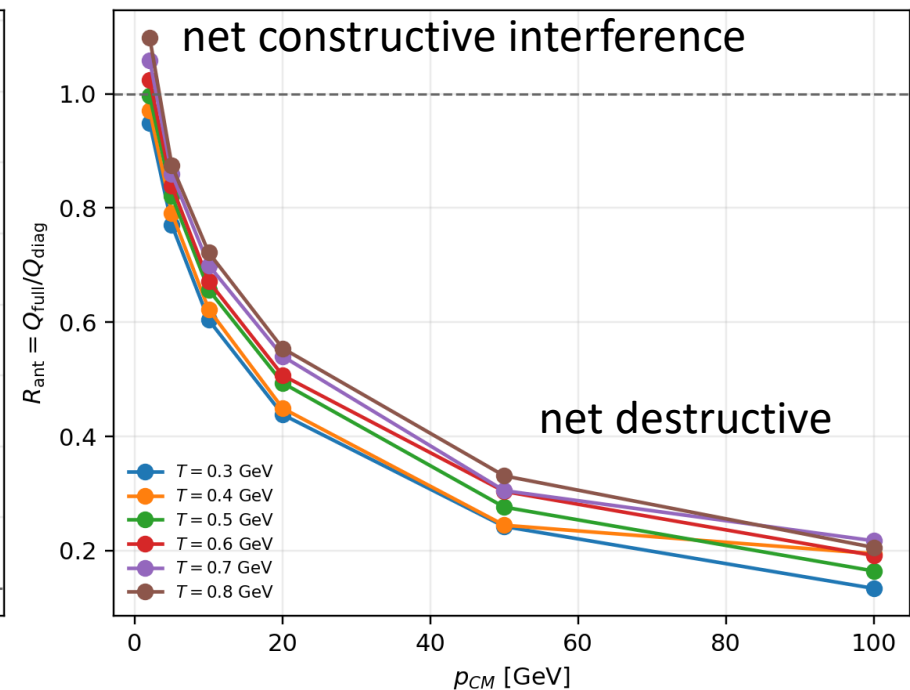
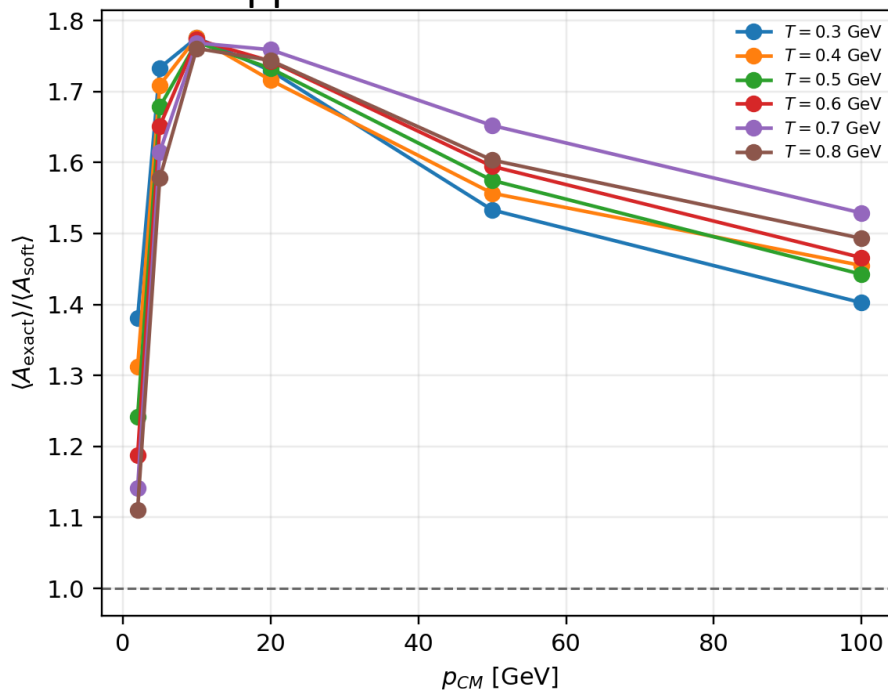
$$\mathcal{M} = \sum_i \mathcal{M}_i,$$

$$|\mathcal{M}|^2 = \underbrace{\sum_i |\mathcal{M}_i|^2}_{Q_{\text{diag}}} + \underbrace{\sum_{i < j} 2 \text{Re}(\mathcal{M}_i \mathcal{M}_j^*)}_{Q_{\text{int}}}.$$

$$R_{\text{ant}} = \frac{Q_{\text{full}}}{Q_{\text{diag}}} = \frac{Q_{\text{diag}} + Q_{\text{int}}}{Q_{\text{diag}}}.$$

Soft approximation

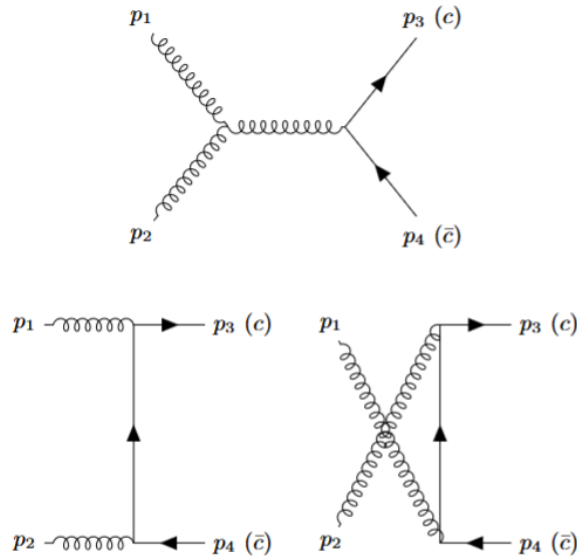
Exact DQPM antenna scan, $\mu_B = 0$ GeV



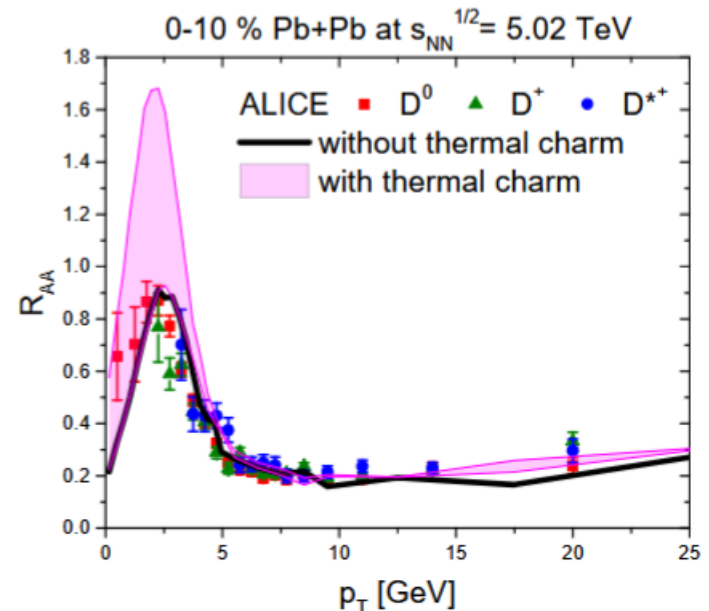
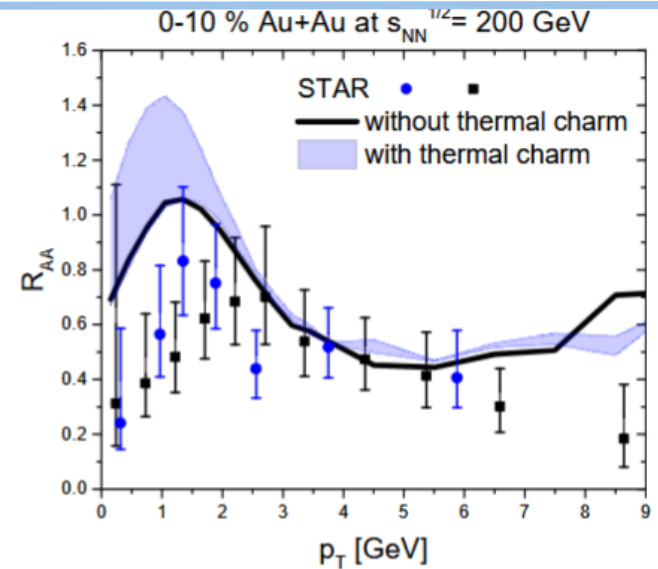
At high momentum, **destructive interference** cancels roughly 80% or more of the incoherent diagonal contribution.

From heavy-quark diffusion to thermal charm production in PHSD

Charm quark pair production



- Thermal production is concentrated in the earliest, hottest partonic stage
- Comparison to D-meson RAA favors charm quarks gaining additional thermal mass in the QGP
- This connects open heavy flavour directly to microscopic QGP quasiparticle properties

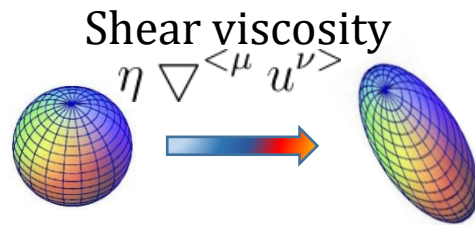


Properties of QGP: transport coefficients

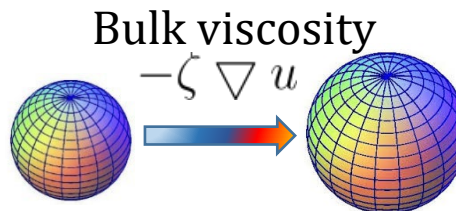
Hydrodynamics

$$\begin{cases} \partial_\mu T^{\mu\nu} = 0 & T^{\mu\nu} = -Pg^{\mu\nu} + wu^\mu u^\nu + \Delta T^{\mu\nu} \\ \partial_\mu J_B^\mu = 0 & J_B^\mu = n_B u^\mu + \Delta J_B^\mu \end{cases}$$

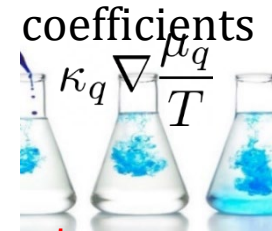
$\eta (D^\mu u^\nu + D^\nu u^\mu + \frac{2}{3} \Delta^{\mu\nu} \partial_\rho u^\rho) - \zeta \Delta^{\mu\nu} \partial_\rho u^\rho$
 $\Delta J_B^\mu = \kappa_B D^\mu (\frac{\mu_B}{T})$
 input for hydro



Transport coefficients:

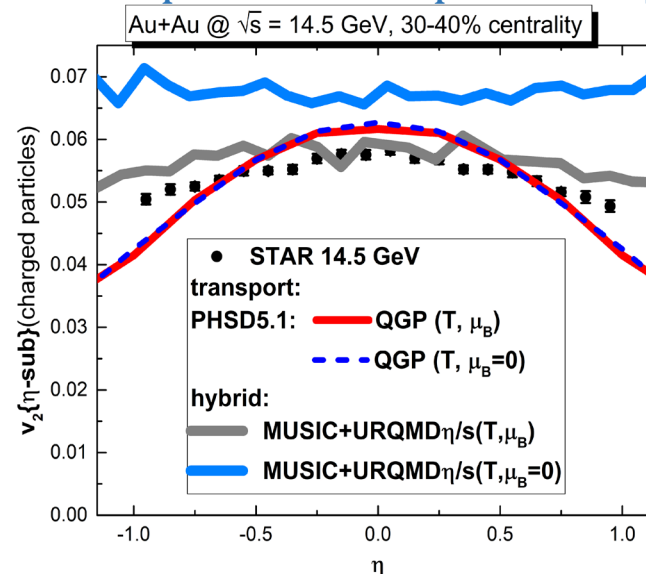
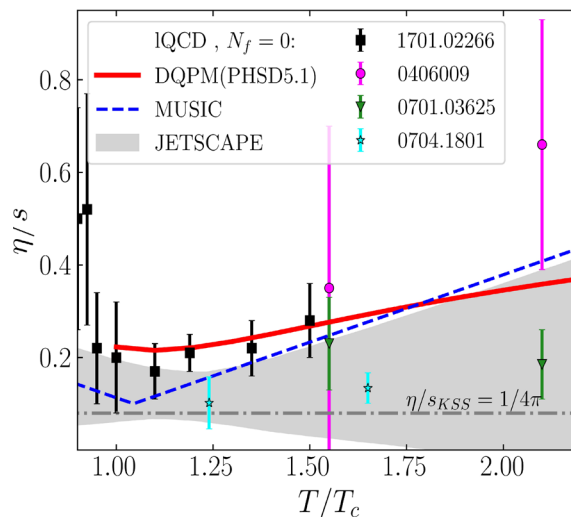


(B, Q, S) diffusion



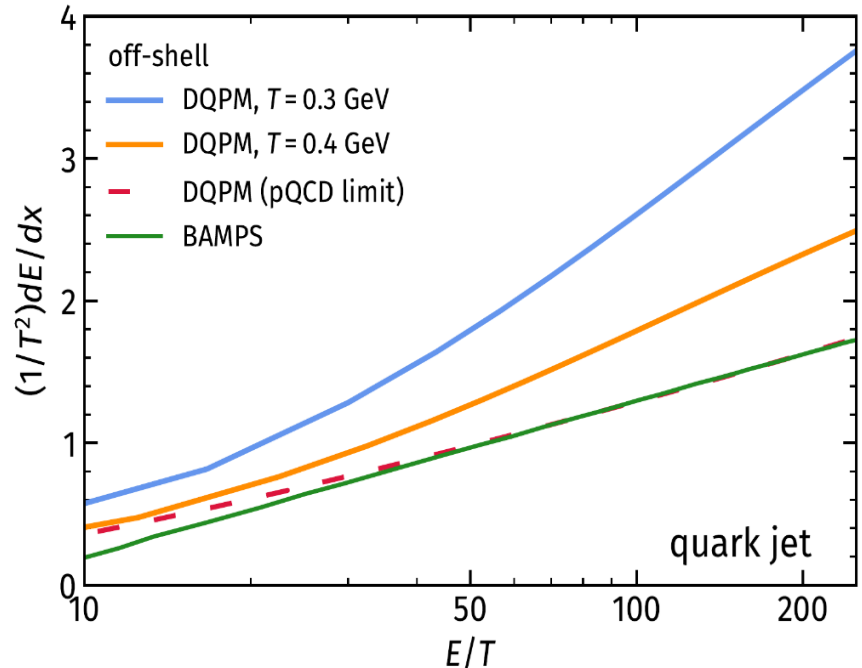
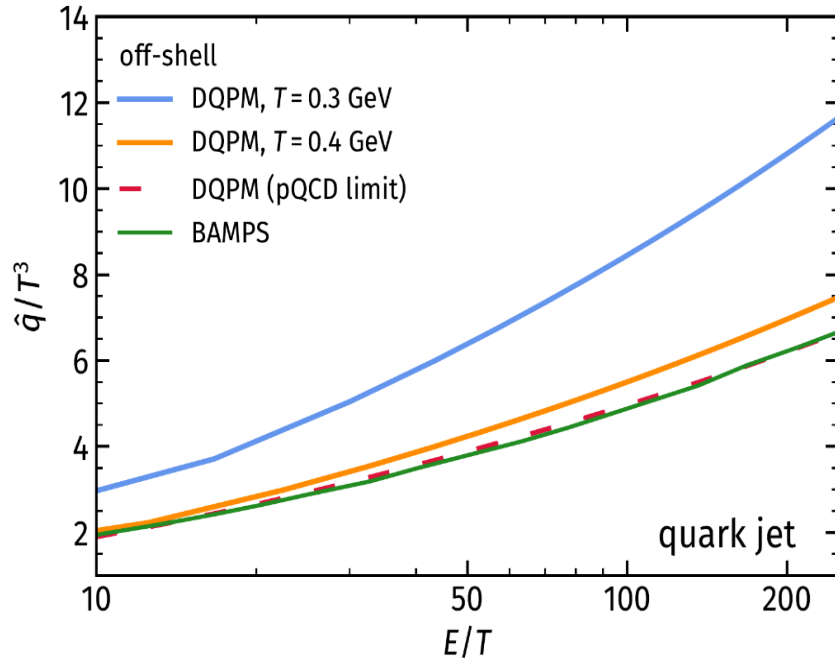
Model predictions for QGP: ! same EoS but different transport coefficients

Transport coefficients can serve a bridge for comparison transport and hydro



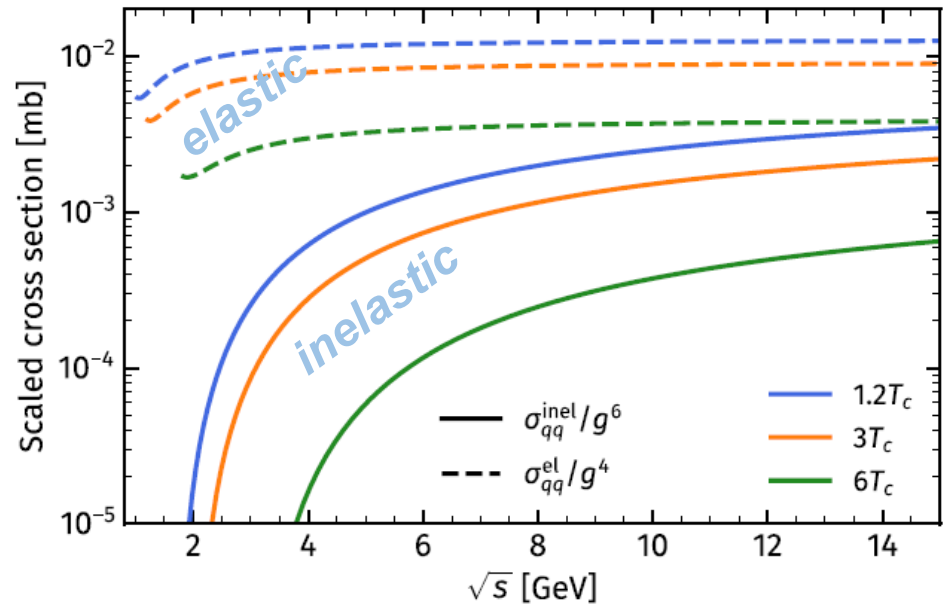
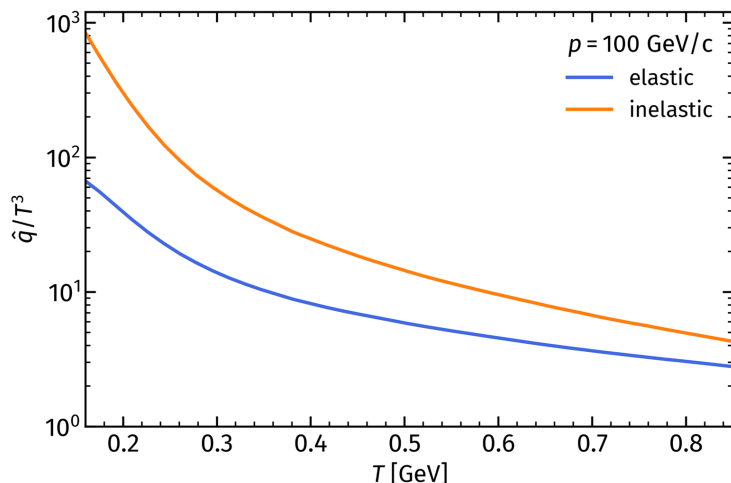
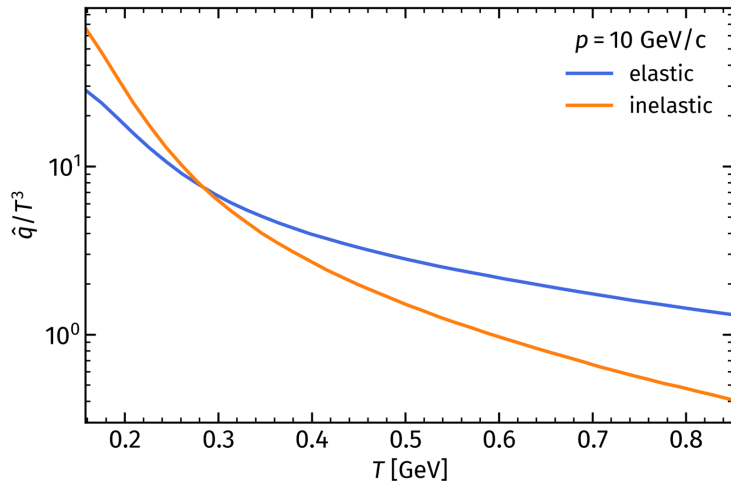
MUSIC:
C. Shen,
S. Alzhrani, PRC
102 (2020) 1,
014909

Properties of QGP: transport coefficients



- **Logarithmic growth** of $\hat{q}$ and energy loss dE/dx with **jet energy E**
- DQPM predicts **stronger suppression than pQCD**
- **Aligning** with **pQCD-based** calculations in the **pQCD-limit**

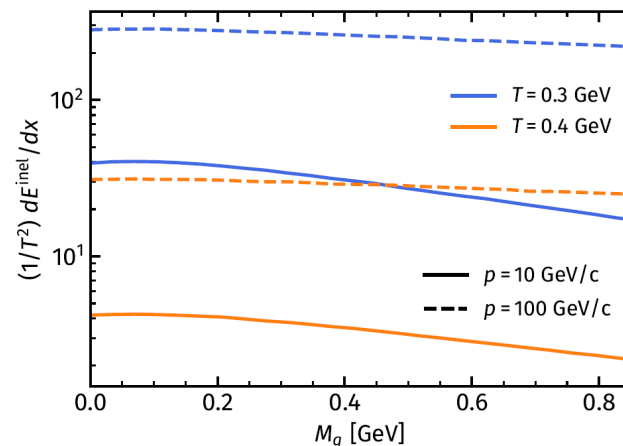
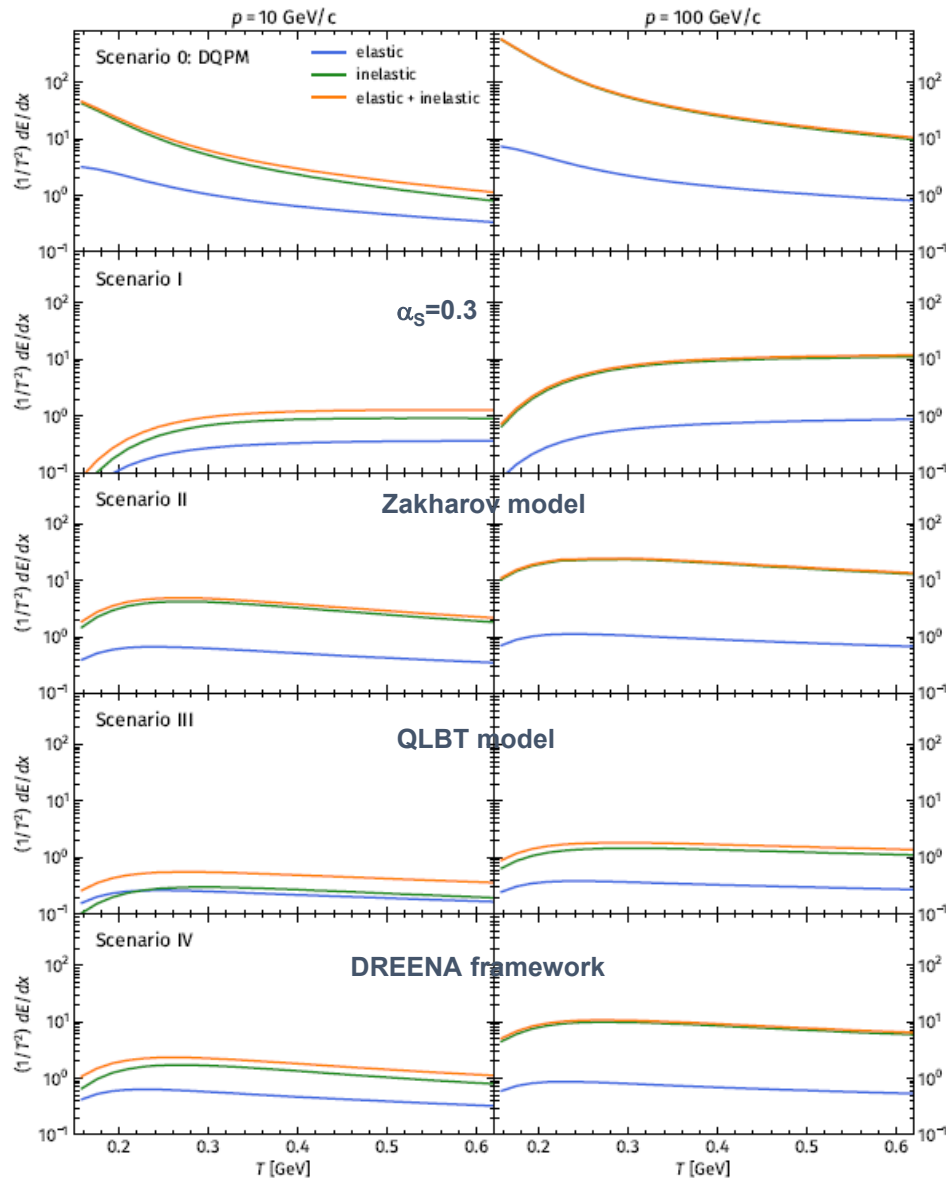
$\hat{q}$ from elastic ($2 \rightarrow 2$) vs. inelastic ($2 \rightarrow 3$) scattering



→ Strong **dependence** of **elastic** ($\sim g^4$) and **inelastic** ($\sim g^6$) cross sections on **strong coupling g**

- Temperature and momentum **dependence** is stronger for **inelastic** reactions
- **Stronger energy loss** at large jet energies and small temperatures
- ? '**questionable**' **suppression** of jets in heavy-ion collisions

dE/dx from elastic + inelastic scattering



- **High sensitivity** to the **choice** of the **strong coupling** (“scenario”)
- The **“default” DQPM** with the thermal couplings produces the **highest energy loss**
- **Energy loss** is decreasing with increasing m_g

Connection to SUBA-Jet

SUBA-Jet :

elastic/inelastic collision $\rightarrow$ preformed virtual gluon $\rightarrow$ additional medium kicks
 $\rightarrow$ phase accumulation $\rightarrow$ real emitted gluon.

Present DQPM work

SUBA-Jet

arxiv:2404.14579

Gluon is generated as part of a complete 2 $\rightarrow$ 3 event

Starts from a preformed virtual gluon

Formation time evaluated locally

Formation evolves after successive kicks

No propagation during formation

Explicit evolution during formation

Coherence encoded by $S_{\text{LPM}}(r)$

Coherence encoded through accumulated phase

Exact off-shell DQPM matrix element

Gunion-Bertsch radiation seed

Main results: $\hat{q}_{\text{rad}}$, dE/dx

Main benchmark: $dN_g/d\omega$, $dN_g/dk_{\perp}$

Natural PHSD embedding

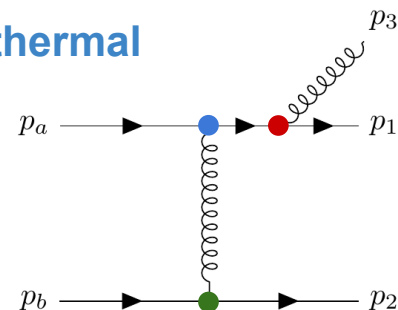
Full jet Monte Carlo architecture

Different scenarios for strong couplings

- Jet is **not a part** of the QGP medium $\rightarrow$ strong coupling is **not thermal**

$\rightarrow$ consider **different strong couplings**

in *thermal* (●), *jet* (●), and *radiative* (●) vertices



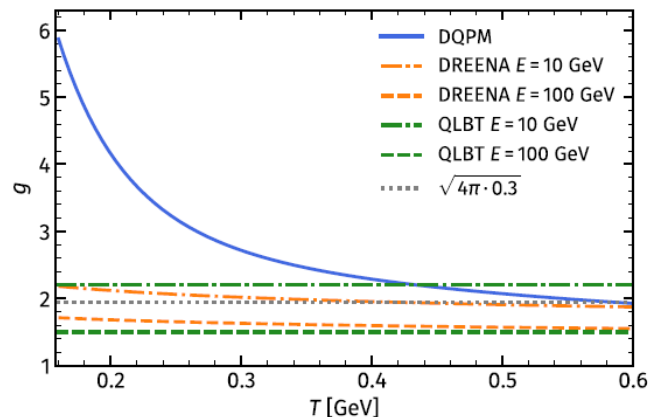
	Vertex		
Model	● medium parton	● jet parton	● emitted gluon
Scenario 0	$g^{\text{DQPM}}(T)$		
Scenario I	$g = \sqrt{4\pi \times 0.3}$		
Zakharov model	Scenario II	$g^{\text{DQPM}}(T)$	$g(Q^2)$
	Scenario III	$g^{\text{DQPM}}(T)$	$g(E)$
QLBT model	Scenario IV	$g^{\text{DQPM}}(T)$	$g(ET)$
			$g(Q^2)$

$\rightarrow$ “default” DQPM (thermal coupling)

$\rightarrow$ constant strong coupling

$$g^2(t) = \frac{48\pi^2}{(11N_c - 2N_f)} \frac{1}{\ln\left(\frac{t}{\Lambda^2}\right)}$$

Zakharov model
QLBT model
DREENA framework

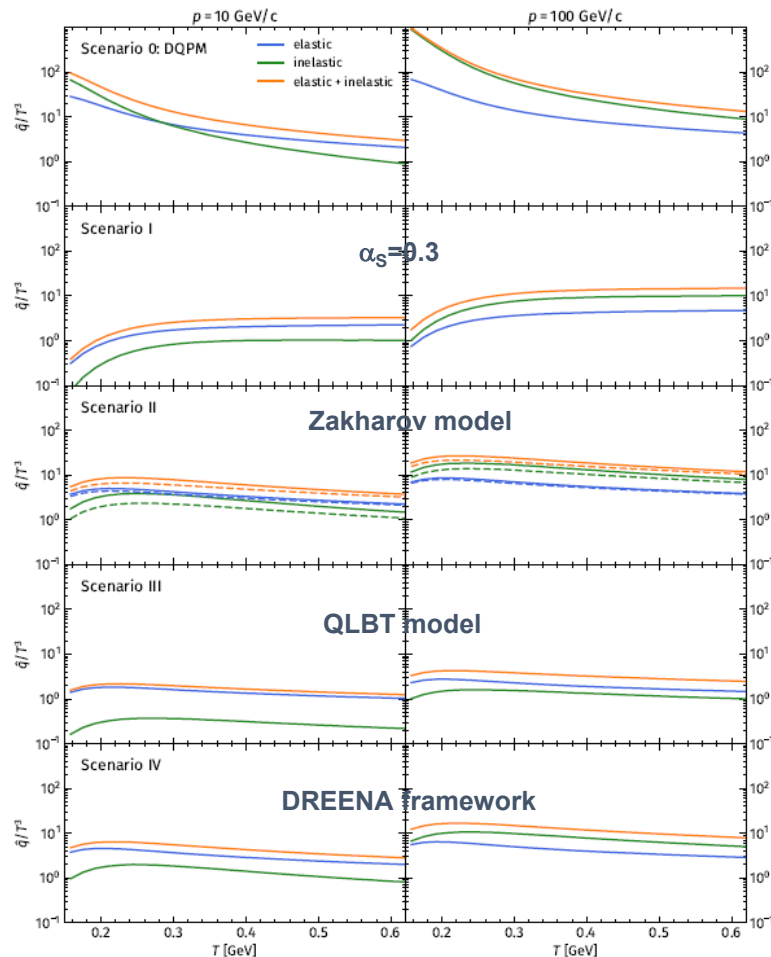
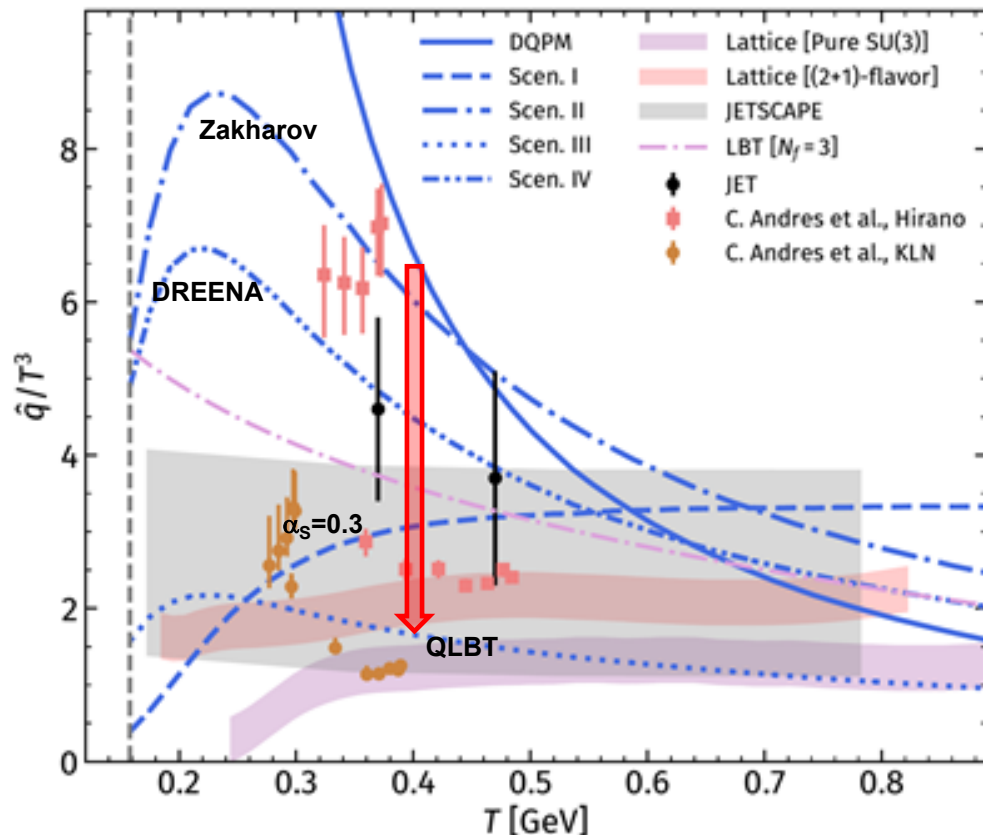


[II] B. G. Zakharov, JETP Lett. 112, 681 (2020).

[III] F.-L. Liu et al., Eur. Phys. J. C 82, 350 (2022).

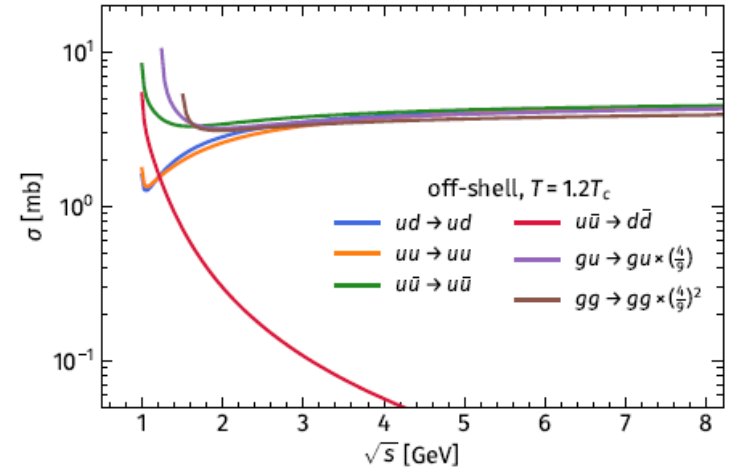
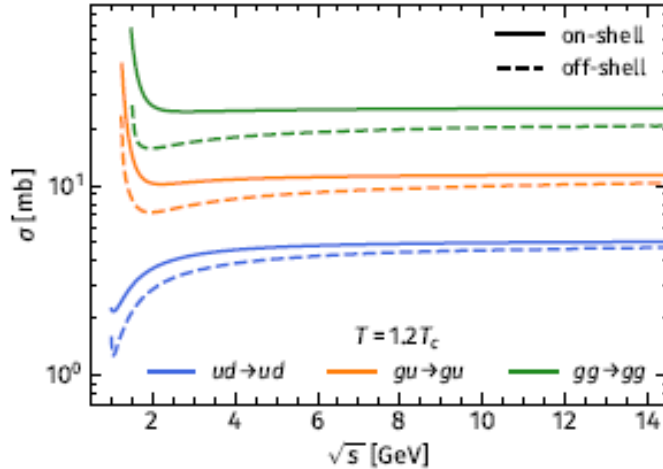
[IV] D. Zigic et al., Front. Phys. 10, 957019 (2022).

$\hat{q}$ from elastic + inelastic scattering



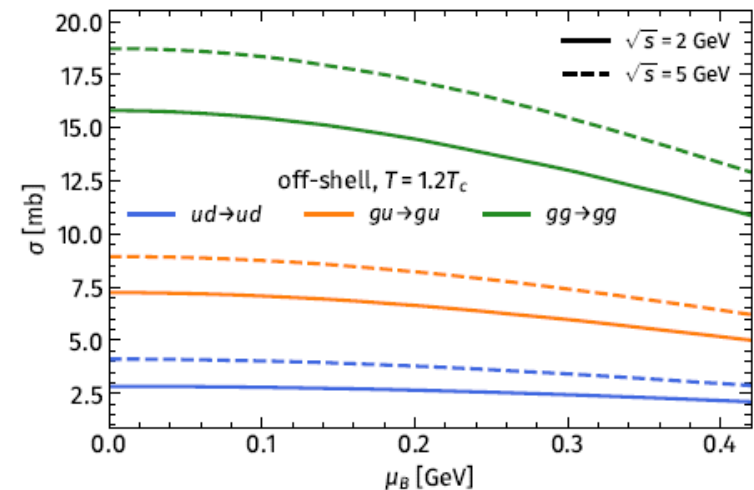
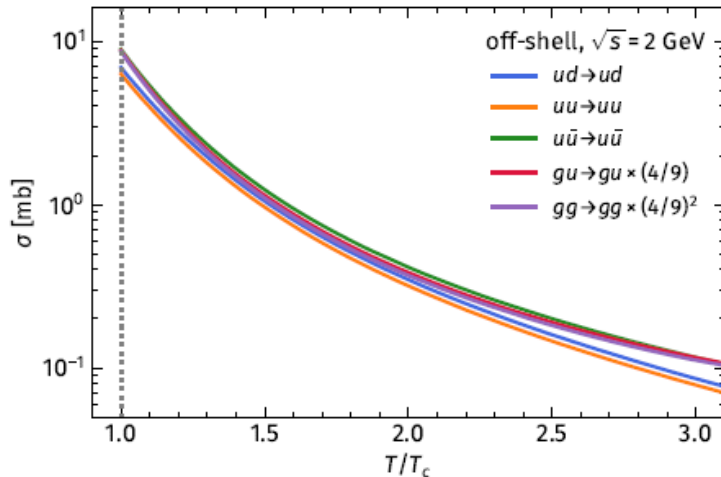
- **High sensitivity** to the **choice** of the **strong coupling** (“scenario”)
- The “**default**” **DQPM** with the thermal couplings produces the **highest values** of the transport coefficients

Total elastic cross sections



□ off-shell effects are stronger at low $s^{1/2}$

□ strong channel dependence at low $s^{1/2}$



□ strong T dependence

□ weak μ_B dependence

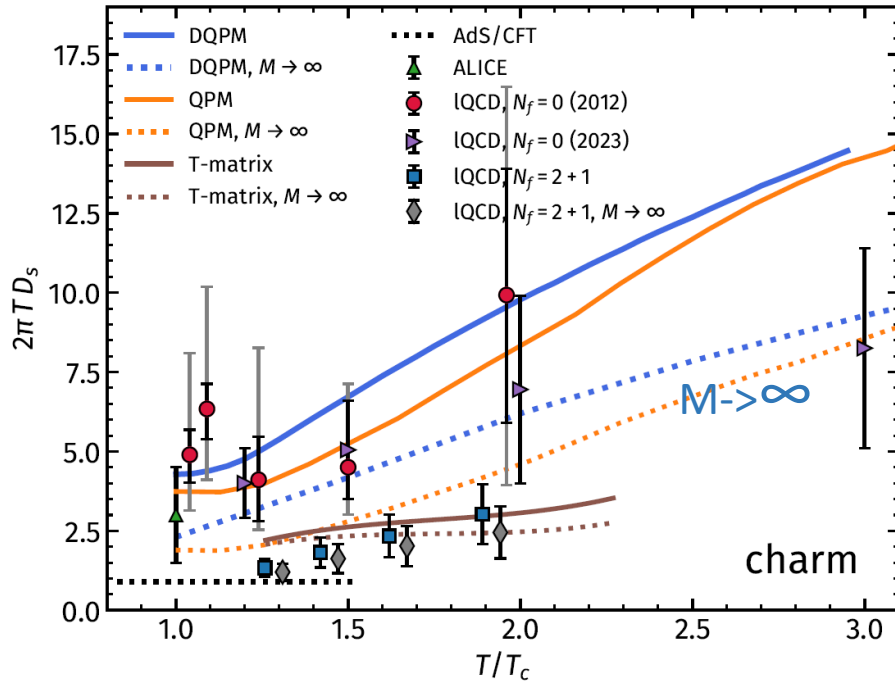
□ ~ scaling with color ratio

$$|\overline{\mathcal{M}}_{gg}|^2 \approx \frac{C_g}{C_q} |\overline{\mathcal{M}}_{gq}|^2 \approx \left(\frac{C_g}{C_q} \right)^2 |\overline{\mathcal{M}}_{qq}|^2$$

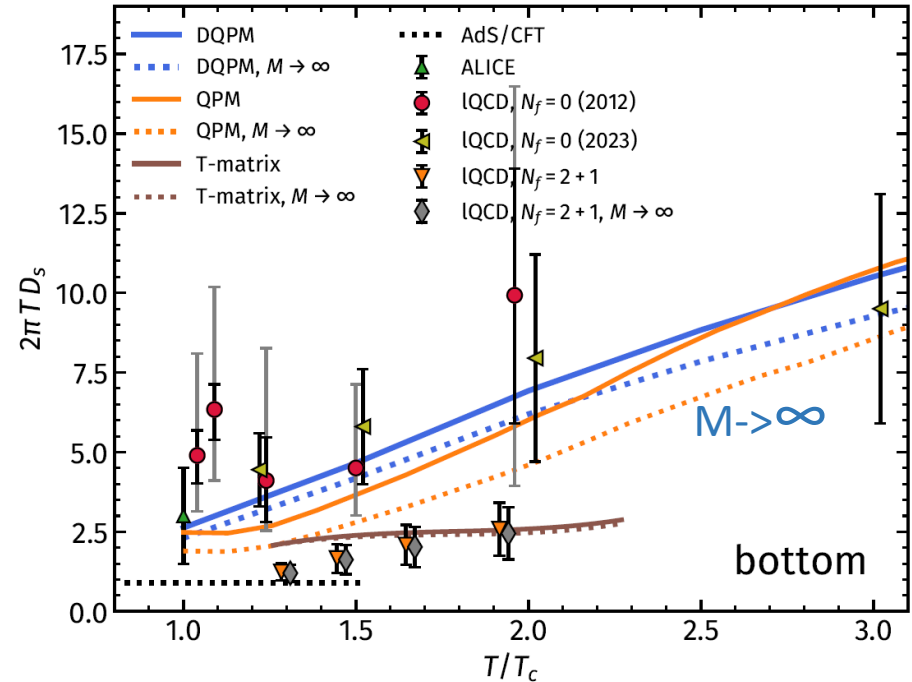
I. Grishmanovskii, O.S. et al., PRC 110, 014908 (2024)
plots from PhD Thesis

Spatial diffusion coefficient $D_s(T)$ for heavy quarks

Charm quark



Bottom quark



→ **Inelastic reactions play a minor role** (vs. elastic reactions) in the determination of D_s which is obtained for $p \rightarrow 0$ limit

$$D_s = \lim_{p \rightarrow 0} \frac{T}{(\mathcal{A}/p)M}$$

→ **D_s decreases with increasing mass** of the heavy quark

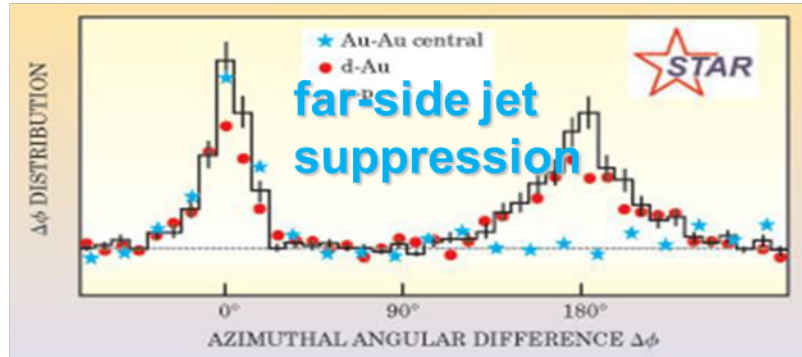
Summary

- DQPM provides a **self-consistent approach** to study **partonic interactions** and **transport properties** of the sQGP
- **Inelastic** interactions are **suppressed** in a thermalized QGP medium, but are **crucial** in the context of **jet attenuation**
- **Jet energy loss** of hard jet partons is **larger** within the DQPM compared to the pQCD-based calculations
- **transport coefficients** are highly **sensitive** to the choice of the **strong coupling**
- $D_s(T)$ for heavy quarks is not sensitive to inelastic reactions

Thank you for your attention!

Jet suppression

C. Adler et al. [STAR], Phys. Rev. Lett. 90 (2003) 082302



STAR (2003):

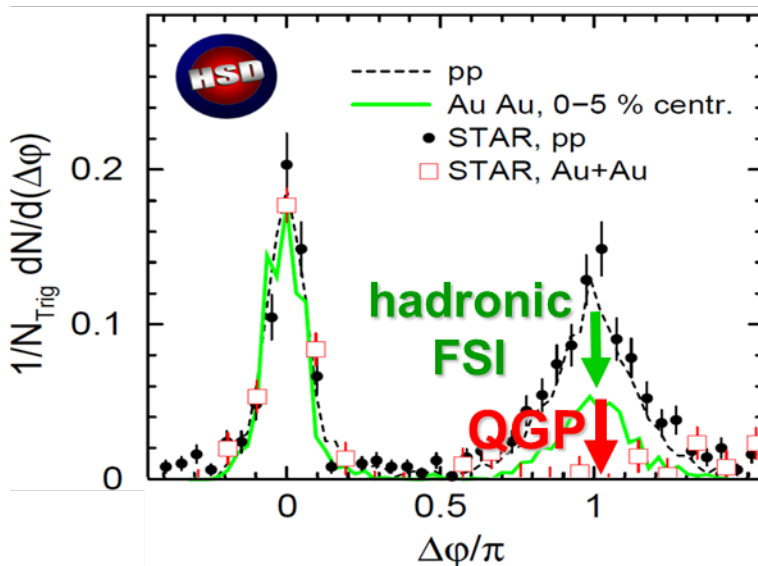
Discovery of “**Jet Quenching**”:
strong suppression of far-side
jet in A+A vs p+p and d+A



HSD (Hadron-String Dynamics)

microscopic transport approach (2004):

W. Cassing, K. Gallmeister, C. Greiner,
J.Phys.G30 (2004) S801; NPA 748 (2005) 41



- good description of jet quenching in p+p, d+A and near-side jet angular correlation for central Au+Au
 - strong underestimation of suppression of the far-side jet due to hadronic interaction
- indication for **strong jet suppression in QGP**

Modeling of interaction
of jet partons with sQGP

DQPM: EoS

Entropy and baryon density in the quasiparticle limit
(G. Baym 1998, Blaizot et al. 2001):

$$s^{dqp} = - \int \frac{d\omega}{2\pi} \frac{d^3p}{(2\pi)^3} \left[d_g \frac{\partial n_B}{\partial T} (\text{Im}(\ln -\underline{\Delta}^{-1}) + \text{Im} \underline{\Pi} \text{Re} \underline{\Delta}) \right. \\ \left. + \sum_{q=u,d,s} d_q \frac{\partial n_F(\omega - \mu_q)}{\partial T} (\text{Im}(\ln -\underline{S}_q^{-1}) + \text{Im} \underline{\Sigma}_q \text{Re} \underline{S}_q) \right. \\ \left. + \sum_{\bar{q}=\bar{u},\bar{d},\bar{s}} d_{\bar{q}} \frac{\partial n_F(\omega + \mu_q)}{\partial T} (\text{Im}(\ln -\underline{S}_{\bar{q}}^{-1}) + \text{Im} \underline{\Sigma}_{\bar{q}} \text{Re} \underline{S}_{\bar{q}}) \right] \\ n^{dqp} = - \int \frac{d\omega}{2\pi} \frac{d^3p}{(2\pi)^3} \left[\sum_{q=u,d,s} d_q \frac{\partial n_F(\omega - \mu_q)}{\partial \mu_q} (\text{Im}(\ln -\underline{S}_q^{-1}) + \text{Im} \underline{\Sigma}_q \text{Re} \underline{S}_q) \right. \\ \left. + \sum_{\bar{q}=\bar{u},\bar{d},\bar{s}} d_{\bar{q}} \frac{\partial n_F(\omega + \mu_q)}{\partial \mu_q} (\text{Im}(\ln -\underline{S}_{\bar{q}}^{-1}) + \text{Im} \underline{\Sigma}_{\bar{q}} \text{Re} \underline{S}_{\bar{q}}) \right]$$

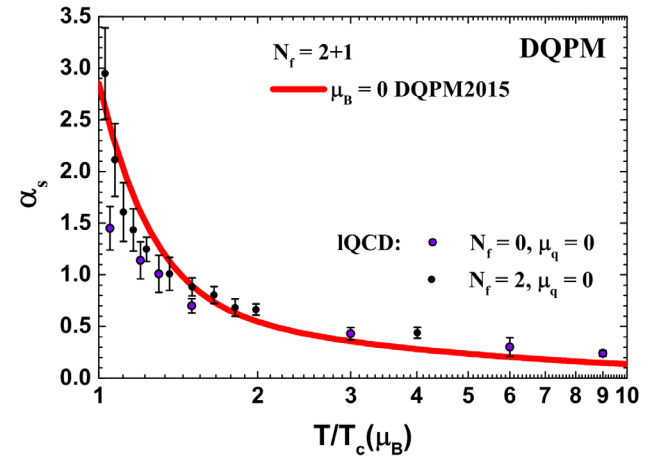
➤ Input: entropy density as a $f(T, \mu_B = 0)$

$$g^2(s/s_{SB}) = d((s/s_{SB})^e - 1)^f \quad \text{fix the parameters}$$

$$s^{DQPM}(\Pi, \Delta, S_q, \Sigma) = s^{lattice}$$

➤ Scaling hypothesis for the **crossover region** at finite μ_B

$$g^2(T/T_c, \mu_B) = g^2\left(\frac{T^*}{T_c(\mu_B)}, \mu_B = 0\right) \quad \text{with} \quad T^* = \sqrt{T^2 + \mu_q^2/\pi^2}$$



Properties of QGP: transport coefficients



$EoS(\epsilon, n)$
 $\sigma(\sqrt{s}, m_q, m_q, T, \mu_B)$
 $m(T, \mu_B)$

On practice: effective models
for QGP

Today:

Transport coefficients at finite T and μ_B

- 1.) crossover, CEP and 1st order phase transition ($N_f = 3$ PNJL model)
- 2.) crossover + CEP ($N_f = 3$ DQPM)



– Dynamical QPM for partonic phase

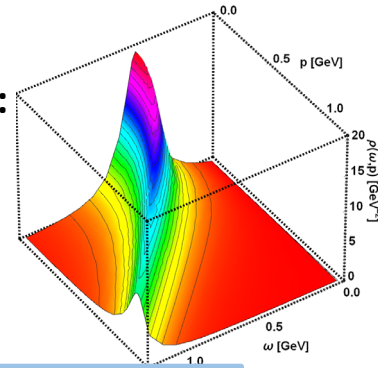
W. Cassing, E. Bratkovskaya, PRC 78 (2008) 034919
P. Moreau, O. S , L. Oliva, T. Song, W. Cassing, E. Bratkovskaya,
PRC 100 (2019) , 014911;
O. S, P. Moreau, L. Oliva, V. Voronyuk, V. Kireyeu, T. Song,
E. Bratkovskaya, Particles 3 (2020), 178-192

Dynamical Quasi-Particle Model

The QGP phase is described in terms of strongly-interacting quasiparticles - quarks and gluons with Lorentzian spectral functions:

$$\rho_j(\omega, \mathbf{p}) = \frac{\gamma_j}{\tilde{E}_j} \left(\frac{1}{(\omega - \tilde{E}_j)^2 + \gamma_j^2} - \frac{1}{(\omega + \tilde{E}_j)^2 + \gamma_j^2} \right)$$

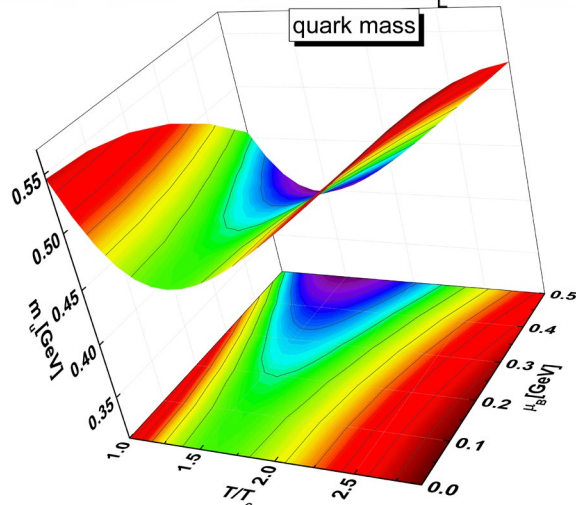
$$\equiv \frac{4\omega\gamma_j}{(\omega^2 - \mathbf{p}^2 - M_j^2)^2 + 4\gamma_j^2\omega^2}$$



resummed propagators: $\Delta_i(\omega, \mathbf{p}) = \frac{1}{\omega^2 - \mathbf{p}^2 - \Pi_i}$ & self-energies: $\Pi_i = m_i^2 - 2i\gamma_i\omega$

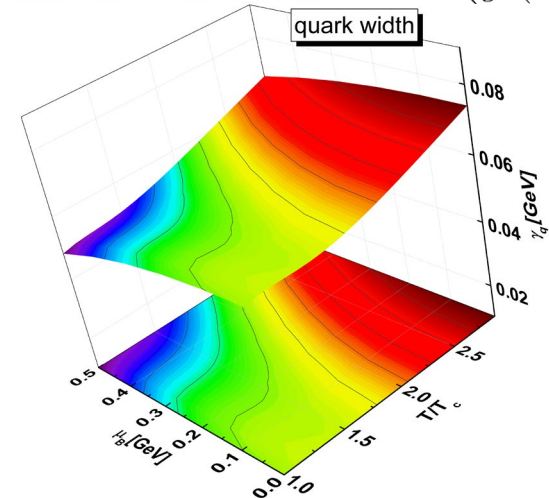
Re Π_i : thermal mass (M_g, M_q)

$$m_{q(\bar{q})}^2(T, \mu_B) = C_q \frac{g^2(T, \mu_B)}{4} T^2 \left[1 + \left(\frac{\mu_B}{3\pi T} \right)^2 \right]$$



Im Π_i : interaction width (γ_g, γ_q)

$$\gamma_j(T, \mu_B) = \frac{1}{3} C_j \frac{g^2(T, \mu_B) T}{8\pi} \ln \left(\frac{2c_m}{g^2(T, \mu_B)} + 1 \right)$$



DQPM: EoS

Entropy and baryon density in the quasiparticle limit
(G. Baym 1998, Blaizot et al. 2001):

$$s^{dqp} = - \int \frac{d\omega}{2\pi} \frac{d^3p}{(2\pi)^3} \left[d_g \frac{\partial n_B}{\partial T} (\text{Im}(\ln -\underline{\Delta}^{-1}) + \text{Im} \underline{\Pi} \text{Re} \underline{\Delta}) \right. \\ \left. + \sum_{q=u,d,s} d_q \frac{\partial n_F(\omega - \mu_q)}{\partial T} (\text{Im}(\ln -\underline{S}_q^{-1}) + \text{Im} \underline{\Sigma}_q \text{Re} \underline{S}_q) \right. \\ \left. + \sum_{\bar{q}=\bar{u},\bar{d},\bar{s}} d_{\bar{q}} \frac{\partial n_F(\omega + \mu_q)}{\partial T} (\text{Im}(\ln -\underline{S}_{\bar{q}}^{-1}) + \text{Im} \underline{\Sigma}_{\bar{q}} \text{Re} \underline{S}_{\bar{q}}) \right] \\ n^{dqp} = - \int \frac{d\omega}{2\pi} \frac{d^3p}{(2\pi)^3} \left[\sum_{q=u,d,s} d_q \frac{\partial n_F(\omega - \mu_q)}{\partial \mu_q} (\text{Im}(\ln -\underline{S}_q^{-1}) + \text{Im} \underline{\Sigma}_q \text{Re} \underline{S}_q) \right. \\ \left. + \sum_{\bar{q}=\bar{u},\bar{d},\bar{s}} d_{\bar{q}} \frac{\partial n_F(\omega + \mu_q)}{\partial \mu_q} (\text{Im}(\ln -\underline{S}_{\bar{q}}^{-1}) + \text{Im} \underline{\Sigma}_{\bar{q}} \text{Re} \underline{S}_{\bar{q}}) \right]$$

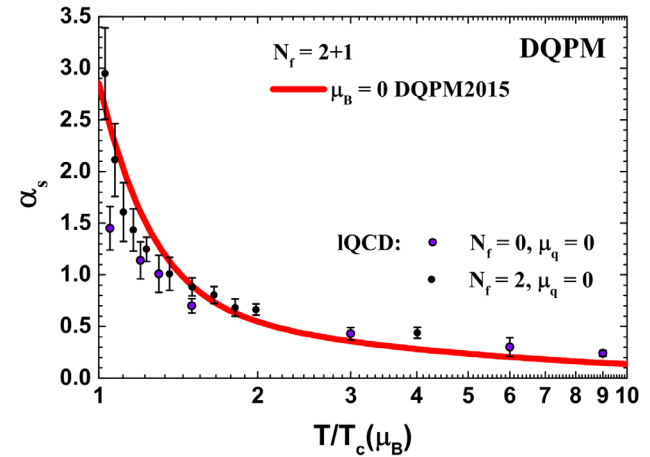
➤ Input: entropy density as a $f(T, \mu_B = 0)$

$$g^2(s/s_{SB}) = d((s/s_{SB})^e - 1)^f \quad \text{fix the parameters}$$

$$s^{DQPM}(\Pi, \Delta, S_q, \Sigma) = s^{lattice}$$

➤ Scaling hypothesis for the **crossover region** at finite μ_B

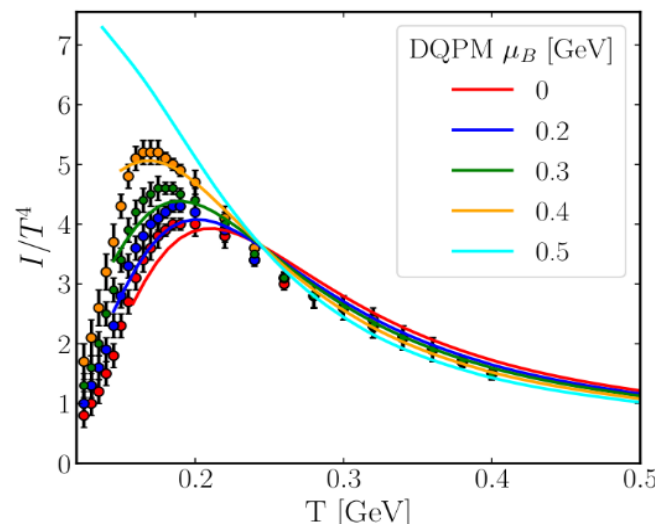
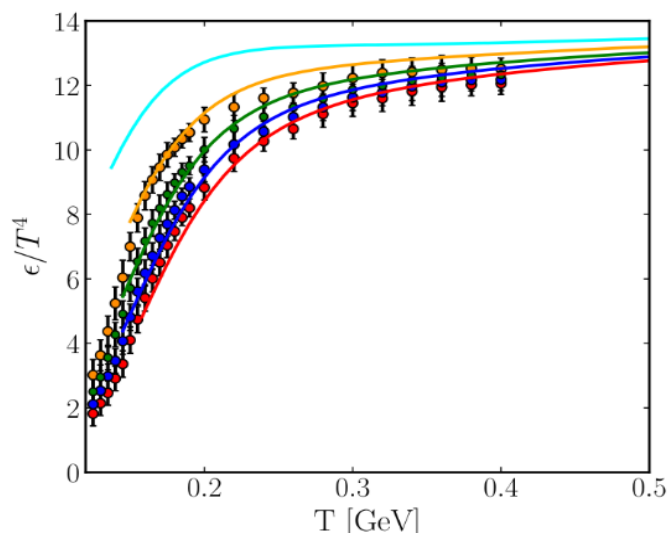
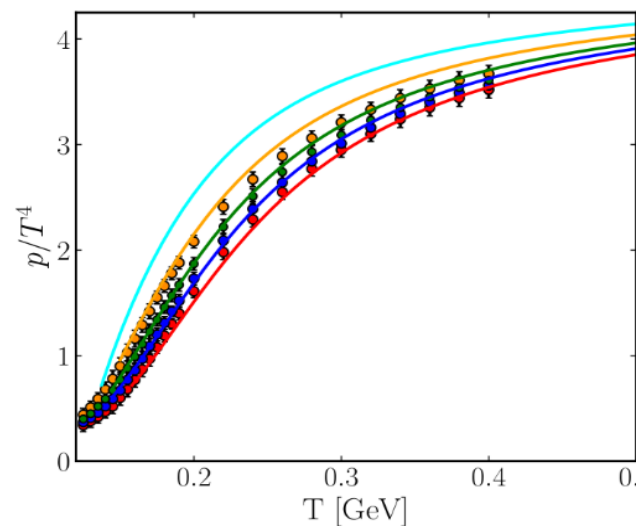
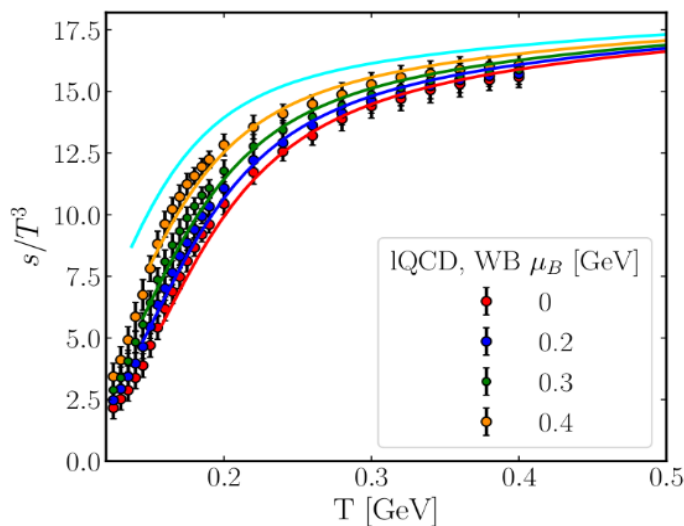
$$g^2(T/T_c, \mu_B) = g^2\left(\frac{T^*}{T_c(\mu_B)}, \mu_B = 0\right) \quad \text{with} \quad T^* = \sqrt{T^2 + \mu_q^2/\pi^2}$$



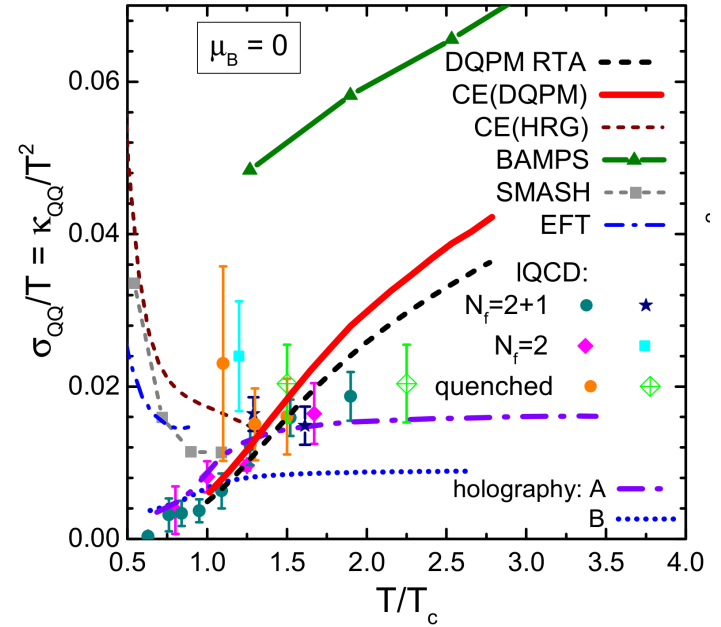
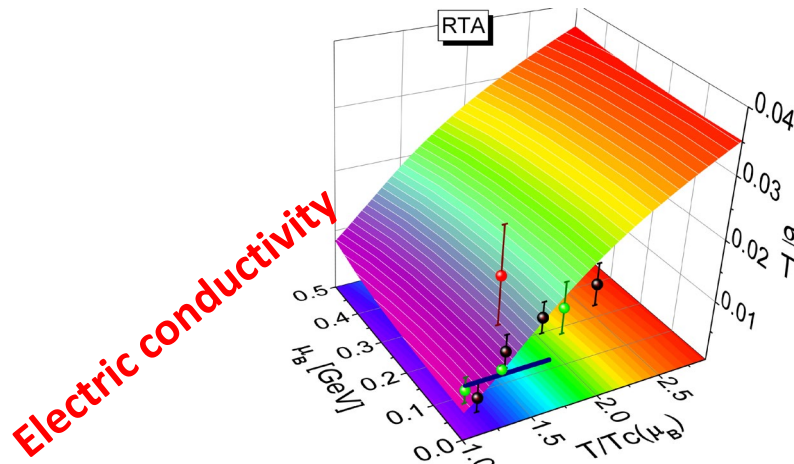
DQPM: EoS

Input:
lattice EoS
 $\mu_B = 0$
(red dots)

Output:
DQPM EoS
 $\mu_B \geq 0$
(lines)



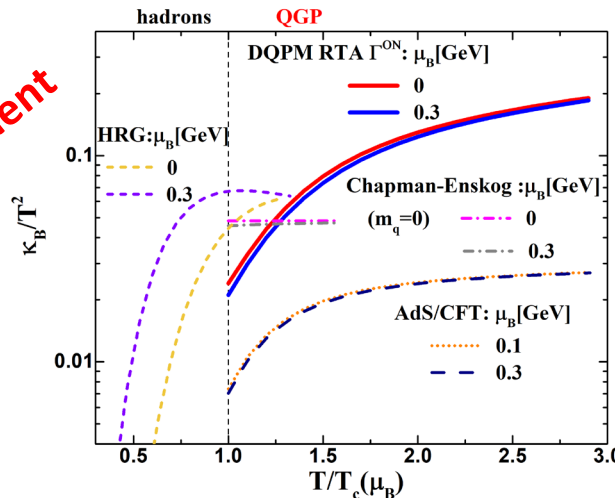
Transport coefficients at finite μ_B



+ Full diffusion coefficient matrix
has been evaluated $\sigma_q = \kappa_q/T$

$$\begin{pmatrix} j_B^\mu \\ j_Q^\mu \\ j_S^\mu \end{pmatrix} = \begin{pmatrix} \kappa_{BB} & \kappa_{BQ} & \kappa_{BS} \\ \kappa_{QB} & \kappa_{QQ} & \kappa_{QS} \\ \kappa_{SB} & \kappa_{SQ} & \kappa_{SS} \end{pmatrix} \cdot \begin{pmatrix} \nabla^\mu \alpha_B \\ \nabla^\mu \alpha_Q \\ \nabla^\mu \alpha_S \end{pmatrix}$$

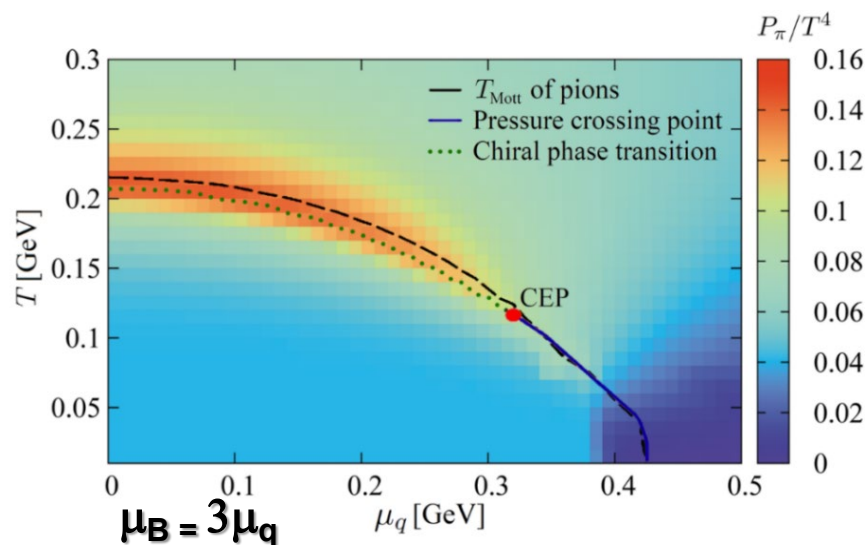
J. A. Fotakis, O. S., C. Greiner, O. Kaczmarek and E. Bratkovskaya PRD 104 (2021) , 034014



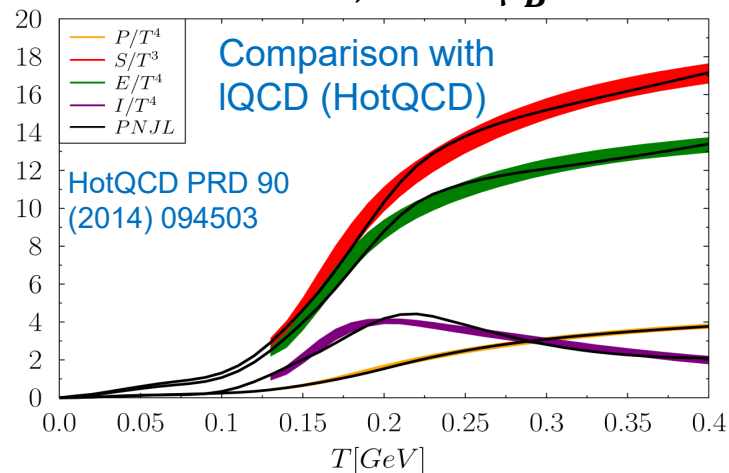
- Light increase with μ_B in the crossover region for shear and bulk viscosities and electric conductivity
- Baryon diffusion coefficients decrease with μ_B

QGP in the Polyakov extended NJL model

- PNJL allows for prediction of macroscopic properties of QGP at finite T and μ_B
- & QGP transport coefficients for $0 \leq \mu_B \leq 1.2$ GeV



➤ Parameters fixed, EoS at $\mu_B = 0$:



➤ **CEP**: $(T, \mu_B) = (110, 960)$ MeV, $\mu_B/T = 8.73$

➤ 1st order PT at high μ_B

➤ **same symmetries** for the quarks as QCD

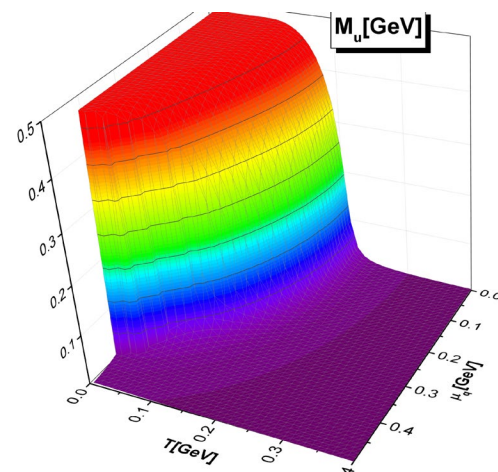
Chiral masses (M_l, M_s)

$$m_i = m_{0i} - 4G\langle\bar{\psi}_i\psi_i\rangle + 2K\langle\bar{\psi}_j\psi_j\rangle\langle\bar{\psi}_k\psi_k\rangle$$

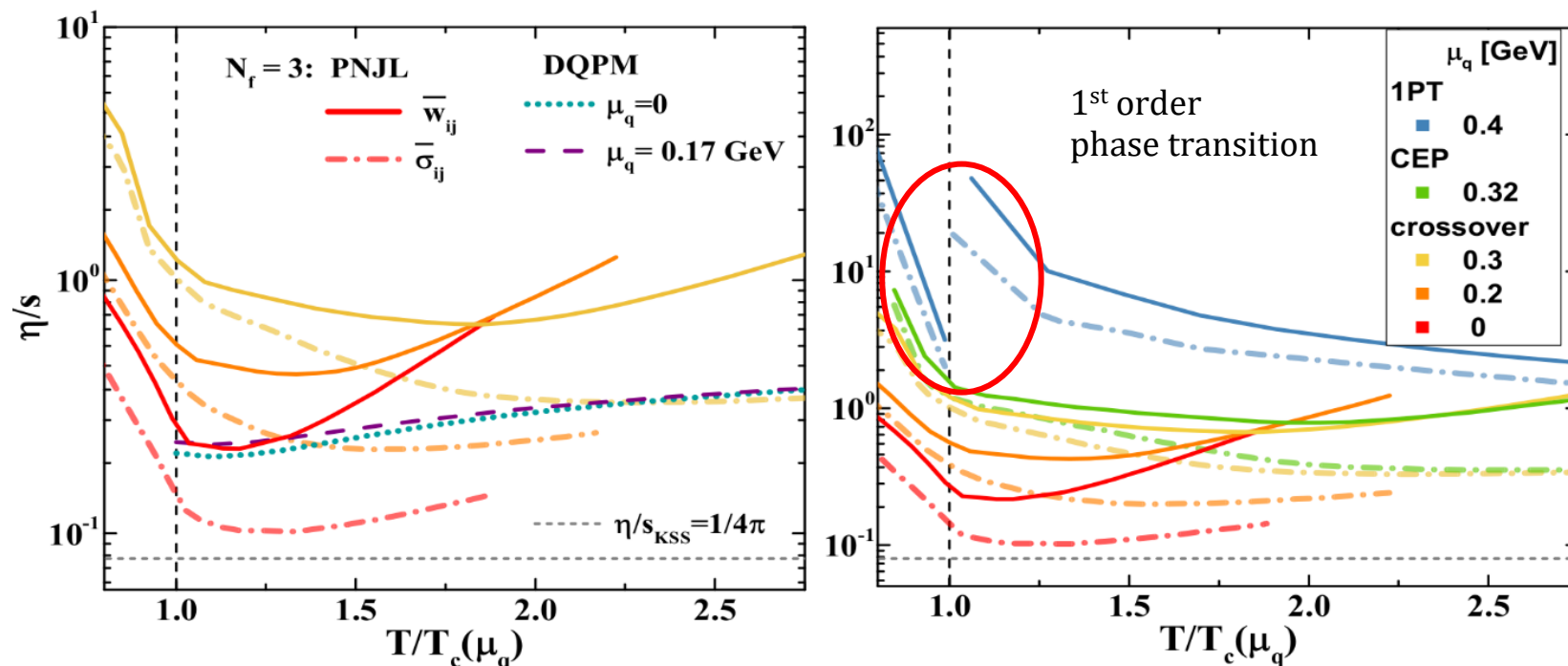
Improved thermodynamics by NNLO in Ω and Polyakov loop

J. M. Torres-Rincon, J. Aichelin PRC 96 (2017) 4 045205

D. Fuseau, T. Steinernert, J. Aichelin PRC 101 (2020) 6 065203



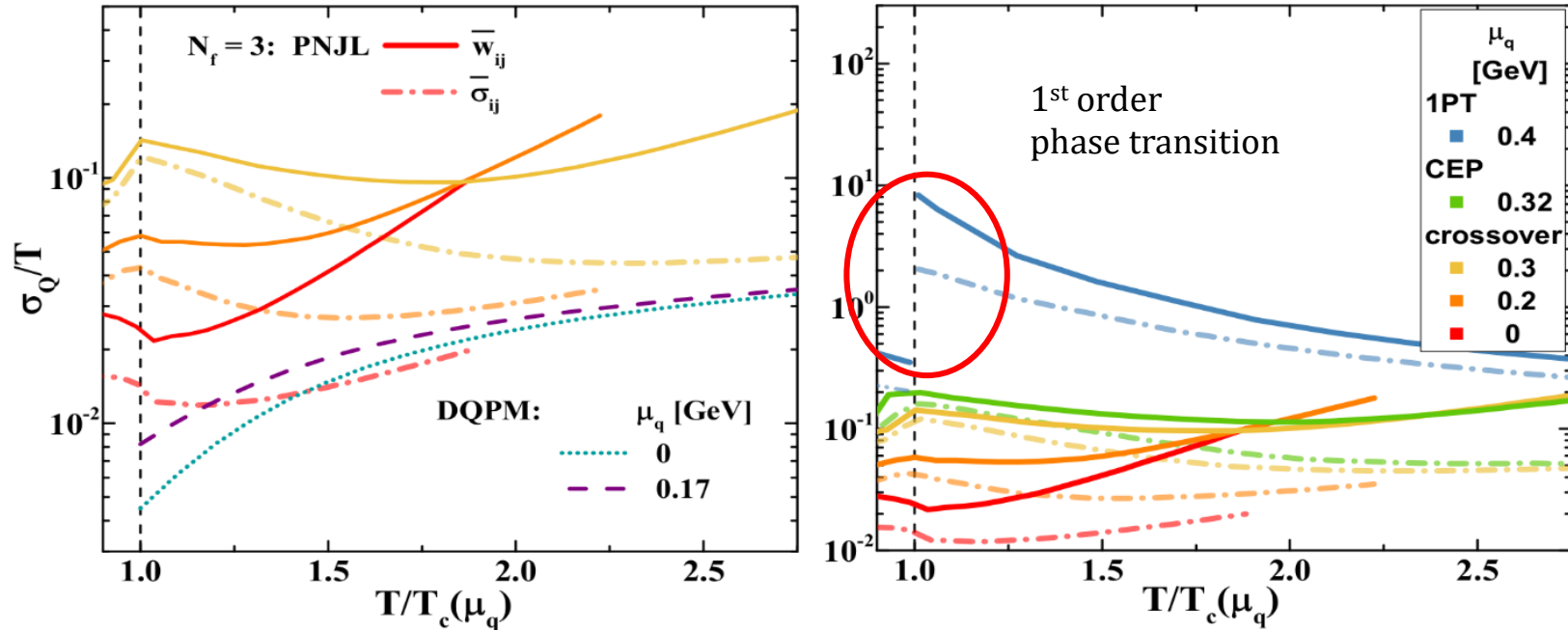
Specific shear viscosity at high μ_B



See also $N_f=2$ NJL results C. Sasaki et al, NPA 832 (2010)

- Two different models have similar increase with μ_B -dependence in the crossover region
- Drastic change of T -dependence for all transport coefficients after 1st order phase transition

Electric conductivity at high μ_B



- Two different models have similar increase with μ_B -dependence in the crossover region
- Drastic change of T-dependence for all transport coefficients after 1st order phase transition

Summary



Local LPM suppression has been embedded consistently into the off-shell DQPM $2 \rightarrow 3$ kernel



Coherence effects increase with jet energy and affect dE/dx more strongly than $\hat{q}$.



The standard massive setup gives moderate suppression; the massless-emission and gluon-jet cases are much more sensitive.



The method is BDMPs-Z-inspired but remains less dynamical than the SUBA-Jet phase-accumulation approach



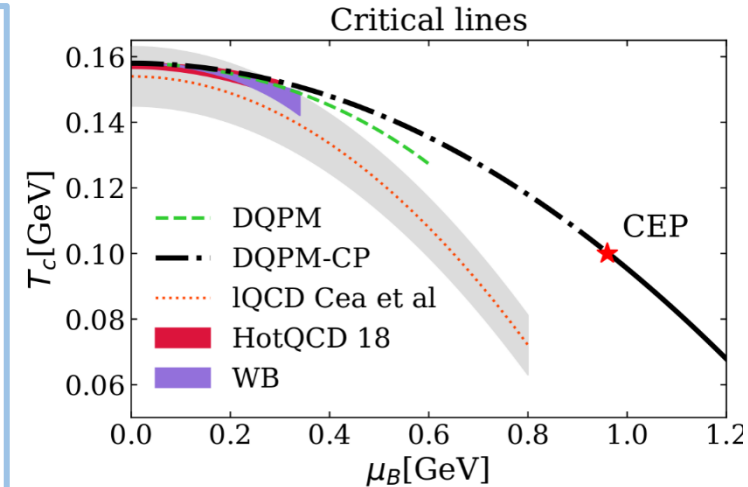
$D_S(T)$ for heavy quarks is not sensitive to inelastic reactions

Antenna interference is substantial and motivates a future unified treatment in thermal transport and PHSD

Thank you for your attention!

Quasiparticle model with CEP at high μ_B

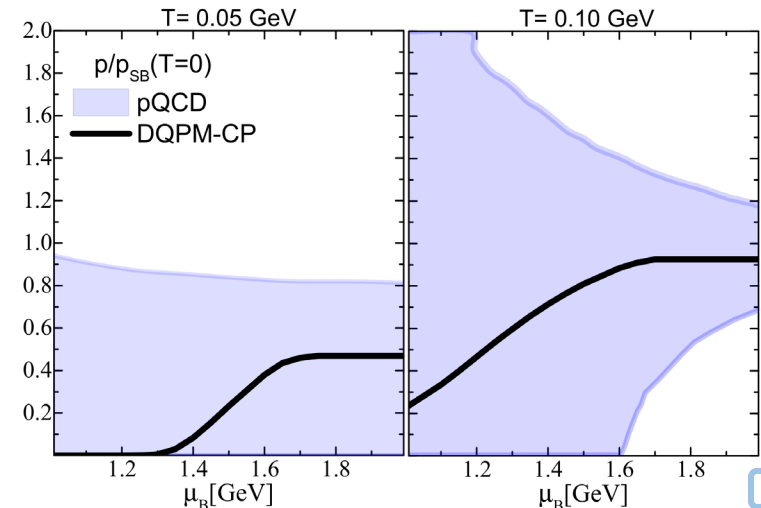
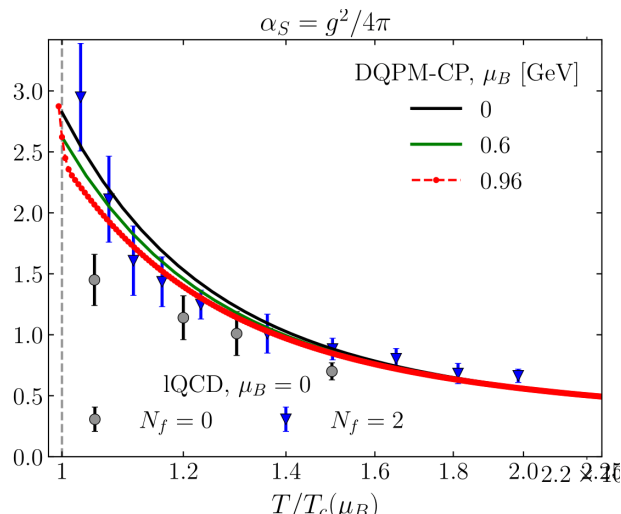
- DQPM-CP for high μ_B , including the CEP region based on the scaling properties of the entropy density from the PNJL model
- DQPM-CP interpolates EoS and microscopic properties between two asymptotics - high $T \gg T_c, \mu_B = 0$ and $T > T_c, \mu_B \gg T$
- EoS and transport coefficients of the QGP phase for the wide range of $T > T_c, \mu_B$



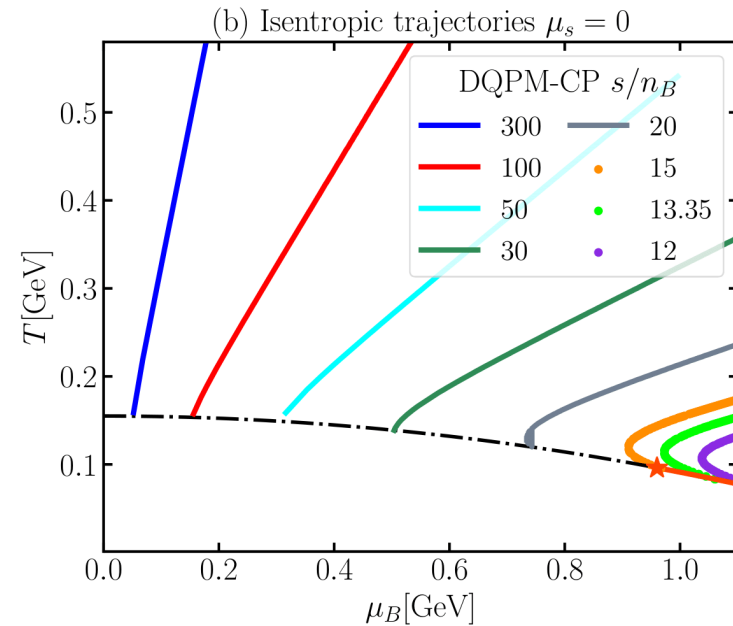
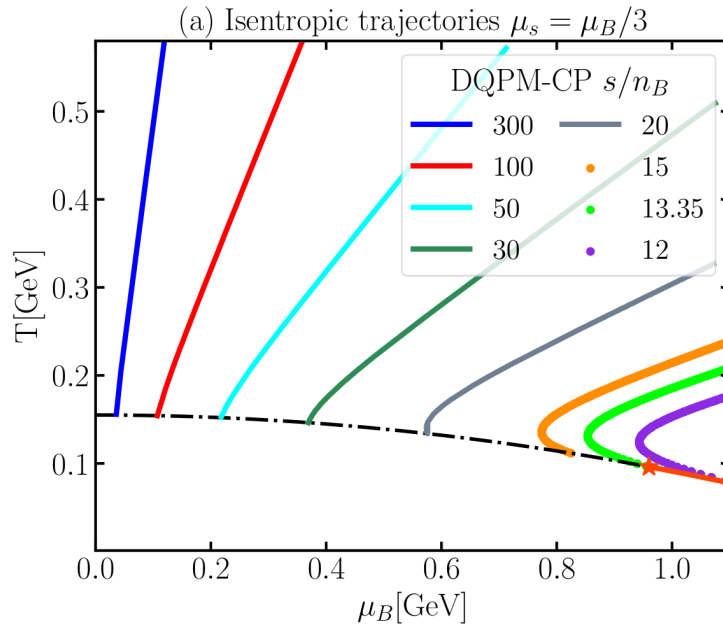
➤ **CEP:** $(T, \mu_B) = (100, 960) \text{ MeV}, \mu_B/T = 9.6$

➤ **EoS:** for $\mu_B/T < 2$ agreement with lQCD for $\mu_B/T > 6$ agreement with pQCD

Near CEP: $g^2 = f(s^{PNJL}(T/T_c)) \rightarrow g^2(T/T_c)$



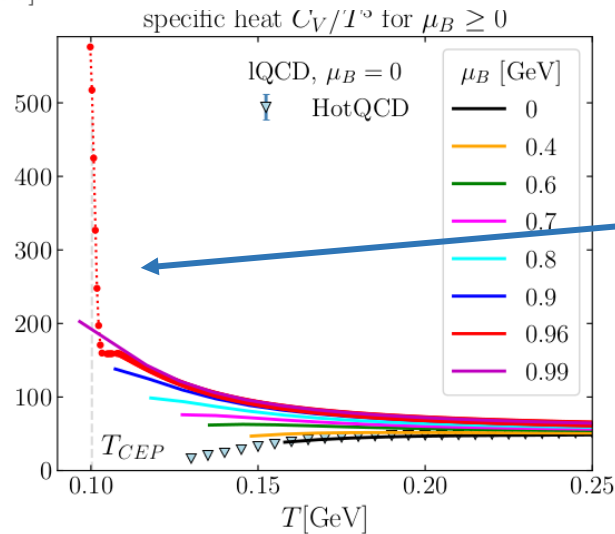
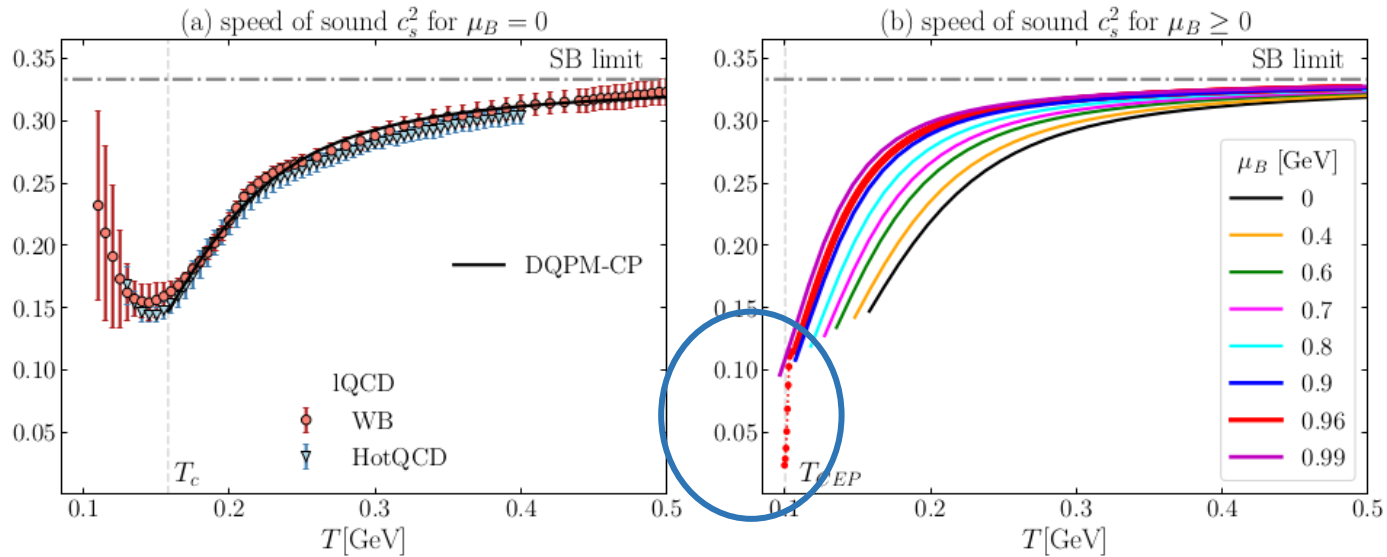
Isentropic trajectories



- CEP acts as an attractor of isentropic trajectories (Chiho Nonaka and Masayuki Asakawa PRC 71 (2005), 044904)
- Trajectories of $s/n_B = \text{const}$ for $\langle n_s \rangle = 0$ are shifted towards higher μ_B

Speed of sound

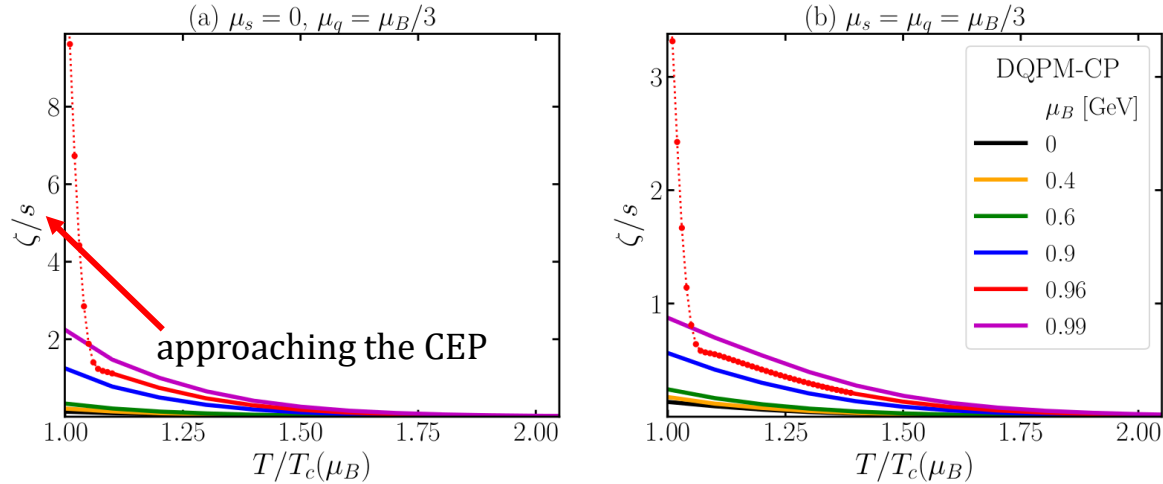
- **EoS**: for $\mu_B/T < 2$ agreement with lQCD for $\mu_B/T > 6$ agreement with pQCD



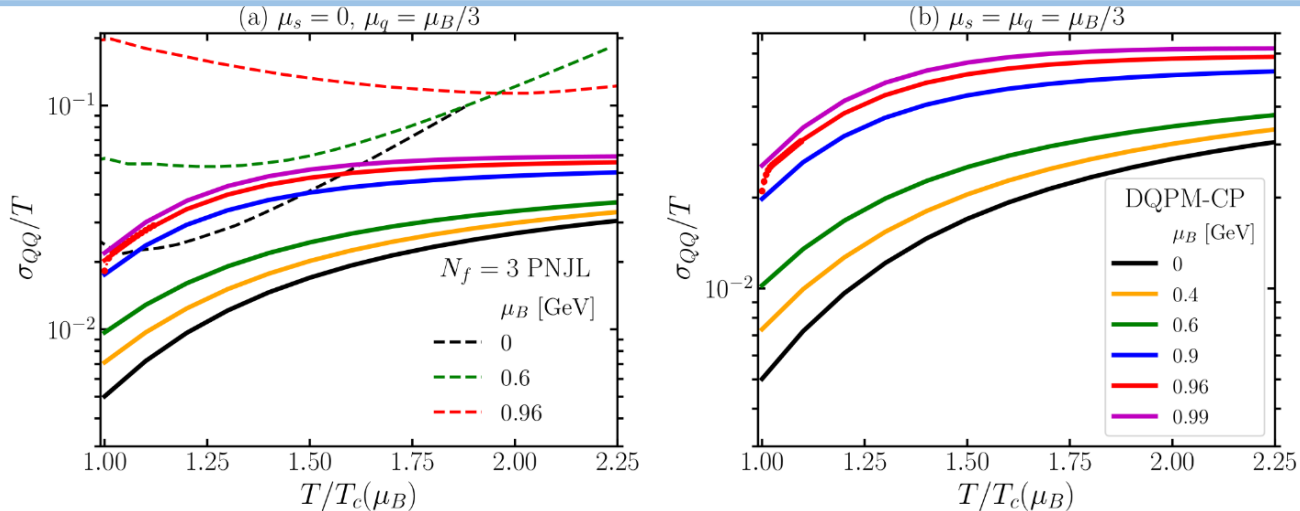
Near CEP critical scaling
can be seen:

$$\ln(C_V) = -\alpha \cdot \ln(T - T_{CEP}) + const$$

Shear and bulk viscosities near the CEP

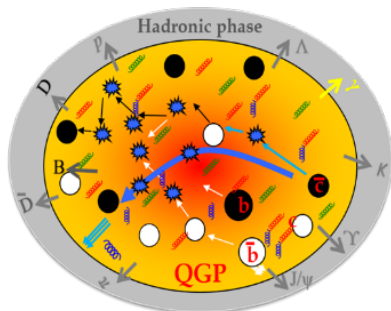


- Sudden rise of specific bulk viscosity approaching the CEP



- B,Q,S diffusion coefficients have pronounced μ_B, μ_S -dependence
- Only small increase approaching the CEP

Modelling HICs: PHSD

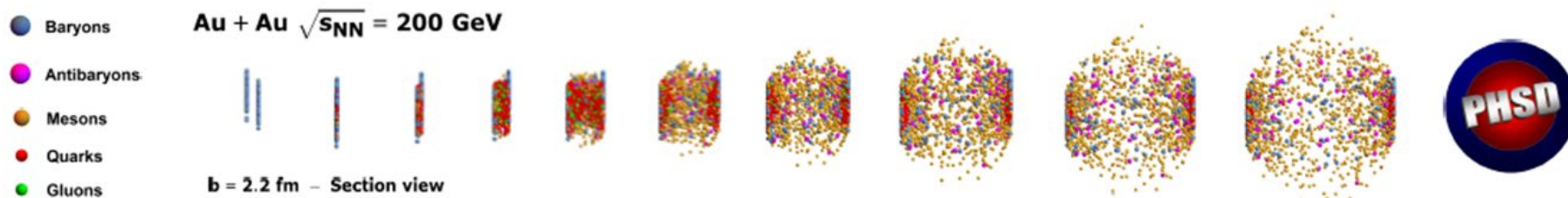


QGP out-of equilibrium $\rightleftharpoons$ HIC

Parton-Hadron-String-Dynamics (PHSD)

Non-equilibrium **microscopic transport approach** for the description of strongly-interacting **hadronic** and **partonic** matter created in heavy-ion collisions

Dynamics: based on the solution of generalized off-shell transport equations derived from Kadanoff-Baym many-body theory



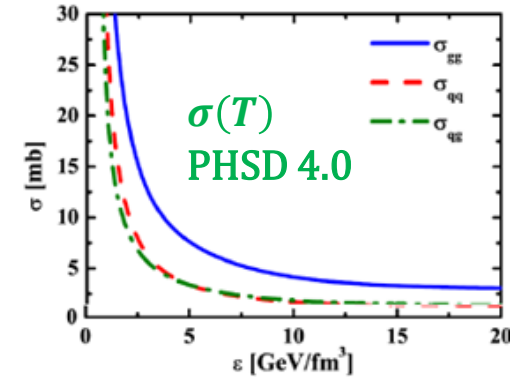
W. Cassing, E. Bratkovskaya, PRC 78 (2008) 034919; NPA831 (2009) 215; W. Cassing, EPJ ST 168 (2009) 3;; P. Moreau, O. Soloveva, L. Oliva, T. Song, W. Cassing, E. Bratkovskaya, PRC 100 (2019), 014911; O. Soloveva, P. Moreau, L. Oliva, V. Voronyuk, V. Kireyeu, T. Song, E. Bratkovskaya, Particles 3 (2020), 178-192;....

PHSD

➤ PHSD 4.0 : only isotropic $\sigma(T)$ and $\rho(T)$

parton cross
sections

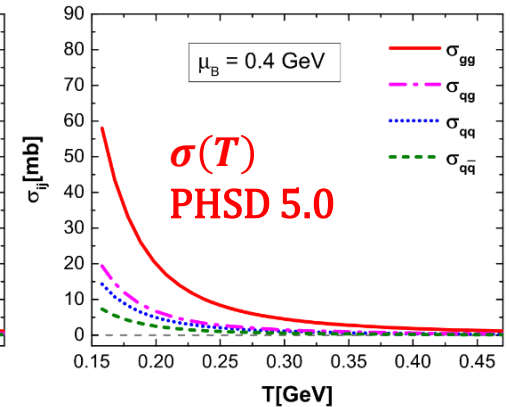
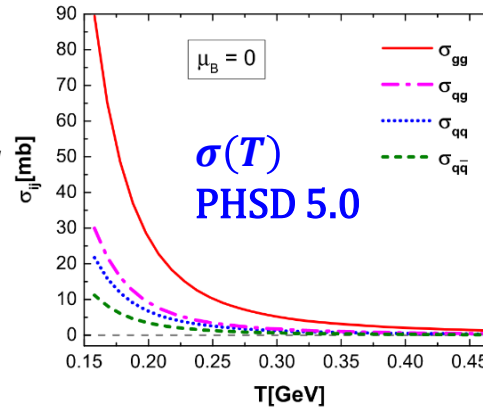
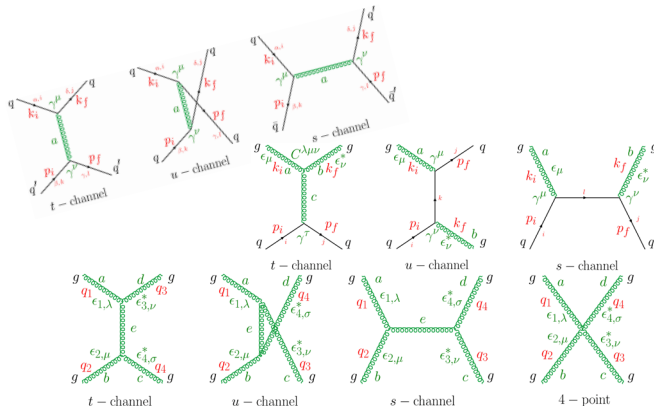
parton
spectral function
(masses and widths)



new PHSD 5 : angular dependence of $d\sigma/d\cos\theta$

➤ PHSD 5.0 : with $\sigma(\sqrt{s}, m_1, m_2, T, \mu_B = 0)$ and $\rho(T, \mu_B = 0)$

➤ PHSD 5.0 : with $\sigma(\sqrt{s}, m_1, m_2, T, \mu_B)$ and $\rho(T, \mu_B)$



PHSD: extraction of T and μ_B

For each space-time cell of the PHSD: $T^{\mu\nu} = \sum_i \frac{p_i^\mu p_i^\nu}{E_i} \rightarrow$ Diagonalize in LRF $\rightarrow \varepsilon^{\text{PHSD}}$

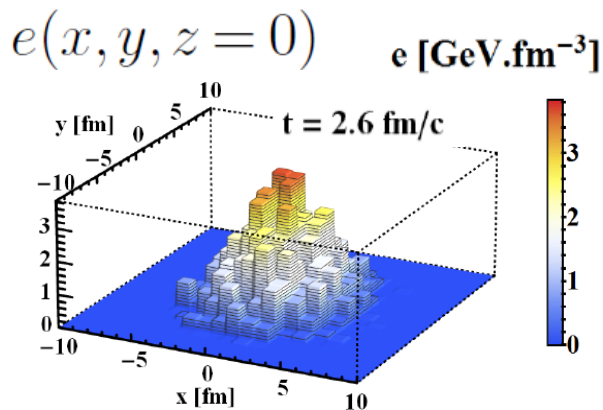
1. Calculate the local energy density $\varepsilon^{\text{PHSD}}$ and baryon density n_B^{PHSD}

2. use IQCD relations
(up to 6th order):

$$\left\{ \begin{array}{l} \frac{n_B}{T^3} \approx \chi_2^B(T) \left(\frac{\mu_B}{T} \right) + \dots \\ \Delta\epsilon/T^4 \approx \frac{1}{2} \left(T \frac{\partial \chi_2^B(T)}{\partial T} + 3\chi_2^B(T) \right) \left(\frac{\mu_B}{T} \right)^2 + \dots \end{array} \right.$$

Use baryon number susceptibilities χ_n from IQCD

Obtain T, μ_B by solving sistem of coupled eqs using ε, n_B



Input:

$\varepsilon^{\text{PHSD}}$ and n_B^{PHSD}



Output:

T, μ_B

for details see P. Moreau, O. S., L. Oliva, T. Song, W. Cassing, E. Bratkovskaya
arXiv:1903.10157, PRC 100 (2019) no. 1, 014911

PHSD: QGP evolution in HICs

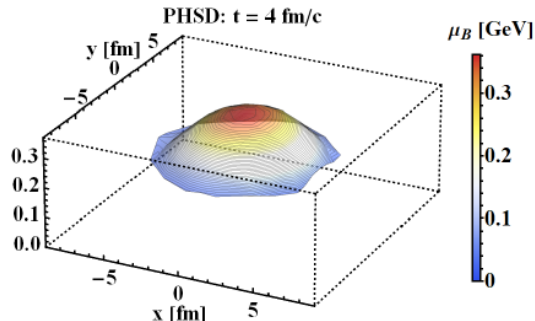
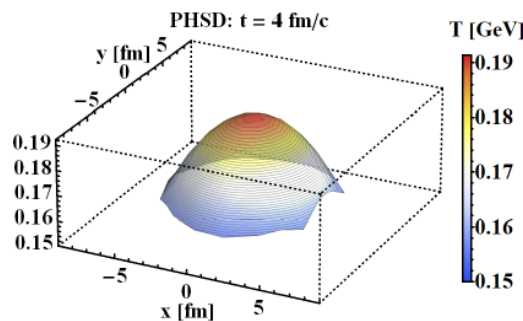
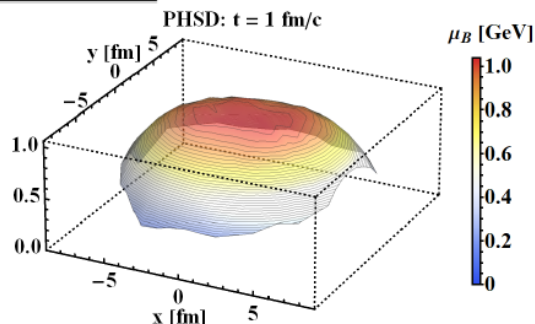
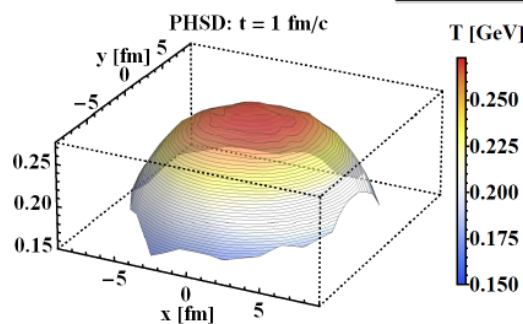
Input:
 ϵ^{PHSD} and n_B^{PHSD}



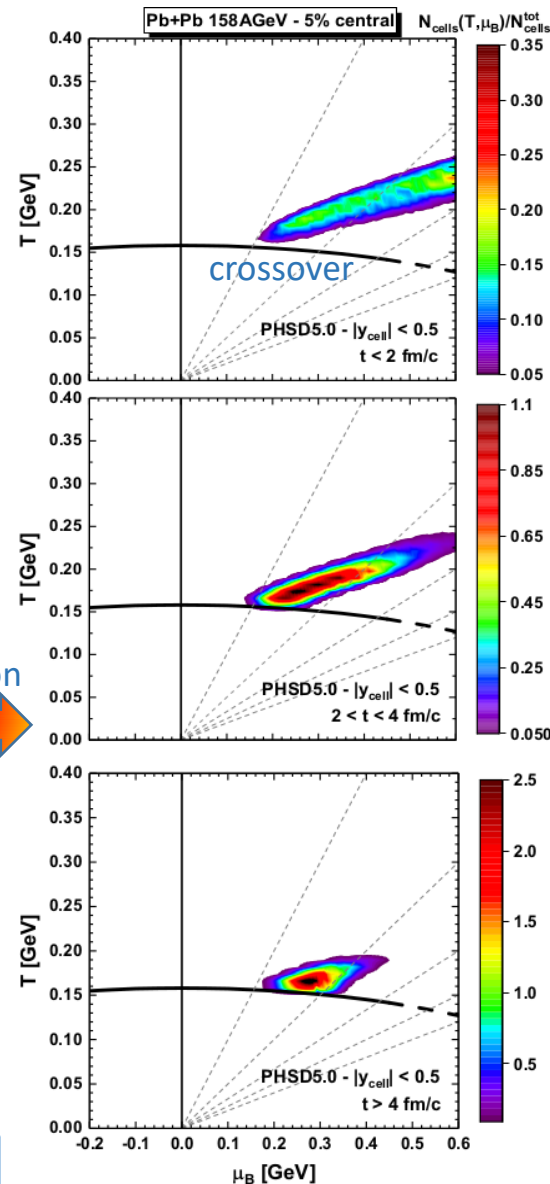
Output:
 T, μ_B

The T -profile in $(x;y)$ & μ_B profile in $(x;y)$
 at midrapidity ($|y_{\text{cell}}| < 1$) at fixed times (1 and 4 fm/c)

Pb+Pb 158A GeV - 5% central



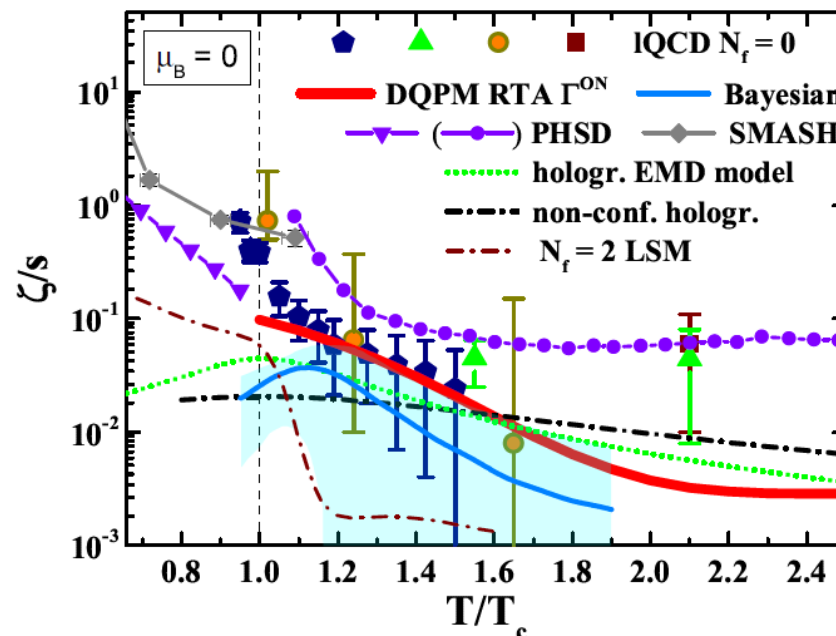
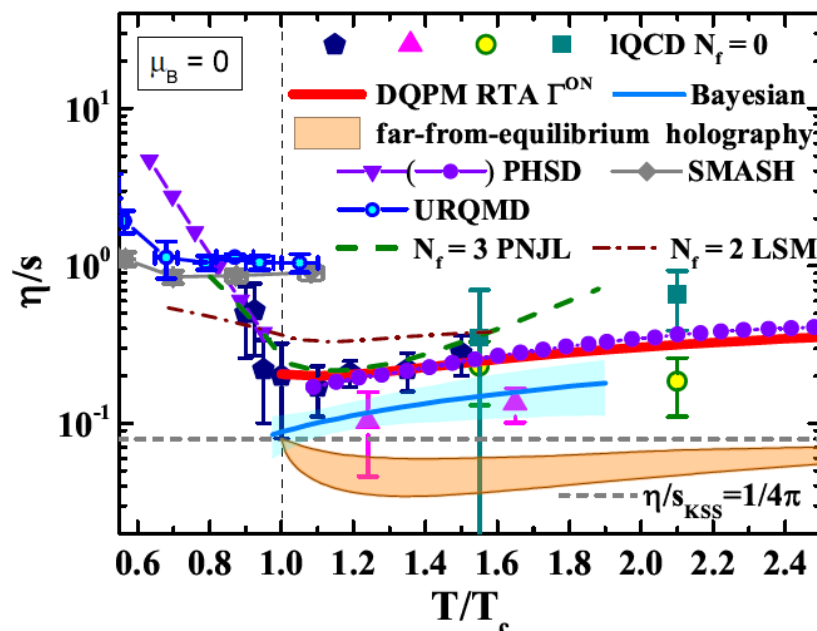
time evolution



Path through the phase diagram is not trivial and
 localized

Specific shear viscosity compilation

Model predictions:



! Different models using the same EoS can have completely different transport coefficients!

Backup slides

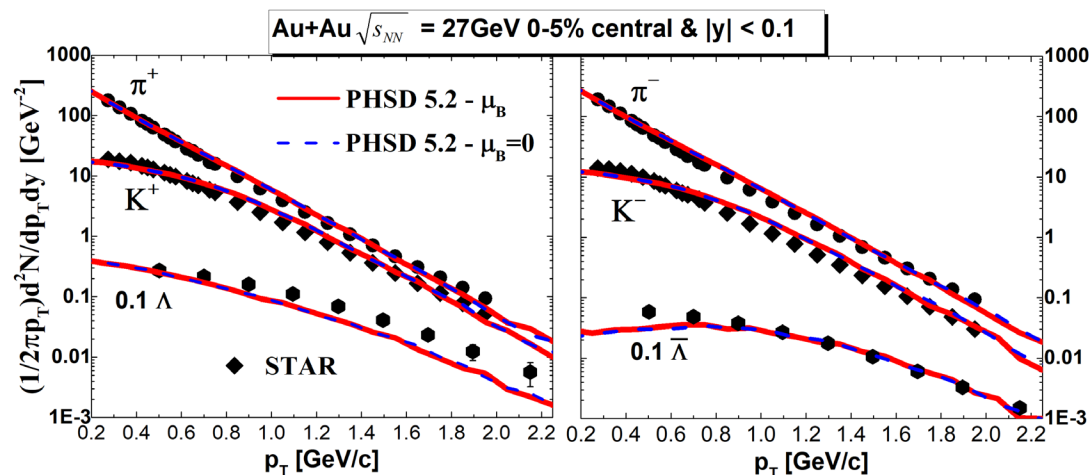
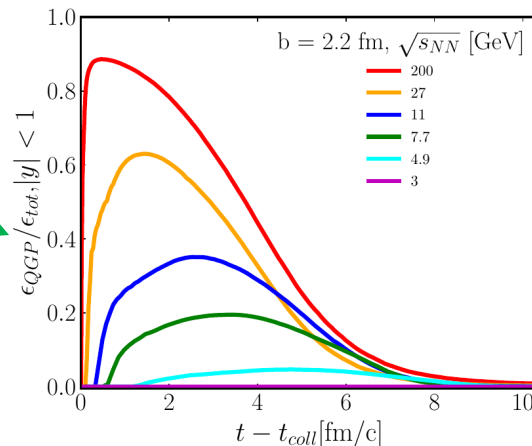
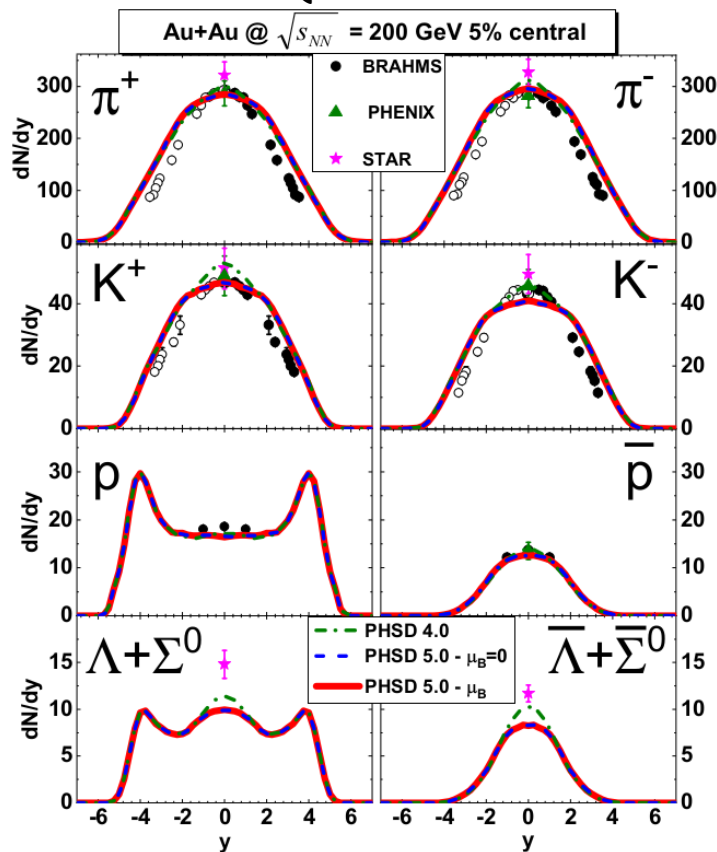
Results for ($\sqrt{s_{NN}} = 200 \text{ GeV} - 7 \text{ GeV}$)



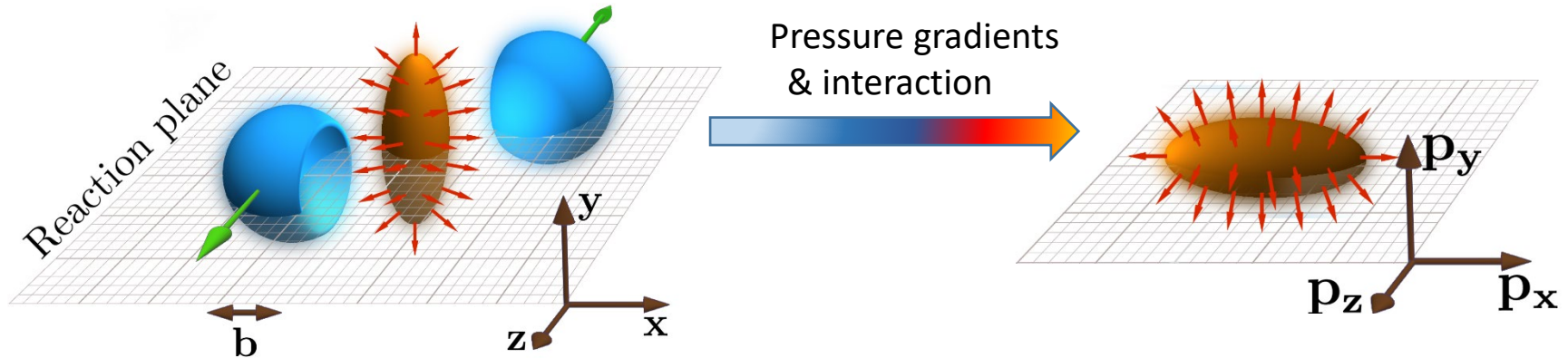
- No visible effects on p_T -spectra, dN/dy of μ_B -dependence
- Small effect of the angular dependence of $d\sigma/d\cos\theta$

at high $\sqrt{s_{NN}}$ - **low** μ_B

! QGP fraction is small at low $\sqrt{s_{NN}}$



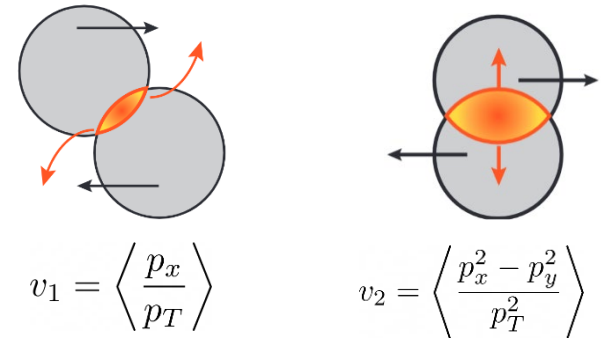
Anisotropic flow coefficients



Quantify the anisotropic flow using Fourier expansion

$$\frac{dN}{d\varphi} \propto \left(1 + 2 \sum_{n=1}^{+\infty} v_n \cos[n(\varphi - \psi_n)] \right)$$

$$v_n = \left\langle \cos n(\varphi - \psi_n) \right\rangle, \quad n = 1, 2, 3, \dots$$



Anisotropic flow

- Assess the transport properties of the QGP
- Sensitive to the QGP EoS and initial state
- Validate models of bulk evolution that are used in the computation of other observables

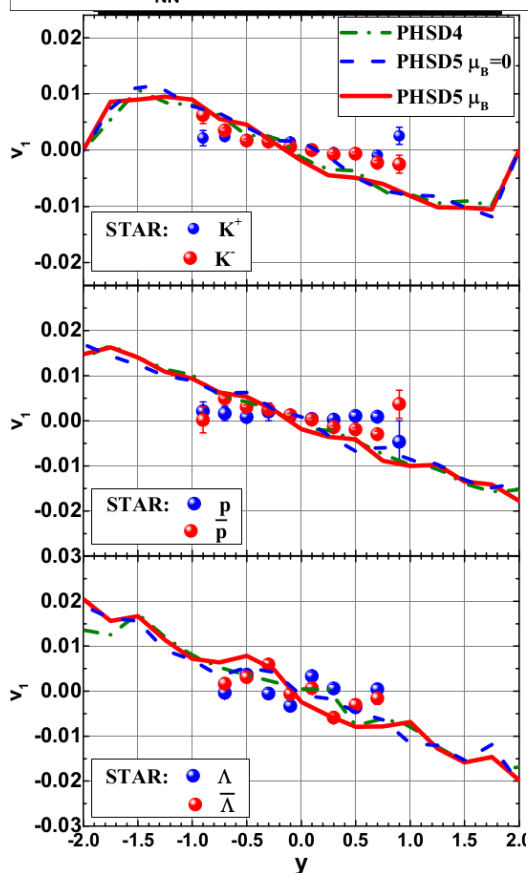
Directed flow ($\sqrt{s_{NN}} = 200 \text{ GeV vs } 27 \text{ GeV}$)

$$v_1 = \left\langle \frac{p_x}{p_T} \right\rangle$$

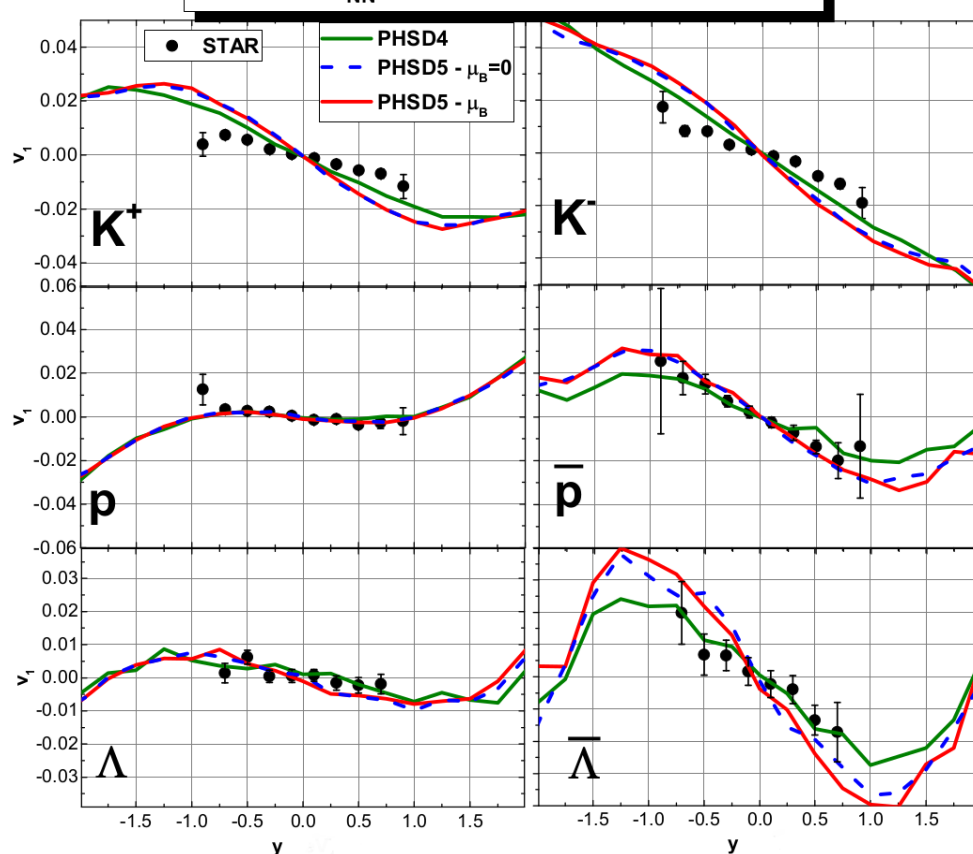
- Influence of the QGP dynamics on final particles observables
- Weak μ_B -dependence - **small** fraction of QGP or **low** μ_B

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no.1, 178-192

Au+Au $\sqrt{s_{NN}} = 200 \text{ GeV}$ 10-40% centrality



Au+Au $\sqrt{s_{NN}} = 27 \text{ GeV}$ 10-40% centrality

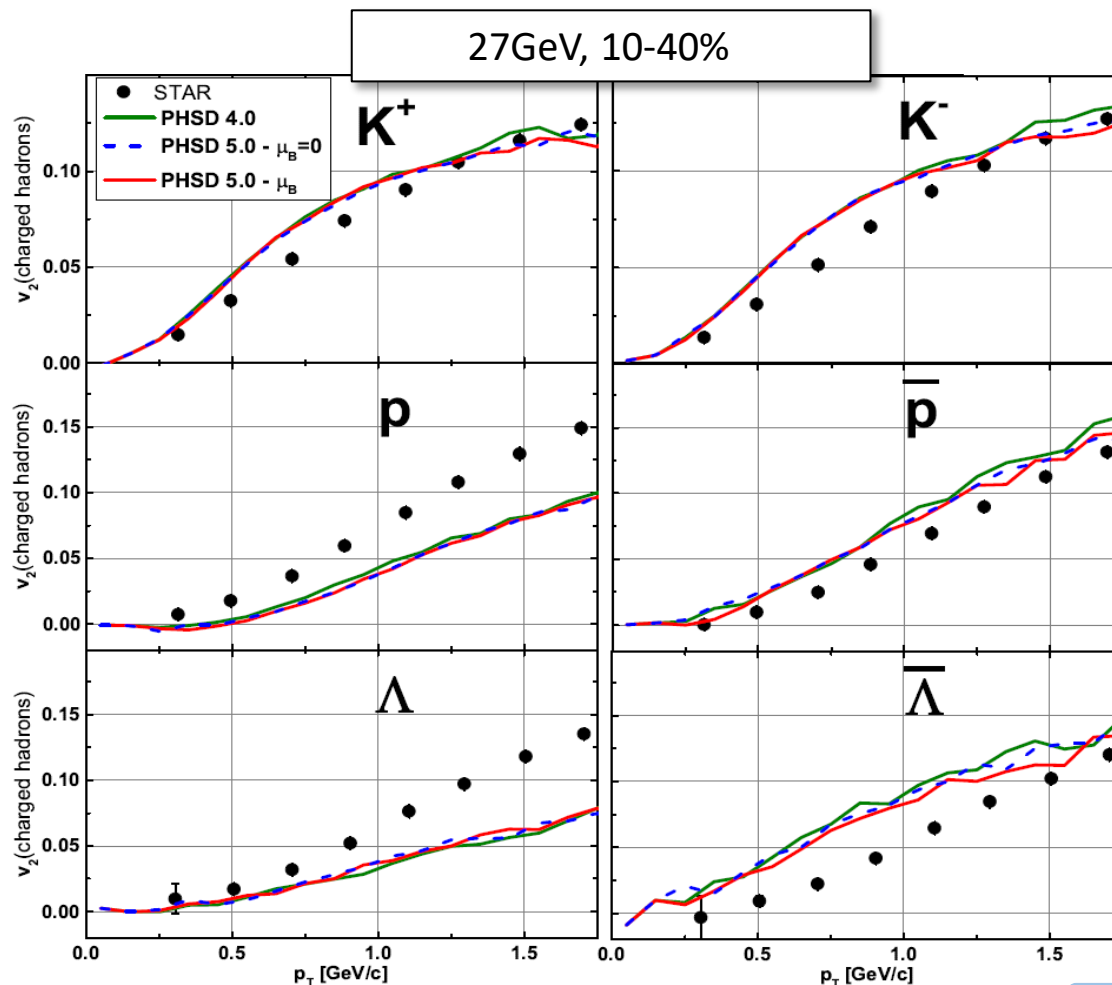
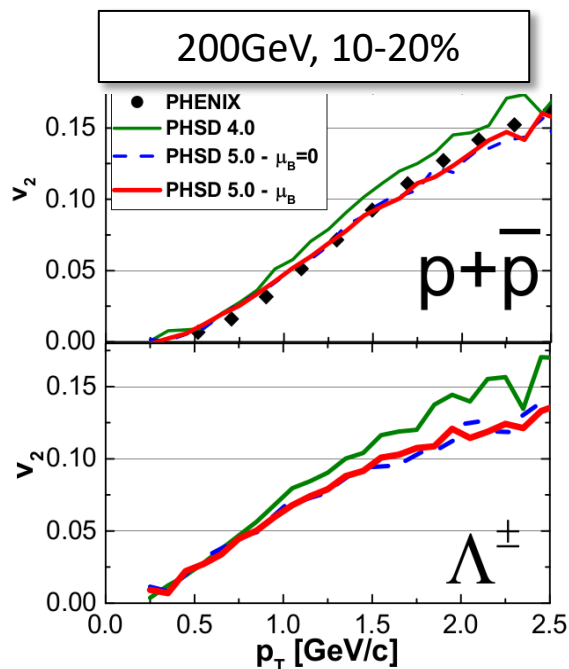


Elliptic flow ($\sqrt{s_{NN}} = 200 \text{ GeV vs } 27 \text{ GeV}$)

$$v_2 = \left\langle \frac{p_x^2 - p_y^2}{p_T^2} \right\rangle$$

- Weak μ_B -dependence
- Small effect of the angular dependence of $d\sigma/d\cos\theta$
- Strong flavor dependence

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Summary

Transport properties of the strongly-interacting QGP matter at finite T and μ_B have been investigated.

Influence of an order of a phase transition on thermodynamic and transport properties has been studied.

- Transport coefficients can differ among the models, which have similar phase structures and EoS
-

Evolution of the QGP matter created in HICs and the sensitivity of the bulk and flow observables on the QGP interactions and transport properties have been explored by the simulations within the PHSD transport approach

- **High- μ_B** regions are probed at **low $\sqrt{s_{NN}}$** or high rapidity regions
Moreover, **QGP** fraction is **small** at **low $\sqrt{s_{NN}}$** : small effect seen in observables
 - **μ_B -dependence** of QGP interactions is more pronounced in observables for strange hadrons and antiprotons
-

Outlook:

- Phase transition in PHSD
- Possible effects of phase transitions on flow/dileptons

Summary

Transport properties of the strongly-interacting QGP matter at finite T and μ_B have been investigated.

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Thank you for your attention!

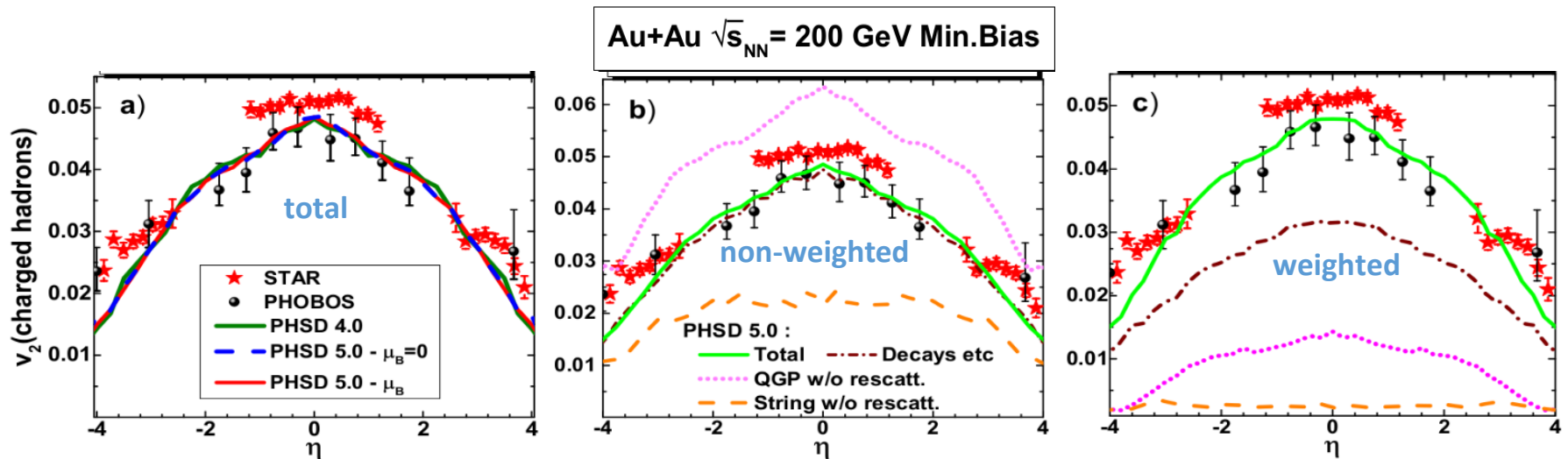
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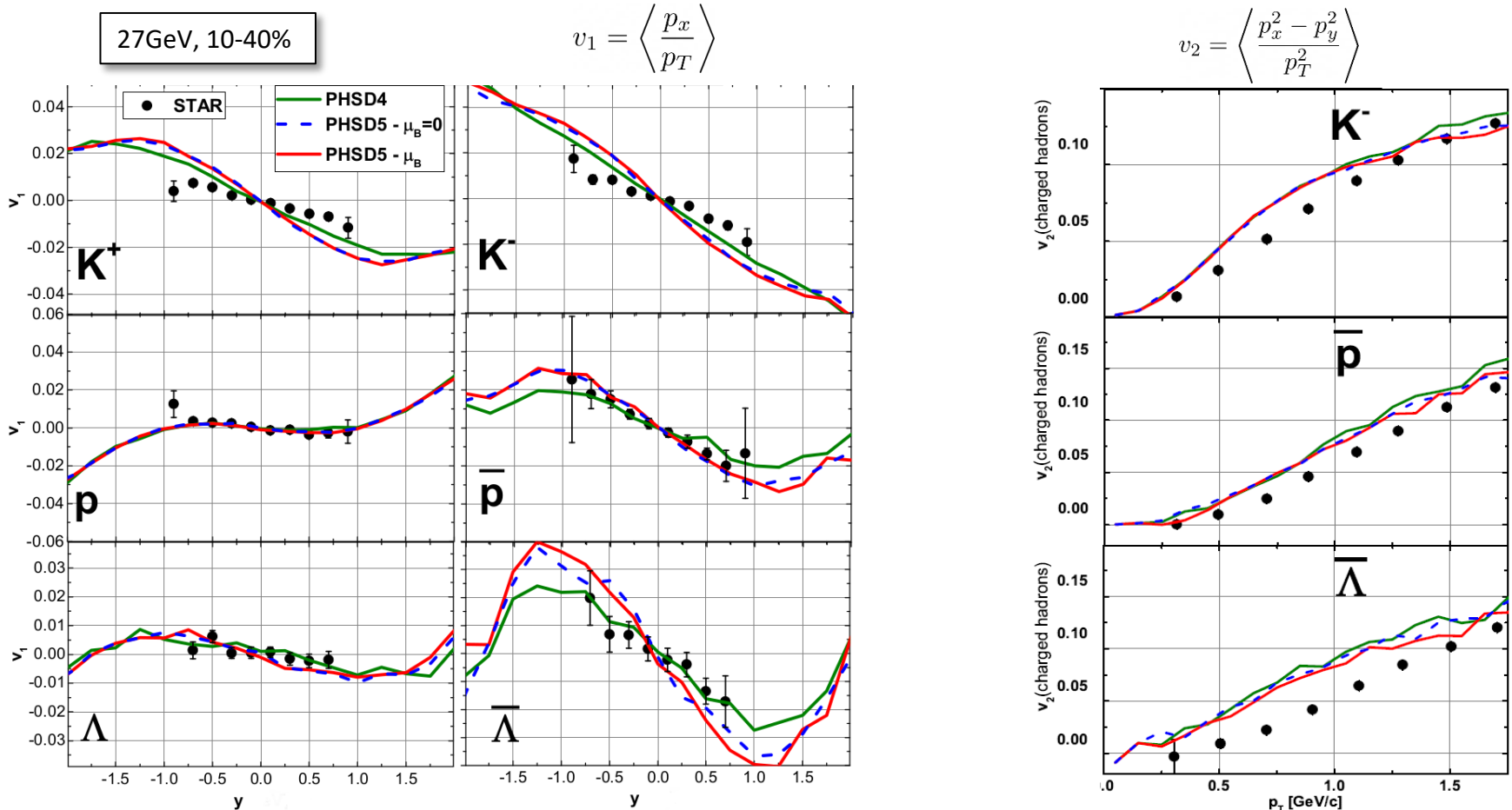
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no.1, 178-192

Channel decomposition :



Elliptic flow ($\sqrt{s_{NN}} = 200 \text{ GeV} - 27 \text{ GeV}$)

- Weak μ_B -dependence – small fraction of QGP or low μ_B
- Small effect of the angular dependence of $d\sigma/d\cos\theta$
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PNJL relaxation times

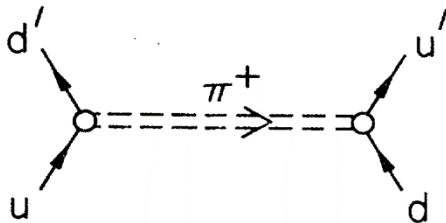
$$\tau_i(\mathbf{p}, T, \mu_B) = \frac{1}{\Gamma_i(\mathbf{p}, T, \mu_B)}$$

$$\Gamma_i^{\text{on}}(\mathbf{p}_i, T, \mu_q) = \frac{1}{2E_i} \sum_{j=q, \bar{q}, g} \int \frac{d^3 p_j}{(2\pi)^3 2E_j} d_j f_j(E_j, T, \mu_q) \int \frac{d^3 p_3}{(2\pi)^3 2E_3} \int \frac{d^3 p_4}{(2\pi)^3 2E_4} (1 \pm f_3)(1 \pm f_4)$$

$$|\bar{\mathcal{M}}|^2(p_i, p_j, p_3, p_4) (2\pi)^4 \delta^{(4)}(p_i + p_j - p_3 - p_4)$$

qq - interactions:

4 point interaction -> meson exchange ($\pi, \sigma, \eta, \eta', K, \dots$ for s, t, u channels)



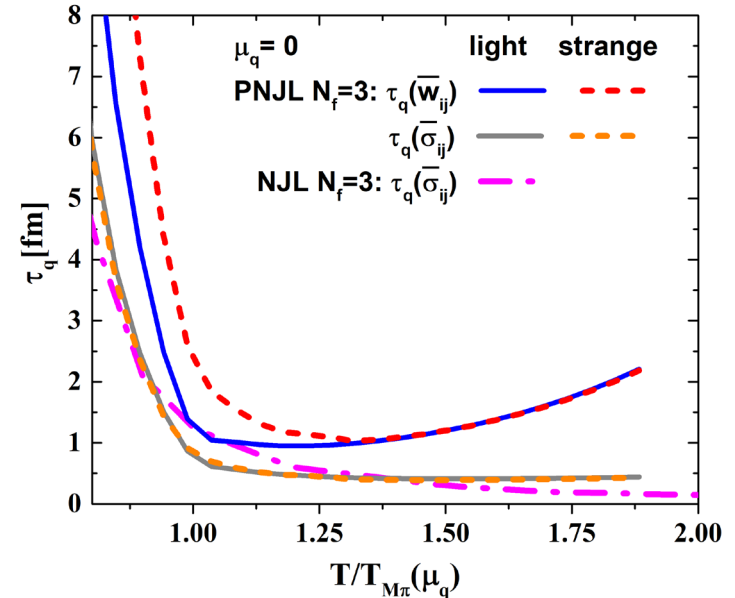
$$\boxed{\text{---}} \Rightarrow \boxed{\text{---}} = (i\gamma_5)\tau^{(-)} \frac{-ig_{\pi qq}^2}{k^2 - m_\pi^2} (i\gamma_5)\tau^{(+)}$$

meson propagator $\mathcal{D} = \frac{2ig_m}{1 - 2g_m \Pi_{ff'}^\pm(k_0, \vec{k})}$

Effective interaction in RPA

$$\text{---} \approx \text{X} + \text{---} + \text{---} + \dots = \frac{\text{X}}{1 - \text{---}}$$

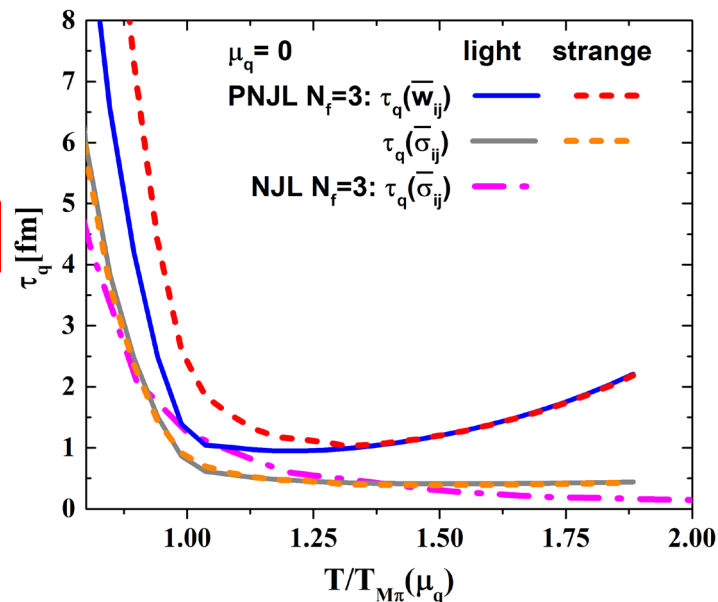
Relaxation times (PNJL vs NJL)



PNJL relaxation times

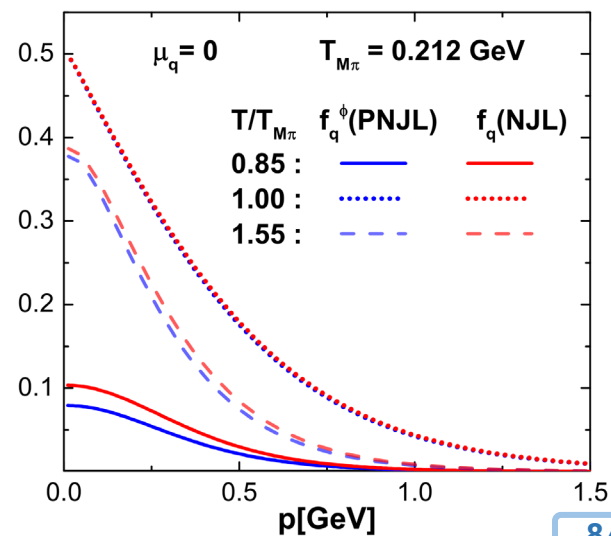
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Modified distribution functions:
Polyakov loop contributions

$$f_q \rightarrow f_q^\Phi(\mathbf{p}, T, \mu) = \frac{(\bar{\Phi} + 2\Phi e^{-(E_{\mathbf{p}} - \mu)/T}) e^{-(E_{\mathbf{p}} - \mu)/T} + e^{-3(E_{\mathbf{p}} - \mu)/T}}{1 + 3(\bar{\Phi} + \Phi e^{-(E_{\mathbf{p}} - \mu)/T}) e^{-(E_{\mathbf{p}} - \mu)/T} + e^{-3(E_{\mathbf{p}} - \mu)/T}},$$



QGP in the Polyakov extended NJL model

- PNJL model based on effective Lagrangian with the same symmetries for the quark dof as QCD

$$\begin{aligned}\mathcal{L}_{PNJL} = & \sum_i \bar{\psi}_i (iD - m_{0i} + \mu_i \gamma_0) \psi_i \\ & + G \sum_a \sum_{ijkl} \left[(\bar{\psi}_i i\gamma_5 \tau_{ij}^a \psi_j) (\bar{\psi}_k i\gamma_5 \tau_{kl}^a \psi_l) + (\bar{\psi}_i \tau_{ij}^a \psi_j) (\bar{\psi}_k \tau_{kl}^a \psi_l) \right] \\ & - K \det_{ij} [\bar{\psi}_i (-\gamma_5) \psi_j] - K \det_{ij} [\bar{\psi}_i (+\gamma_5) \psi_j] \\ & - \mathcal{U}(T; \Phi, \bar{\Phi})\end{aligned}$$

← Polyakov-loop effective potential fitted to the YM

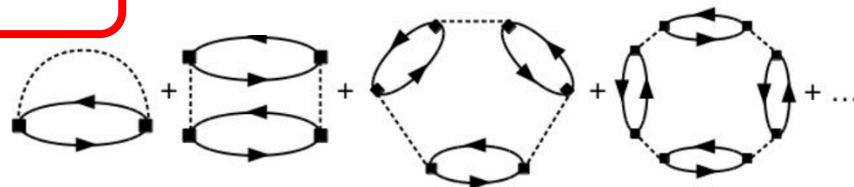
5 parameters fixed by vacuum values K, π masses, η - η' mass splitting, π decay constant, Chiral condensate

Improvements:

- Next to leading order in $N_c(O(1/N_c)^0)$ of the grand-canonical potential : presence of the mesons below T_c

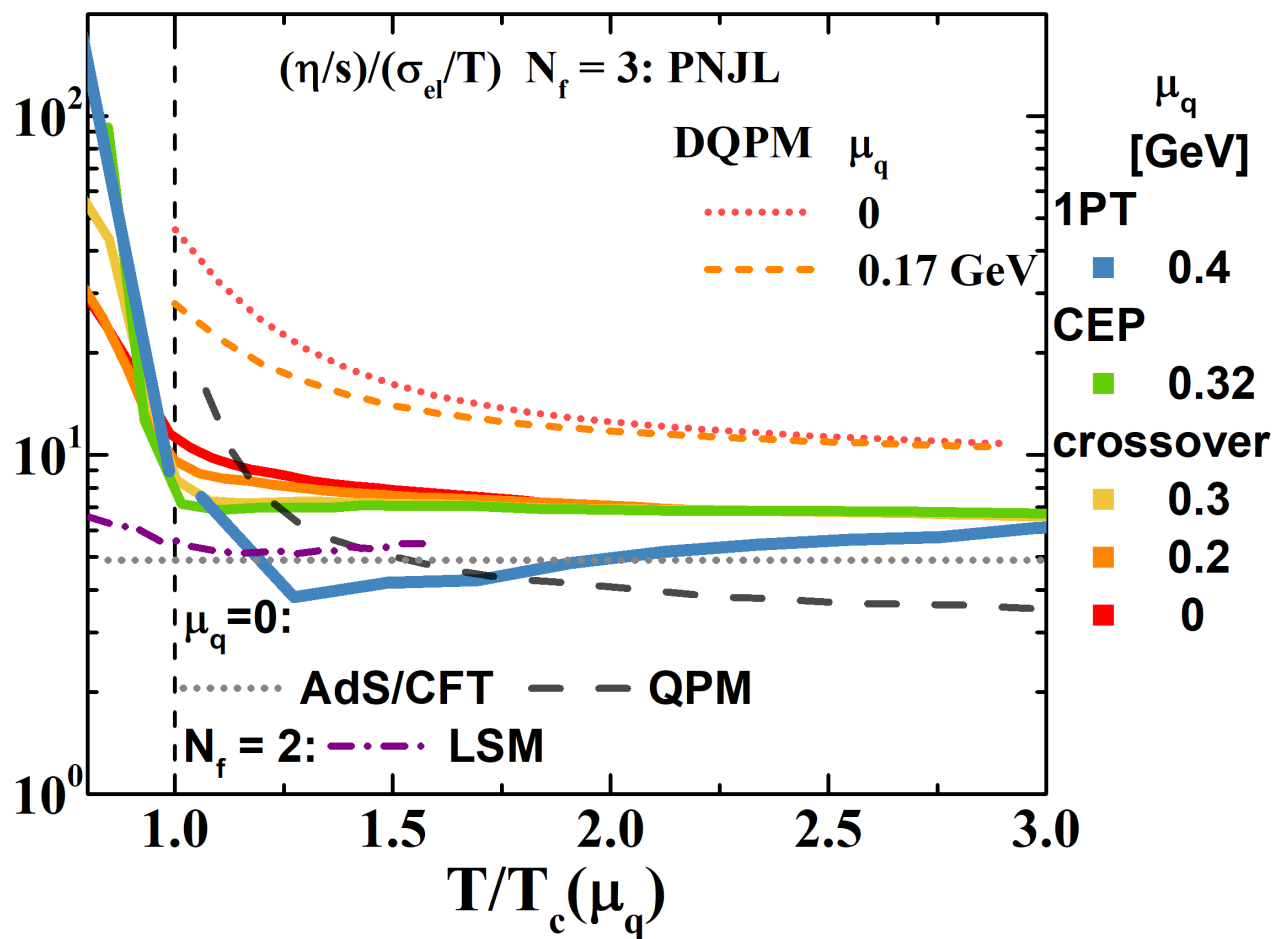
$$\Omega_{PNJL}(T, \mu_i) = \Omega_q^{(-1)}(T, \mu_i) + \sum_{M \in J^\pi = \{0^+, 0^-\}} \Omega_M^{(0)}(T, \mu_M(\mu_i)) + \mathcal{U}_{glue}(T) ,$$

J. M. Torres-Rincon, J. Aichelin PRC 96 (2017) 4 045205
D. Fuseau, T. Steinernert, J. Aichelin PRC 101 (2020) 6 065203



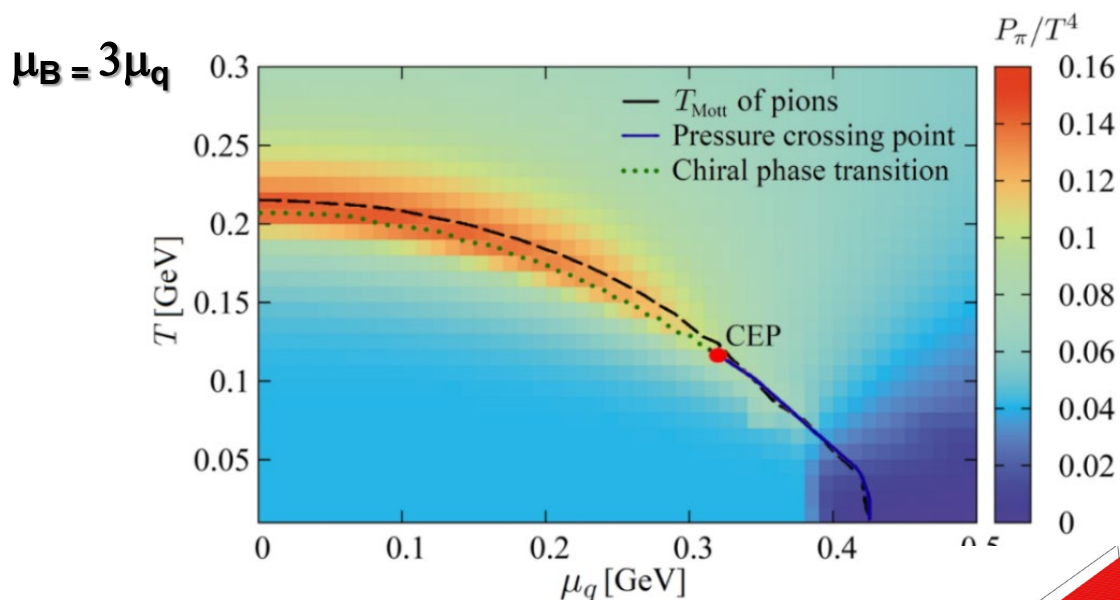
- Modification of the gluon potential due to the presence of the quark

Specific shear viscosity to conductivity



QGP in the Polyakov extended NJL model

- PNJL allows for prediction of macroscopic properties of QGP at finite T and μ_B
- & QGP transport coefficients for $0 \leq \mu_B \leq 1.2$ GeV

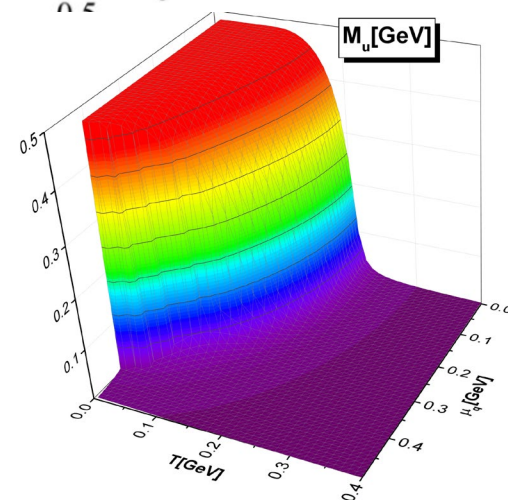


D. Fuseau, T. Steinert, J. Aichelin PRC 101 (2020) 6 065203

- **CEP**: $(T, \mu_B) = (110, 960)$ MeV, $\mu_B/T = 8.73$
- 1st order PT at high μ_B
- **same symmetries** for the quarks as QCD

Chiral masses (M_l, M_s)

$$m_i = m_{0i} - 4G\langle\langle\bar{\psi}_i\psi_i\rangle\rangle + 2K\langle\langle\bar{\psi}_j\psi_j\rangle\rangle\langle\langle\bar{\psi}_k\psi_k\rangle\rangle$$



Specific shear viscosity compilation

- **Kubo formalism: transport coefficients are expressed through correlation functions of stress-energy tensor**

used in **lattice QCD, transport approaches(hadrons), effective models**

$$\eta = \frac{1}{20} \lim_{\omega \rightarrow 0} \frac{1}{\omega} \int d^4x e^{i\omega t} \langle [S^{ij}(t, \mathbf{x}), S^{ij}(0, \mathbf{0})] \rangle \theta(t) \quad S^{ij} = T^{ij} - \delta^{ij} \mathcal{P}$$

$$\zeta = \frac{1}{2} \lim_{\omega \rightarrow 0} \frac{1}{\omega} \int d^4x e^{i\omega t} \langle [\mathcal{P}(t, \mathbf{x}), \mathcal{P}(0, \mathbf{0})] \rangle \theta(t) \quad \mathcal{P} = -\frac{1}{3} T^i_i$$

R. Lang and W. Weise, EPJ. A 50, 63 (2014) (NJL model)

A. Harutyunyan et al, PRD 95, 114021, (2017)

Kinetic theory:

- **Relaxation time approximation(RTA) : consider relaxation time** $\frac{df_a^{\text{eq}}}{dt} = C_a = -\frac{f_a^{\text{eq}} \phi_a}{\tau_a}$

P. Chakraborty and J. I. Kapusta, PRC 83,014906 (2011)

- **Chapman-Enskog : expand the distribution in terms of the Knudsen number**

J. A. Fotakis et al, PRD 101 (2020) 7, 076007 (HRG)

And more!

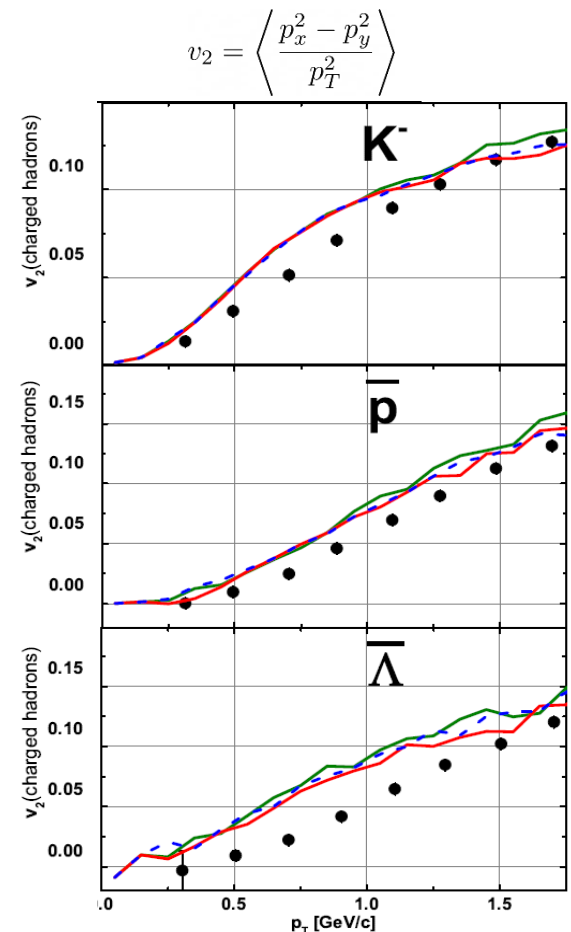
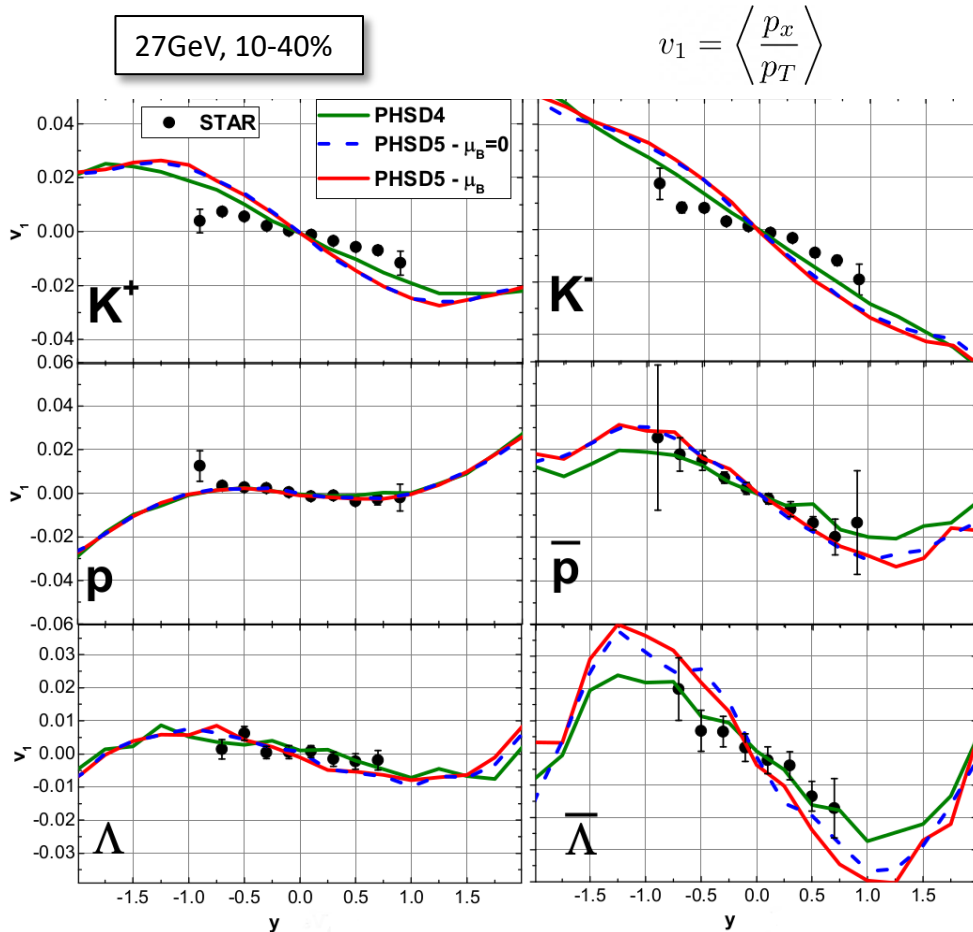
Holographic models: AdS/CFT correspondence

D. T. Son and A. O. Starinets, JHEP 0603, 052 (2006)

M. Attems et al , JHEP 10 (2016), 155.

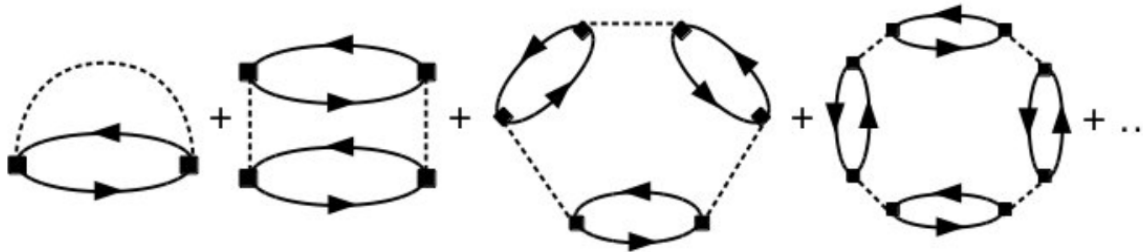
Elliptic flow ($\sqrt{s_{NN}} = 200 \text{ GeV} - 27 \text{ GeV}$)

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PNJL improvements

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presence of the mesons below T_c



J. M. Torres-Rincon, J. Aichelin PRC 96 (2017) 4 045205

- **Modification of the gluon potential due to the presence of the quark**

$$\frac{U(\phi, \bar{\phi}, T)}{T^4} = -\frac{b_2(T)}{2} \bar{\phi}\phi - \frac{b_3}{6} (\bar{\phi}^3 + \phi^3) + \frac{b_4}{4} (\bar{\phi}\phi)^2$$

$$b_2(T) = a_0 + \frac{a_1}{1+\tau} + \frac{a_2}{(1+\tau)^2} + \frac{a_3}{(1+\tau)^3} \quad \text{where} \quad \tau_{\text{phen}} = 0.57 \frac{T - T_{\text{phen}}^{\text{cr}}(T)}{T_{\text{phen}}^{\text{cr}}(T)}$$

$$T_{\text{phen}}(T) = a + bT + cT^2 + dT^3 + \boxed{e \frac{1}{T}}$$

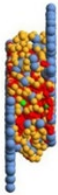
D. Fuseau, T. Steinernert, J. Aichelin PRC 101 (2020) 6 065203

Stages of collisions in PHSD

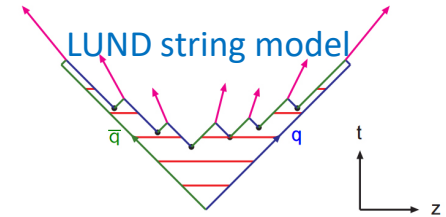
Initial A+A
collision



Partonic phase

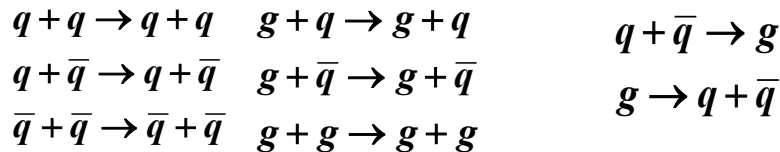


- String formation in primary NN collisions
→ decays to pre-hadrons (baryons and mesons)

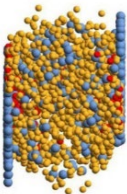


- Formation of a QGP state if $\varepsilon > \varepsilon_{critical}$:
Dissolution of pre-hadrons → DQPM

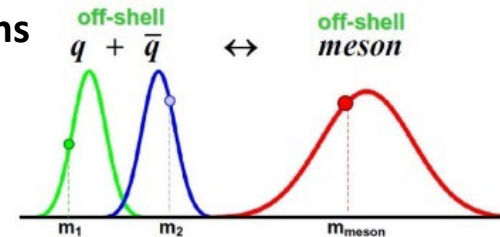
→ massive quarks/gluons and mean-field energy
(quasi-)elastic collisions : inelastic collisions:



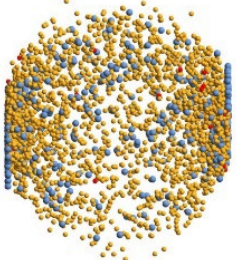
Hadronization



- Hadronization to colorless off-shell mesons and baryons
 $g \rightarrow q + \bar{q}, \quad q + \bar{q} \leftrightarrow \text{meson ('string')}$
 $q + q + q \leftrightarrow \text{baryon ('string')}$



Hadronic phase

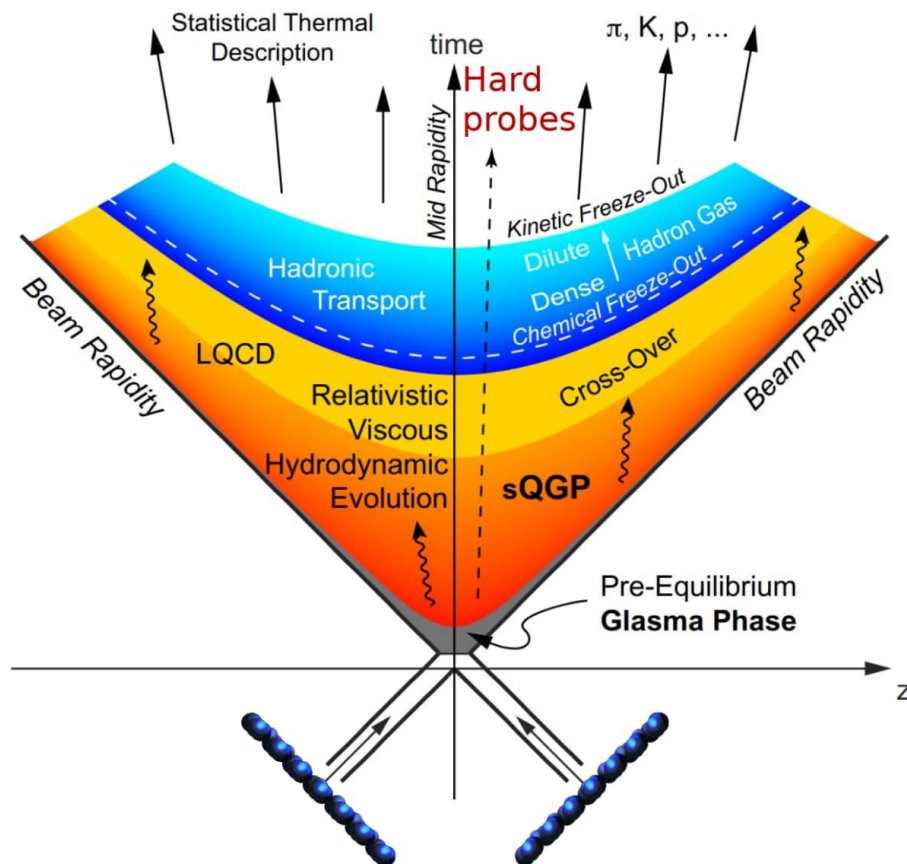


Strict 4-momentum and quantum number conservation

- Hadron-string interactions – off-shell HSD

W. Cassing, E. Bratkovskaya, PRC 78 (2008) 034919; NPA831 (2009) 215;
W. Cassing, EPJ ST 168 (2009) 3

Stages of HIC



$t > 10 \text{ fm}$ - hadronisation and free stream to detectors

$10 \text{ fm} > t > 1 \text{ fm}$ - QGP expansion

$t \approx 1 \text{ fm}$ - Equilibration

$t \ll 1 \text{ fm}$ - Initial state

QGP out-of equilibrium $\leftrightarrow$ HIC

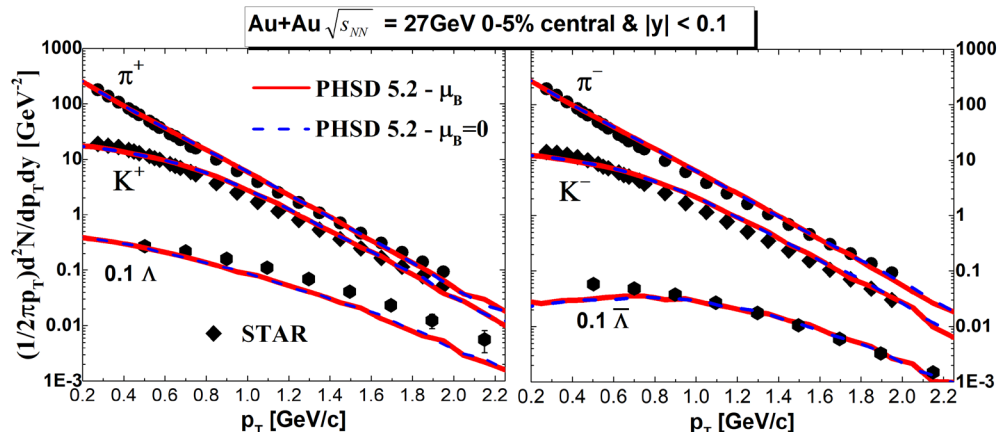
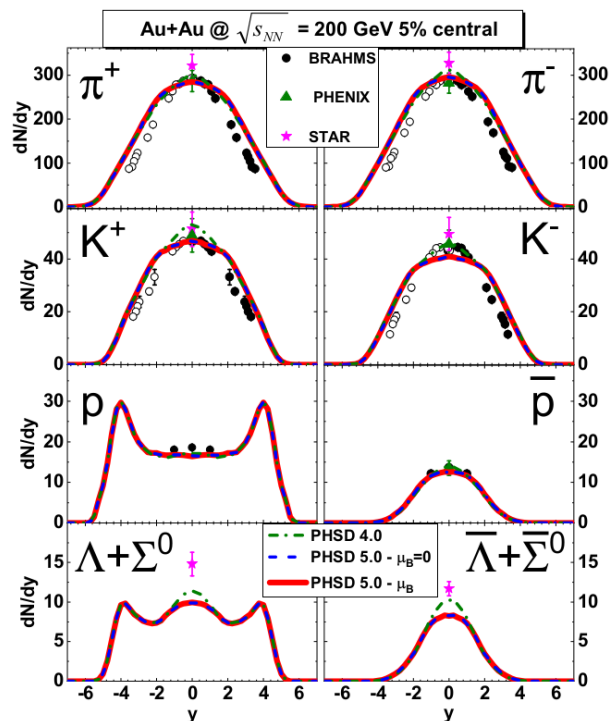
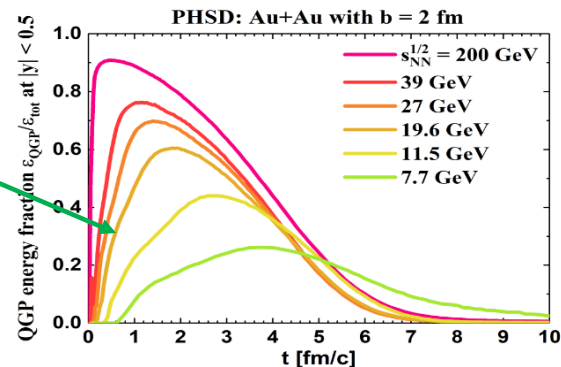
Results for ($\sqrt{s_{NN}} = 200 \text{ GeV} - 7 \text{ GeV}$)



- No visible effects on p_T -spectra, dN/dy of μ_B -dependence
- Small effect of the angular dependence of $d\sigma/d\cos\theta$

at high $\sqrt{s_{NN}}$ - **low** μ_B

! QGP fraction is **small** at low $\sqrt{s_{NN}}$



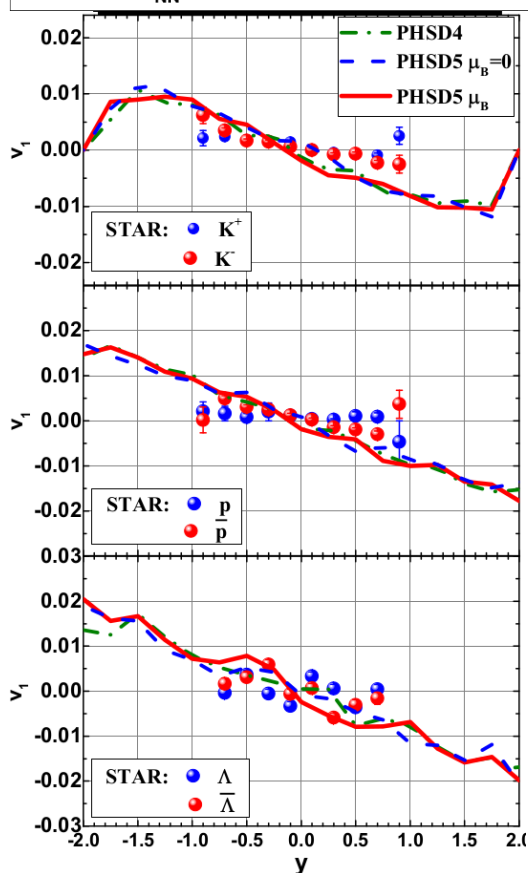
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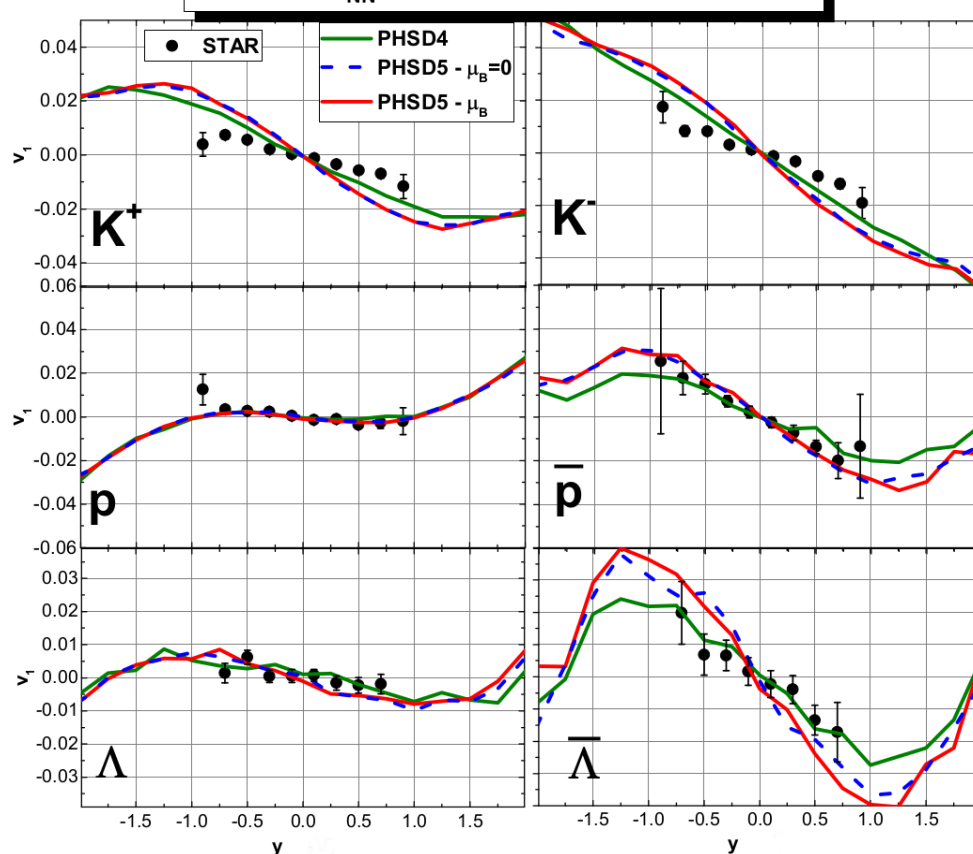
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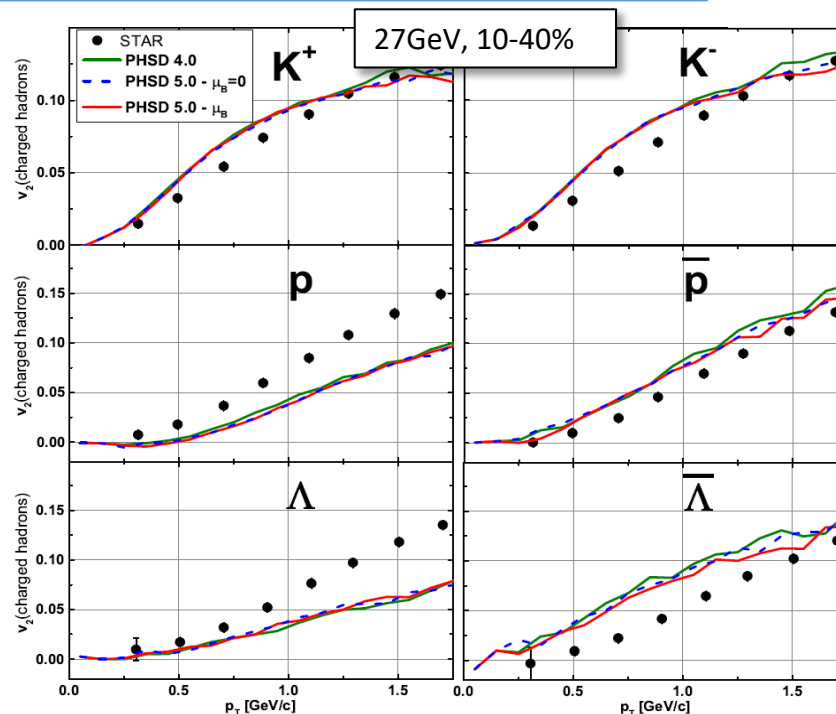
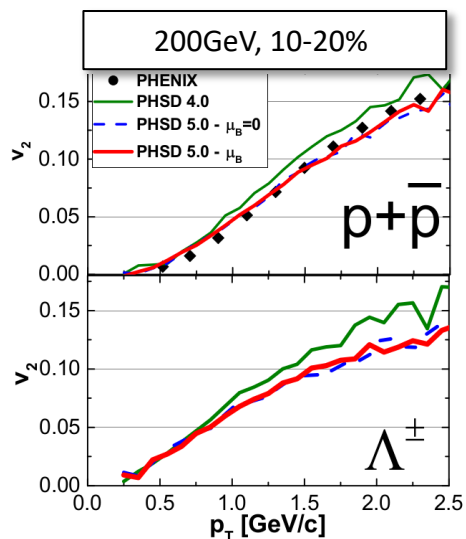


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